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# Flexural Modeling and Design of Fiber Reinforced Concrete Structural Sections

How Fiber Reinforced Concrete can be integrated as an economical, sustainable, and durable aspect of our infrastructure

**PhD Summer School, Politecnico di Milano**

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**Barzin Mobasher**

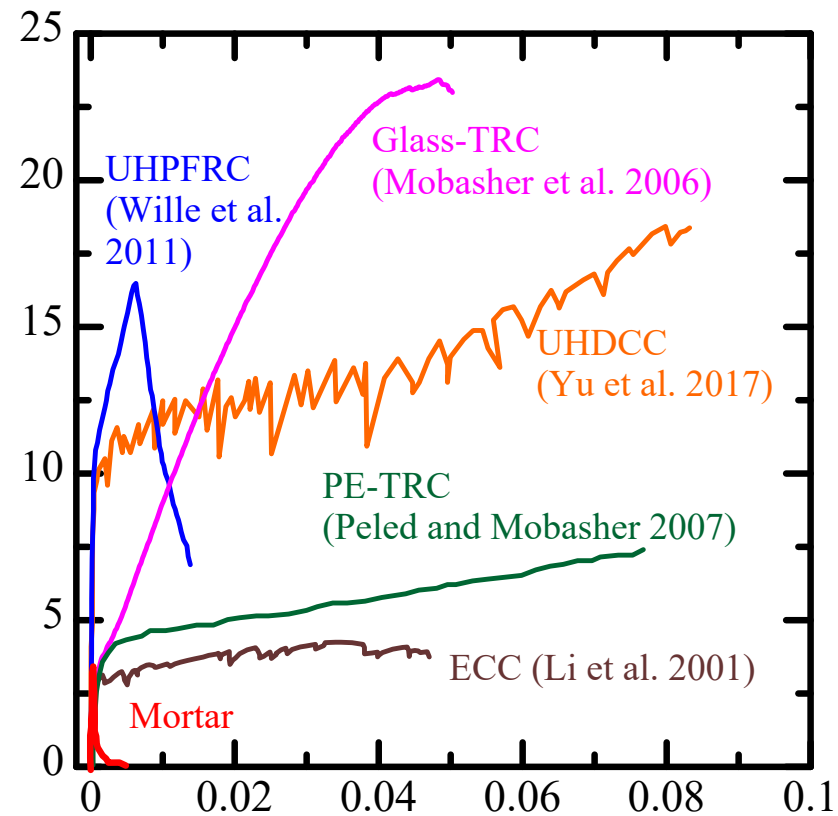
**Arizona State University**

# Outline

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- Objectives
- Overview of FRC material responses
- Parametric Material models
- Algorithm
- Closed form solutions for moment curvature diagram
- Load deflection response
- Simulations
- Conclusions

# Strain Softening and Strain hardening

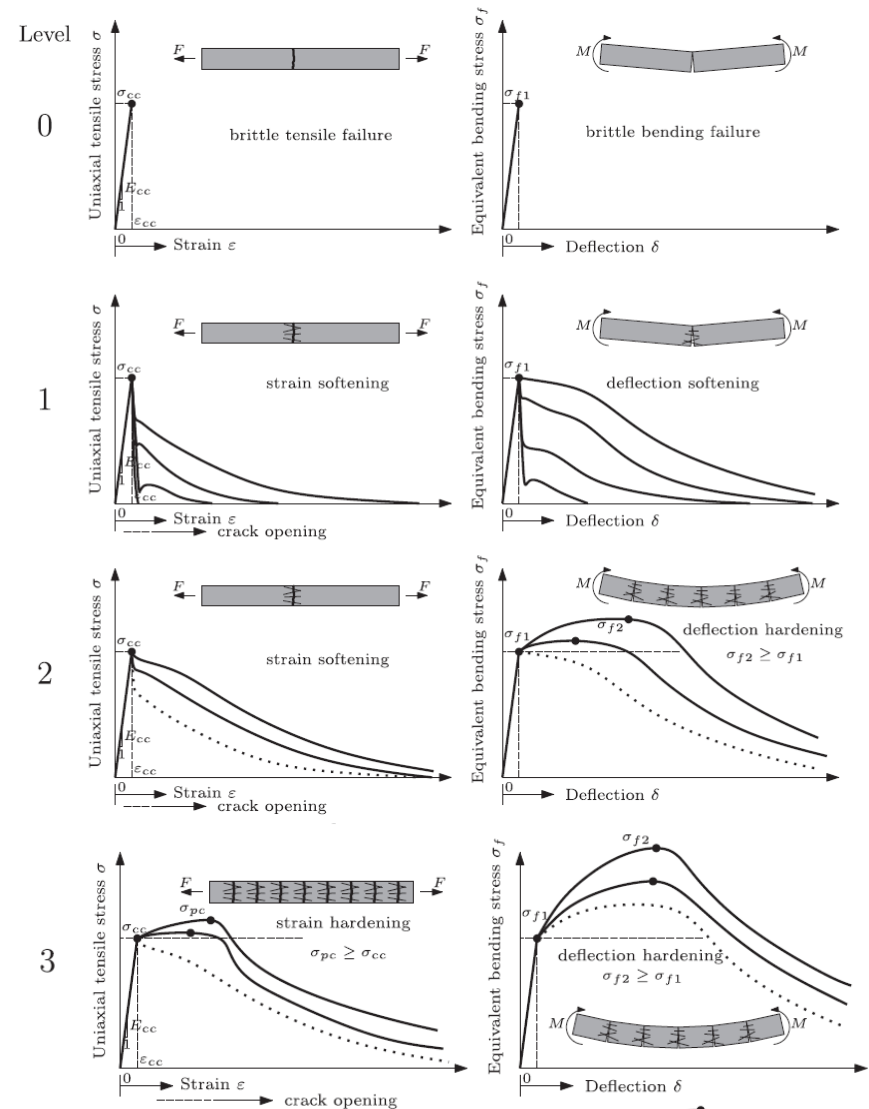


Typical stress-elongation curves in tension of fiber reinforced cement composites: (a) strain-softening.  
(b) strain-hardening.

Tensile stress-strain curves of strain hardening FRCs

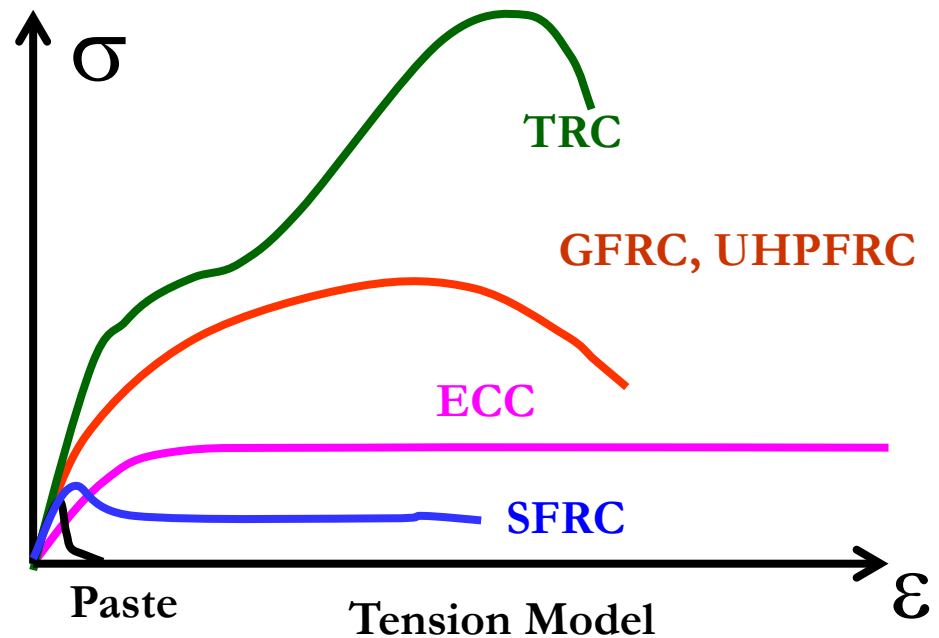
# Unified representation of FRC Composites

- Four Response Systems
  - Strain Softening/Hardening, Deflection Softening/Hardening
- Strain softening behavior
  - Discrete fiber systems
  - SFRC, PFRC, GFRC, SIMCON, SIFCON
- Strain hardening behavior
  - Discrete & continuous fiber systems
  - Textile reinforced cement (TRC), GRFC, Ferrocement (FRC), ECC
- Load-Deflection Calculations



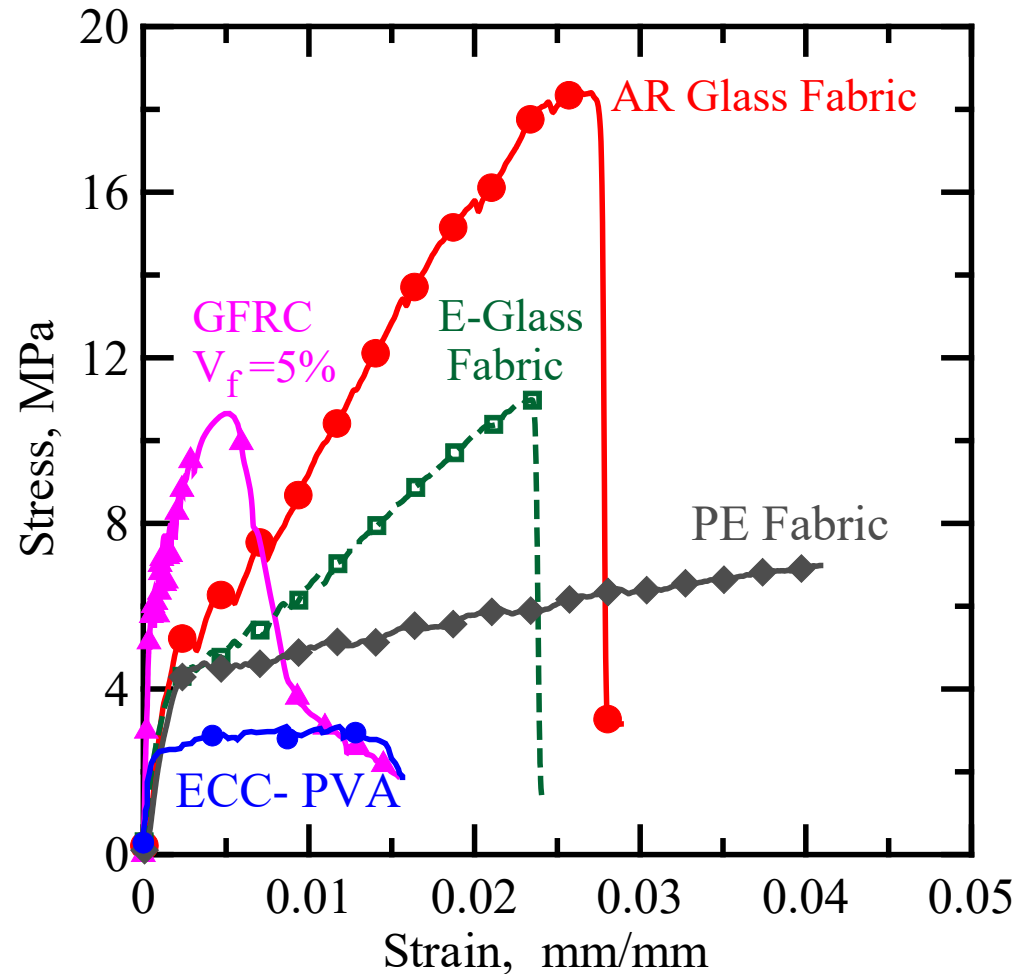
# Materials Responses, Structural Consequences

- FRC materials as structural members
- Similarity in material constitutive models
  - Tensile response ~ various levels of ductility
  - Tensile strength  $\ll$  Compressive strength
- Elastic response
  - Use standard composite materials models
  - Composite laminate theory
  - Finite difference approach for tension stiffening
- Allow damage based models
  - Fracture mechanics
  - Damage based models
  - Plasticity based models

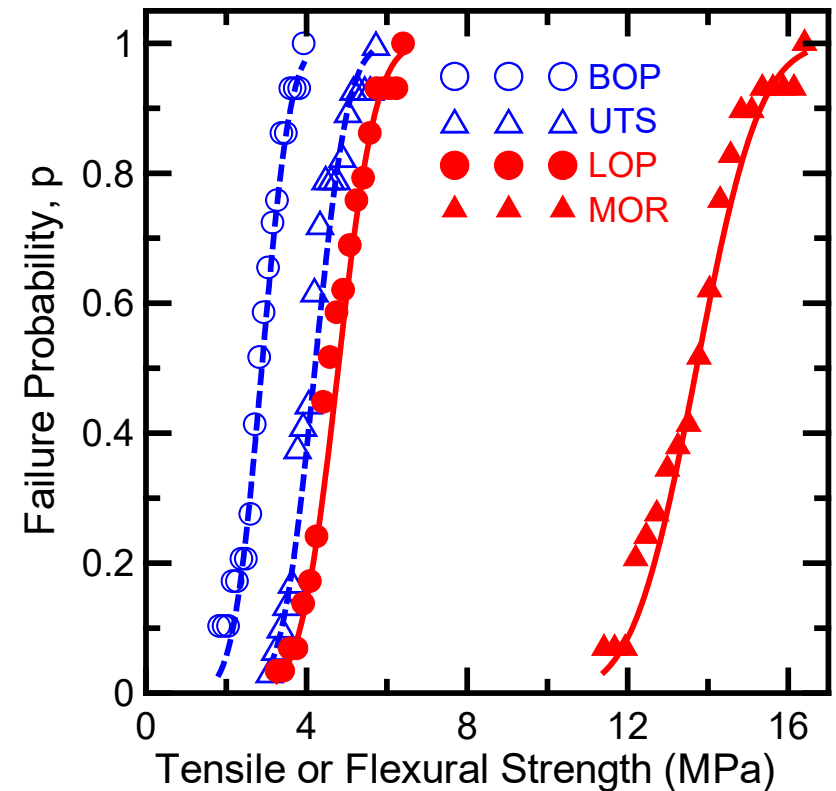
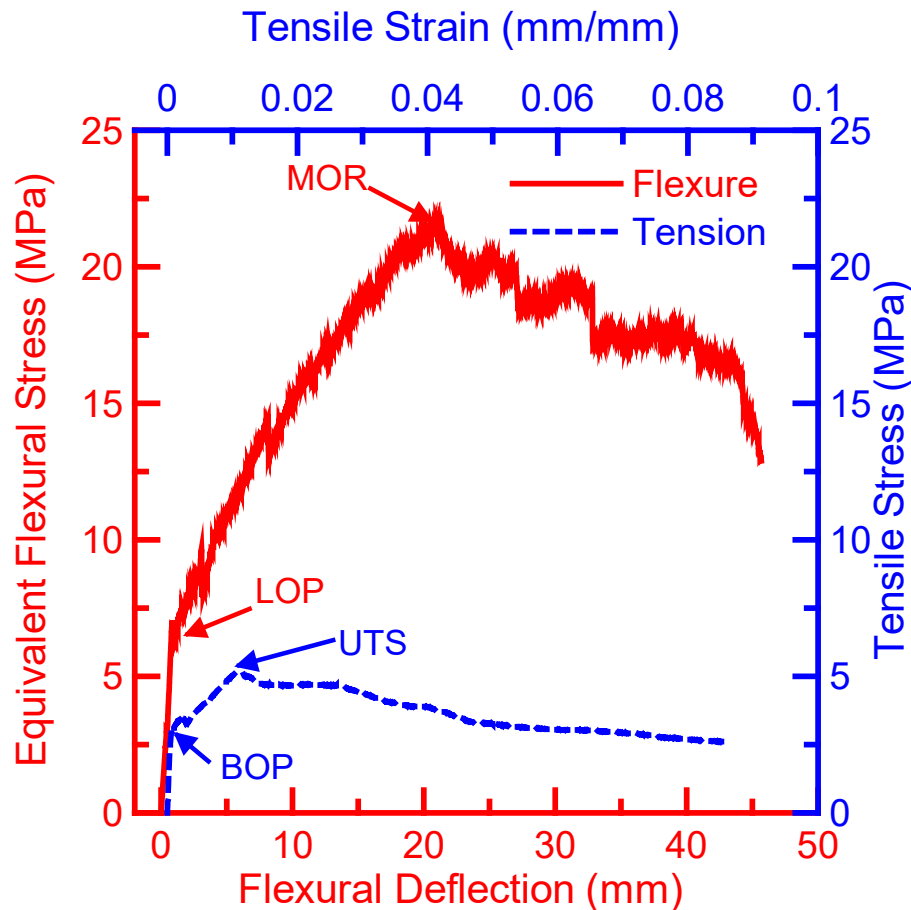


# Strain Hardening Cement Based Composites

- Initial linear elastic response
- First cracking strength a function of fiber content
- Distributed cracking start to saturation
- Saturation crack spacing
- Tension stiffening
- Ultimate strength
- For Strain hardening cement composites, we should ignore the strain softening portion



# Tensile Strength vs. Flexural Strength



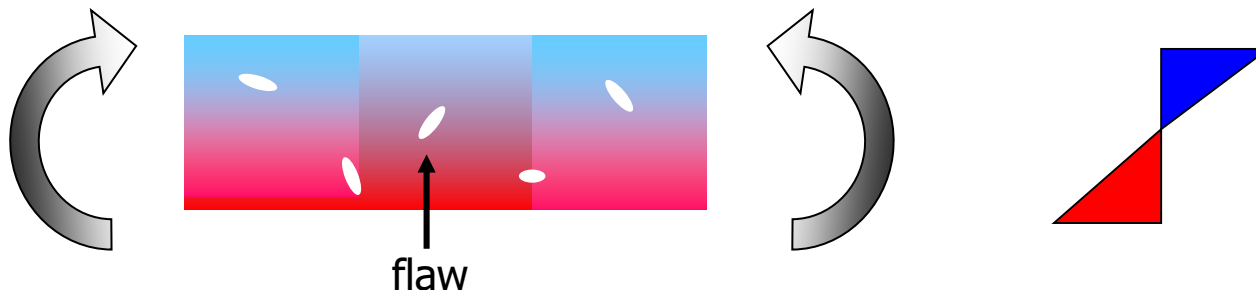
Aldea C. M., Mobasher, B., Jain, N., "Cement-Based Matrix-Grid System For Masonry Rehabilitation," *Textile Reinforced Concrete (TRC) - German/International Experience* symposium, ACI Special Publications, ACI Special Publication, 2007.

# Uniaxial Tension vs. Flexural Bending

- Uniaxial Tension
  - Pro: Constitutive material properties
  - Con: Tension test is difficult to do!

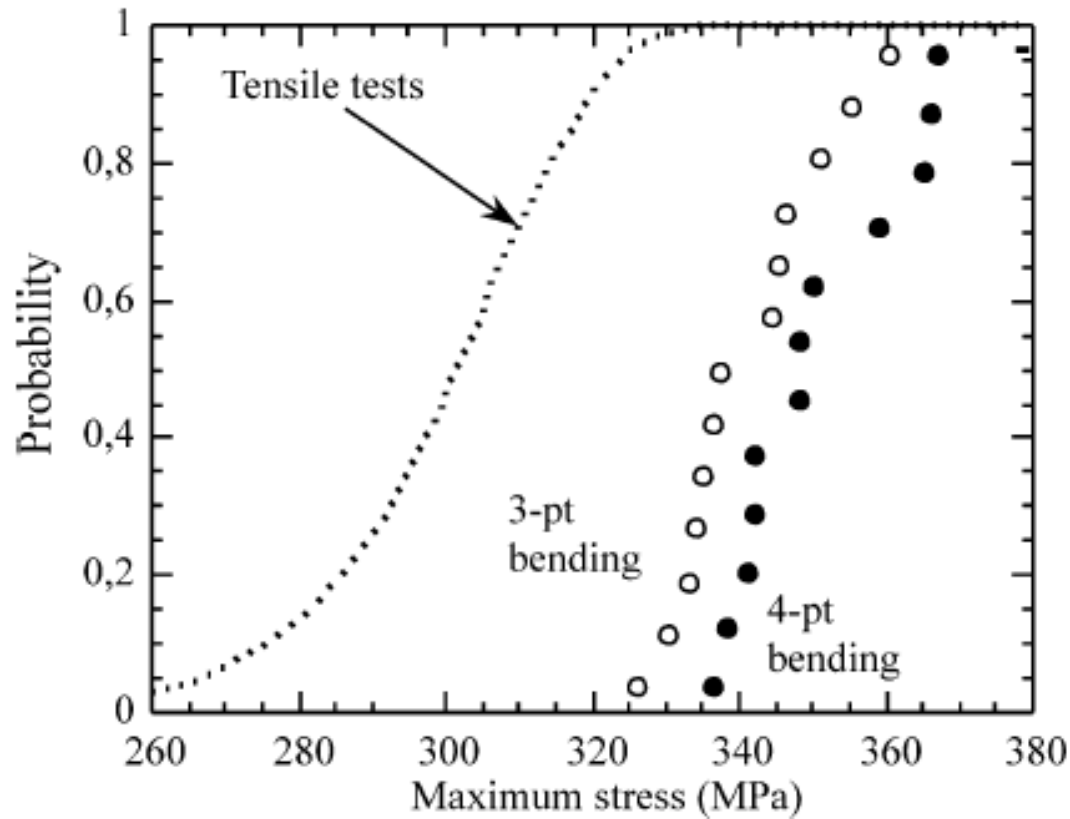


- Flexural Bending (4 point bending)
  - Pro: Bending test is easier to run
  - Con: Unconservative results!





# Size Effect



- V.Calard, J.Lamon, "A probabilistic-statistical approach to the ultimate failure of ceramic-matrix composites—part I: experimental investigation of 2D woven SiC/SiC composites," , Composites Science and Technology 62 (2002) 385–393

# Weibull based approach

- The fracture probability can be given as:

$$F(\sigma) = 1 - \exp \left[ -V \left( \frac{\sigma}{\sigma_0} \right)^m \right]$$

- Mean and COV

$$\bar{\sigma} = \sigma_0 V^{-1/m} \Gamma \left( 1 + \frac{1}{m} \right) \quad CV = \left[ \frac{\Gamma \left( 1 + \frac{2}{m} \right)}{\Gamma^2 \left( 1 + \frac{1}{m} \right)} - 1 \right]^{1/2} \times 100$$

- The effective volume of the four-point bend strength tested with a load span of one-third of a support span is given by:

$$V = \frac{m+3}{6(m+1)} V_0$$

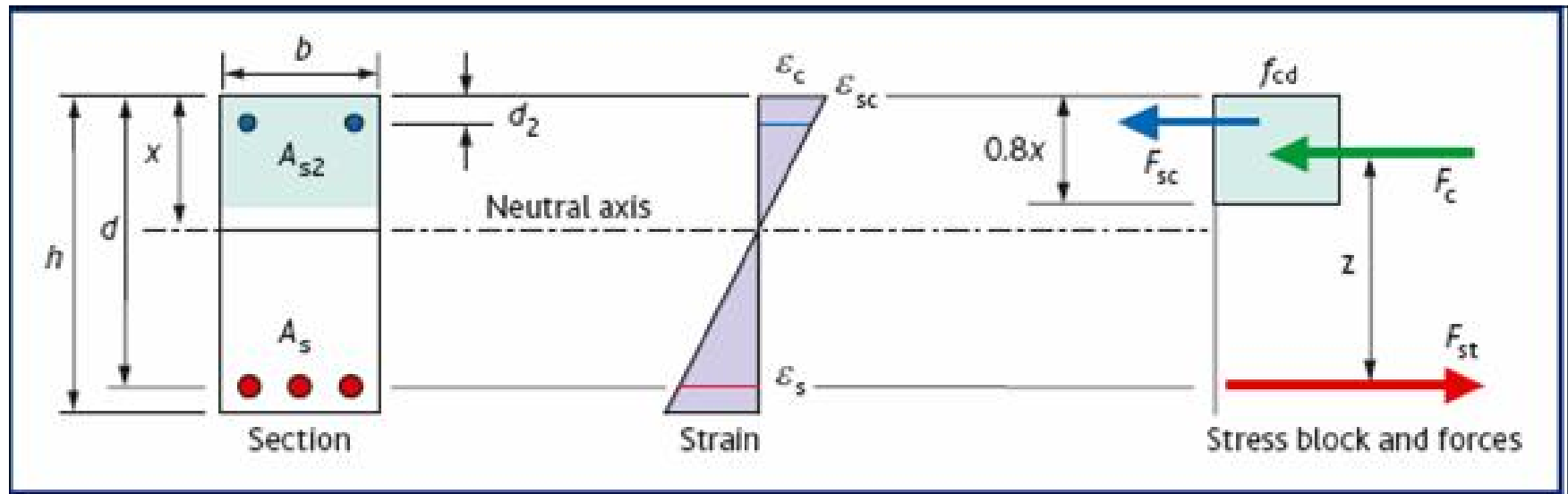
- Comparison of the failure probabilities of the tensile and the 3-pt bending test specimens:

$$\frac{\sigma_b}{\sigma_t} = \left[ 2(m+1) 2 \left( \frac{V_t}{V_b} \right) \right]^{1/m}$$

Assume  $m=7$ ,  $V_t/V_b = 1$ ,  $\sigma_b/\sigma_t = 1.64$

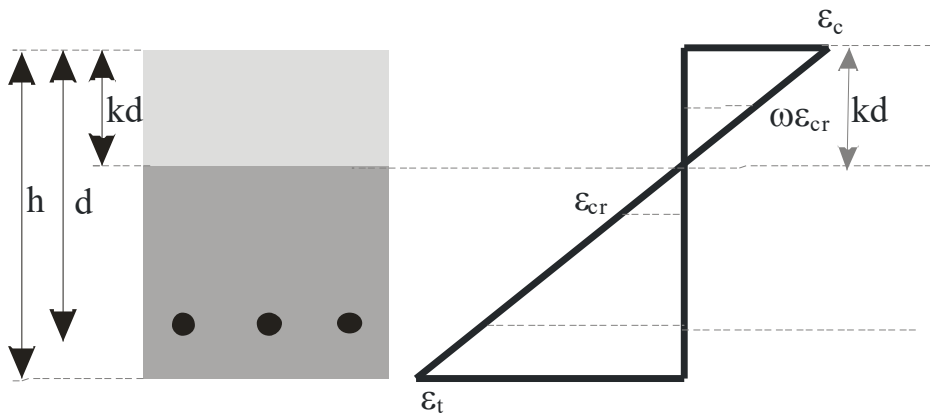
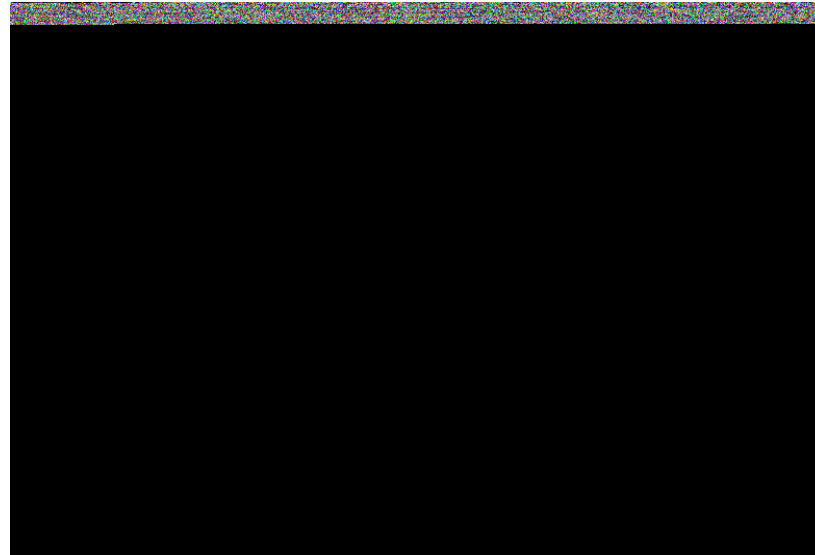
# Flexure design of RC

- Cross sectional analysis
- Plane section remains plane
- Compression: stress block method
- Tension: ignored
- Ultimate limit state: moment capacity

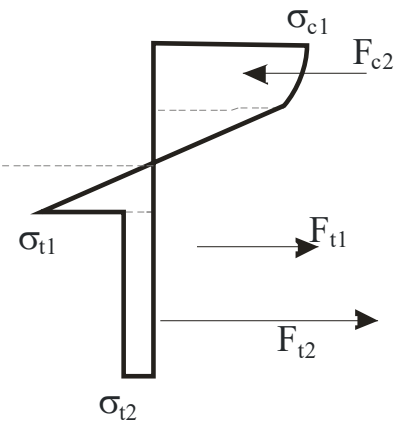


# Testing and Design Concept for FRC

The simplified design assumption for a beam with traditional and fiber and reinforcement.



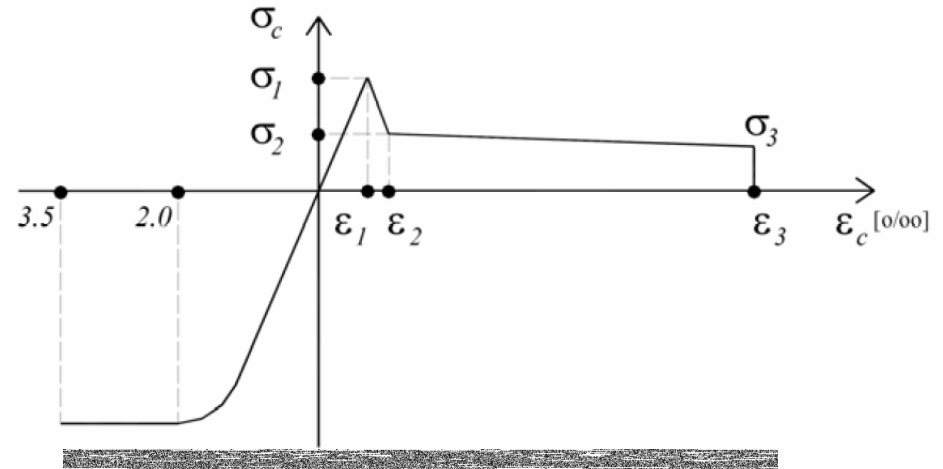
Traditional RC



FRC

# RILEM Stress-Strain Model for FRC

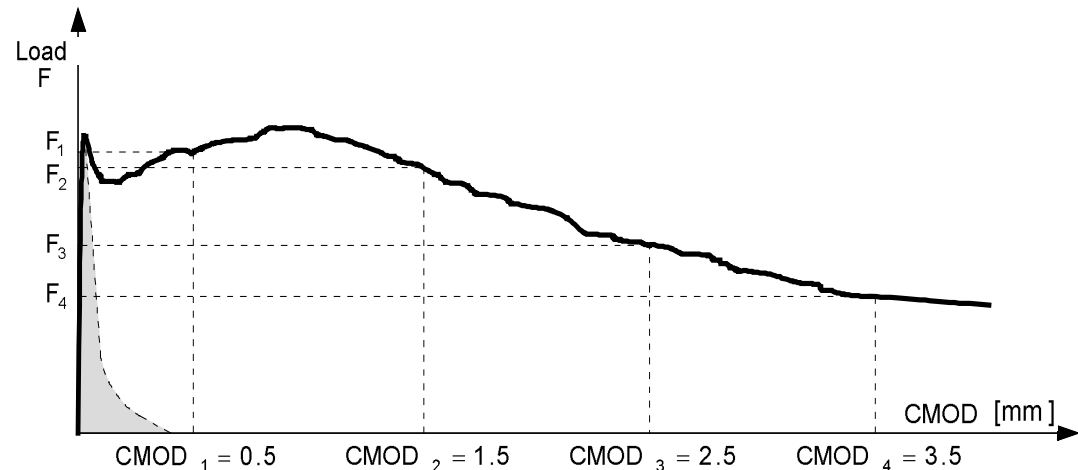
The model stress-strain diagram the same as proposed by RILEM and shown in the figure. The key points of the tension side of the diagram are found in a flexural test of a notched beam.



- $\sigma_1$ , and  $\epsilon_1$  – tensile strength and corresponding strain
- $\sigma_2$ , and  $\epsilon_2$  – stress and strain at the onset of the stable strain softening branch
- $\sigma_3$ , and  $\epsilon_3$  – stress and strain at the end of the softening branch

# RILEM Method for Test and Design

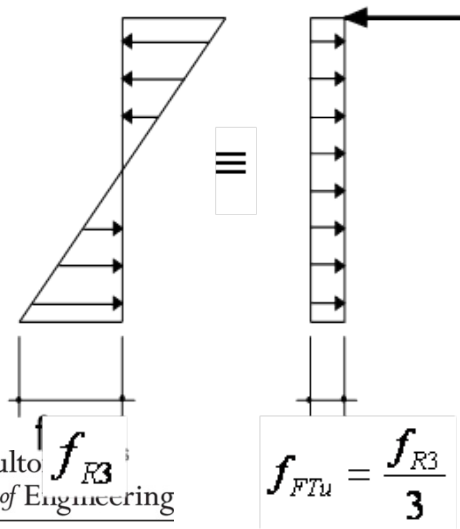
RILEM (EN 14651), a 3-point bending test on a notched beam and using the flexural load vs. crack opening (CMOD) response. Parameters of  $F_{r1}$  to  $F_{r4}$  correspond to apparent flexural load at crack opening of  $\text{CMOD}_1$  (0.5 mm) to  $\text{CMOD}_4$  (3.5 mm)



# RILEM Method for Test and Design

Residual flexural-tensile strength parameters,  $f_{Rj}$ , are calculated from the Load-CMOD curves with regard to the beam dimensions. A simplified model is adopted to compute the equivalent ultimate tensile strength in uniaxial tension  $f_{FTu}$  by means of the residual nominal bending strength  $f_{R3}$  as shown.

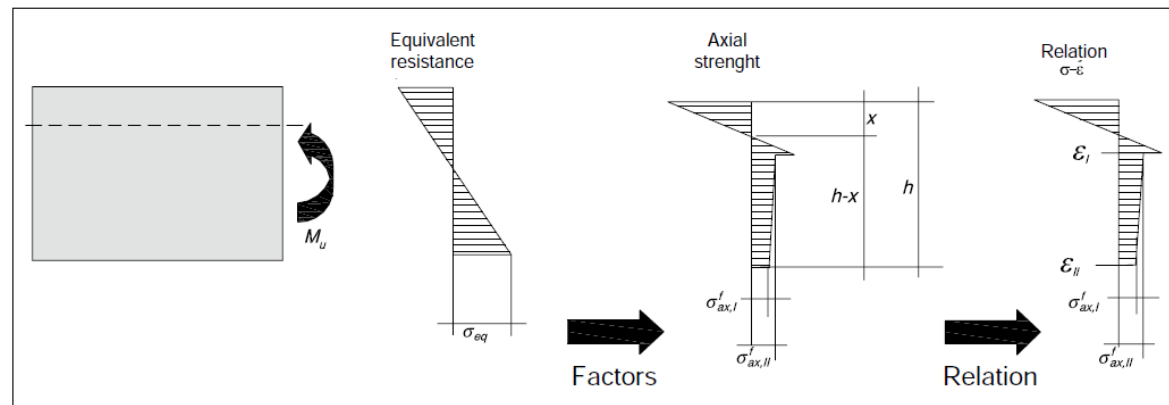
- $F_j$  is the load corresponding with  $CMOD_j$
- $f_{Rj}$  is the residual flexural tensile strength corresponding with  $CMOD_j$
- $l$  is the beam span length [mm];
- $b$  is the width of the specimen [mm];
- $h_{sp}$  is the net height: distance between the notch tip and the top of the specimen [mm].



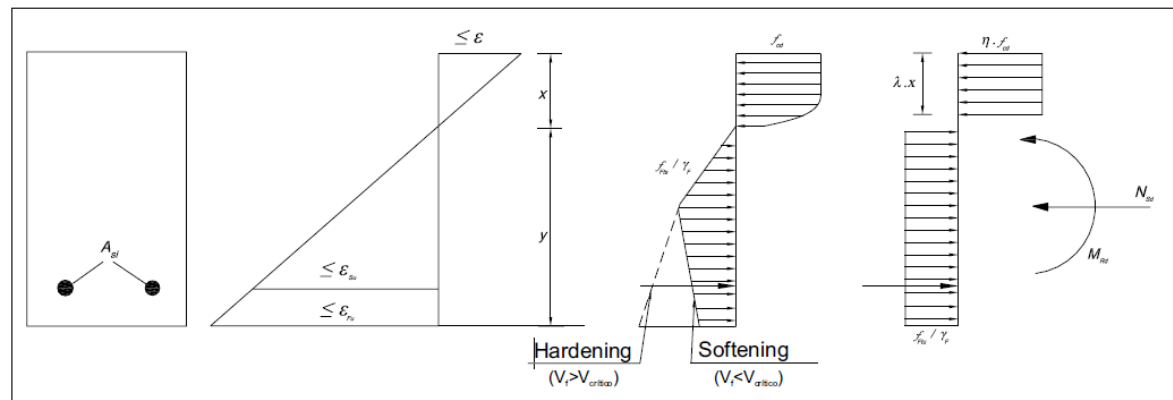
$$f_{R3} = \frac{F_3 \cdot l}{b \cdot h_{sp}^2} \quad \Rightarrow \quad f_{FTU} = \frac{f_{R3}}{3}$$

# RILEM Method for Test and Design

Limit states for the combined compressive and bending stress: use the constituent simplified laws. Both softening and hardening responses can be modeled using a simplified stress block as shown in the figures.



Graphic 3.12 - Limit states for the combined compressive and bending stress: simplified method (stress-block as RILEM TC-162).

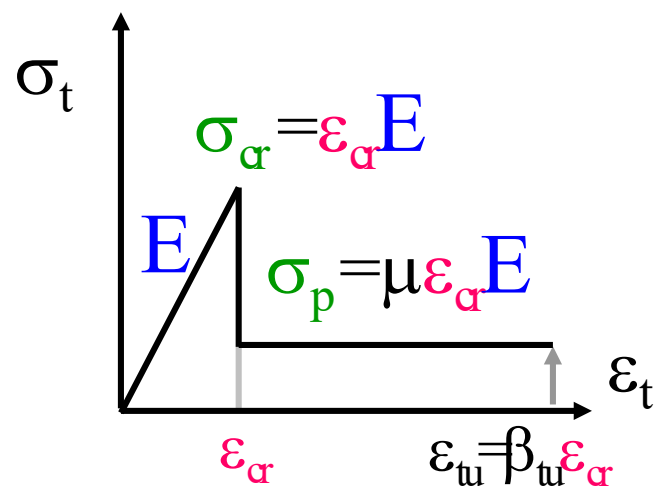
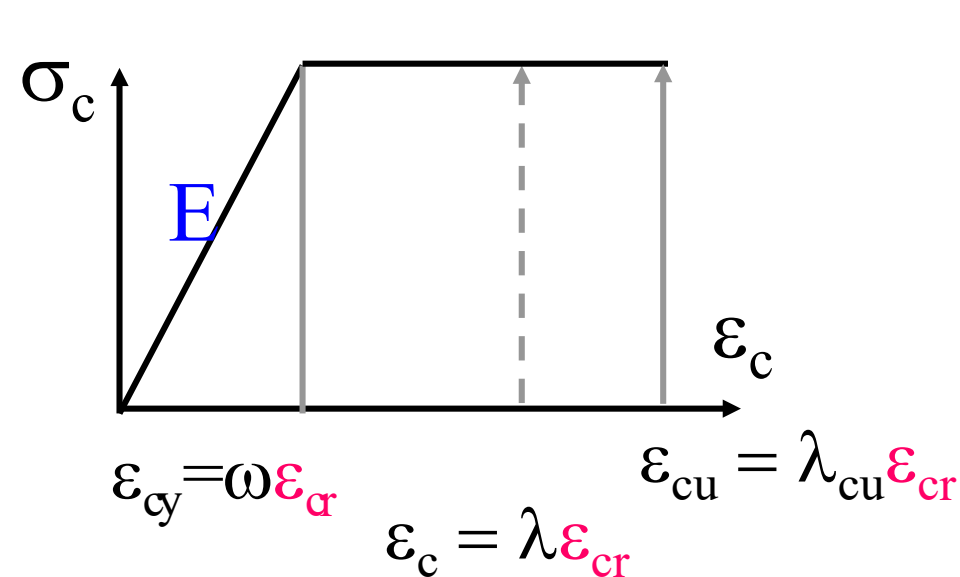


Graphic 3.11 - Limit states for the combined compressive and bending stress: use of the constituent simplified laws (stress-block with coefficients  $\eta$  e  $\lambda$ , as EC2).



# Simplified Material Model for strain softening, (fib model)

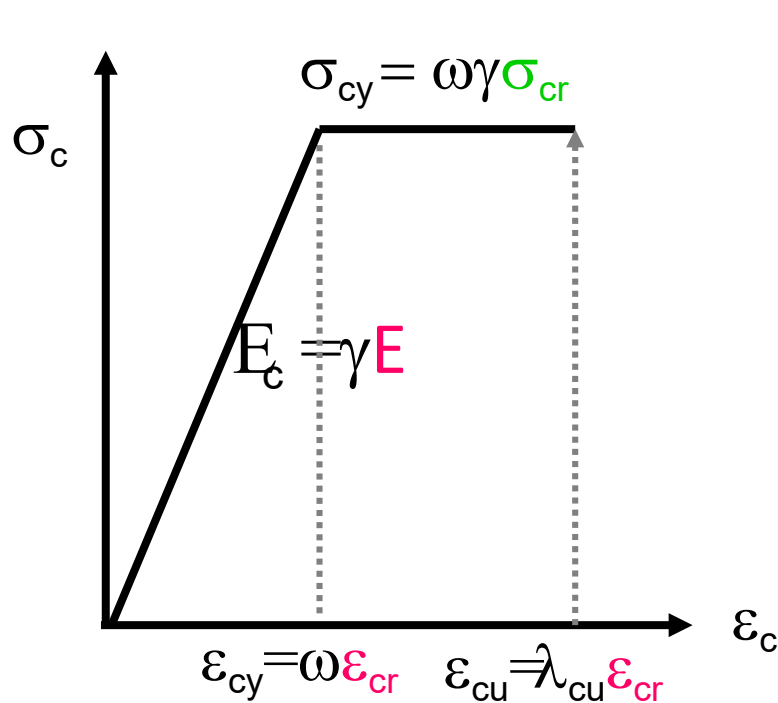
- Two material parameters ( $\varepsilon_{cr}$ ,  $E$ ) and four normalized parameters ( $\omega$ ,  $\mu$ ,  $\lambda_{cu}$ ,  $\beta_{tu}$ )



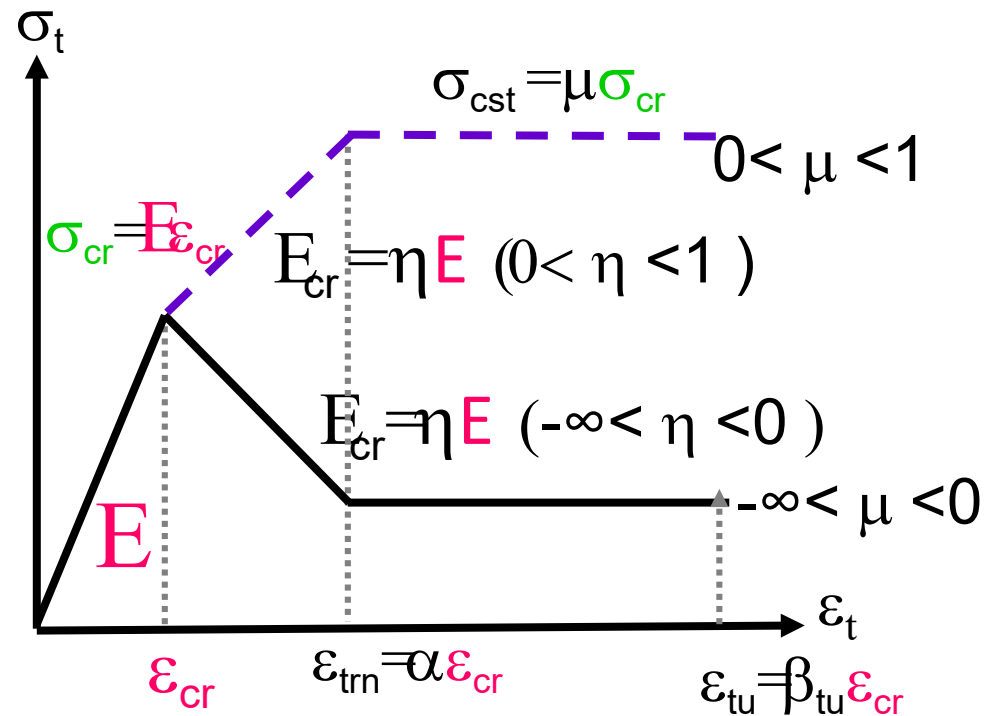
$$M_{cr} = \frac{1}{6} b d^2 E \varepsilon_{cr} \quad \phi_{cr} = \frac{2 \varepsilon_{cr}}{d}$$

# Material Model for Strain Hardening and Softening

- Two material parameters ( $\epsilon_{cr}$ ,  $E$ ) and six normalized parameters ( $\omega$ ,  $\gamma$ ,  $\lambda_{cu}$ ,  $\alpha$ ,  $\beta_{tu}$ ,  $\eta$ ),  $\mu = f(\eta)$



Compression model



Tension model

# Parameters Simplification

- Tension and compression strains:

$$\beta = \frac{\varepsilon_{tbot}}{\varepsilon_{cr}}; \quad \lambda = \frac{\varepsilon_{ctop}}{\varepsilon_{cr}}$$

- Stress and strain limiting parameters:

$$\omega = \frac{\varepsilon_{cy}}{\varepsilon_{cr}}; \quad \alpha = \frac{\varepsilon_{trn}}{\varepsilon_{cr}}; \quad \beta_{tu} = \frac{\varepsilon_{tu}}{\varepsilon_{cr}}; \quad \lambda_{cu} = \frac{\varepsilon_{cu}}{\varepsilon_{cr}}$$

$$\gamma = \frac{E_c}{E}; \quad \eta = \frac{E_{cr}}{E}$$

$$\mu = \frac{\sigma_{cst}}{E\varepsilon_{cr}}$$

- Moment and curvature

$$M_{cr} = \frac{1}{6} b d^2 E \varepsilon_{cr} \quad \phi_{cr} = \frac{2\varepsilon_{cr}}{d}$$

# Derivation of Moment-Curvature Relationship

Incrementally impose

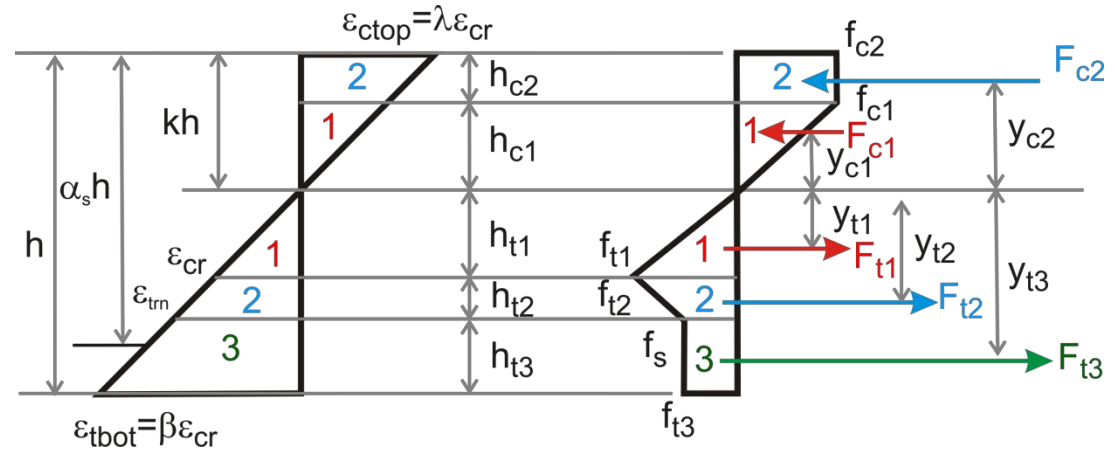
$$0 \leq \varepsilon_t \leq \varepsilon_{tu}$$

Strain Distribution  
Stress Distribution

$$\Sigma F = 0, \text{ determine } k$$

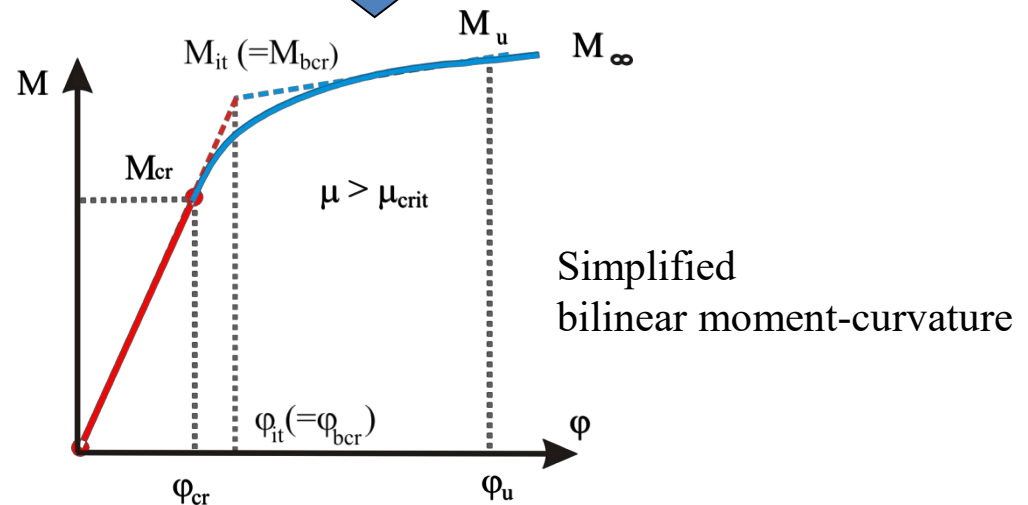
Moment:  $M = \Sigma F_{ci} y_{ci} + \Sigma F_{ti} y_{ti}$   
Curvature:  $\phi = \varepsilon_c / kh$

Stage :  $\lambda > \omega, \beta > \alpha$



Strain

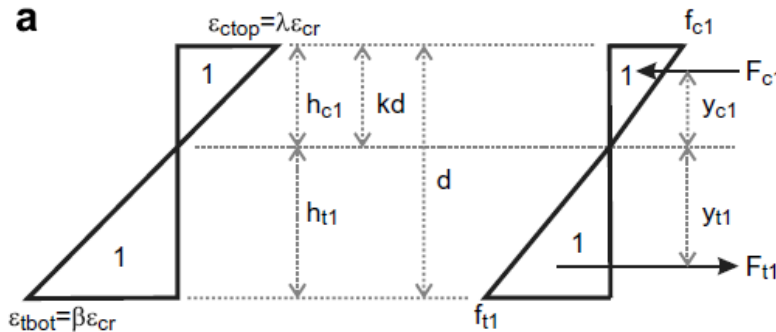
Stress



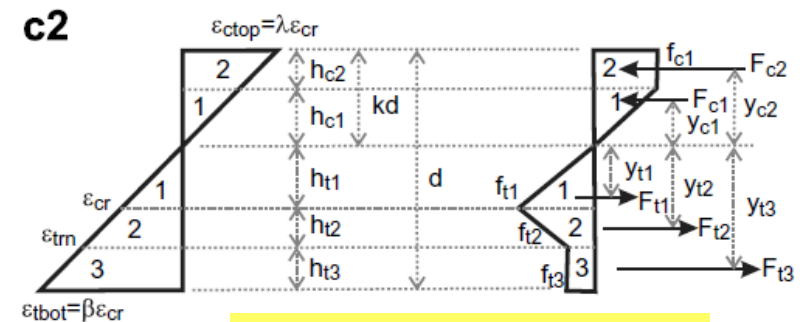
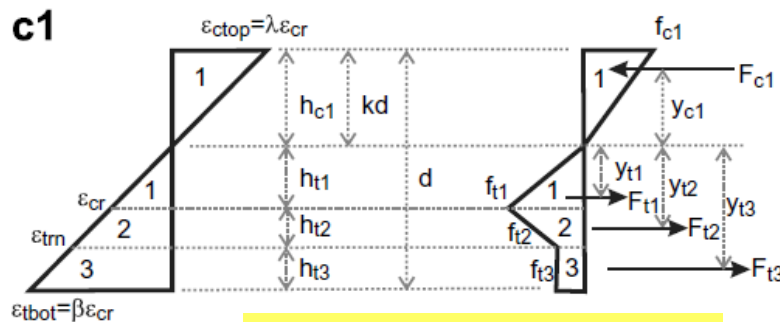
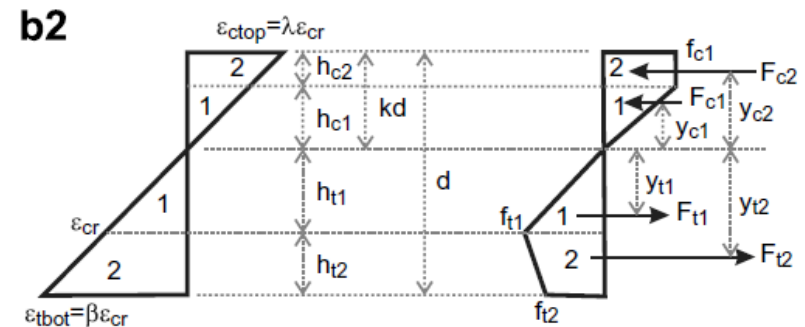
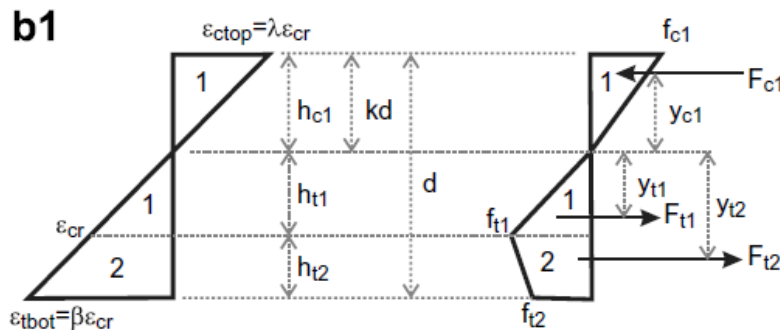
# Stress and Strain Distributions in Different Stages

$$0 < \beta \leq 1 \text{ and } \lambda \leq \omega$$

$$1 < \beta \leq \alpha \text{ and } \lambda \leq \omega$$



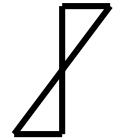
$$1 < \beta \leq \alpha \text{ and } \lambda > \omega$$



$$\alpha < \beta \leq \beta_{tu} \text{ and } \lambda \leq \omega$$

$$\alpha < \beta \leq \beta_{tu} \text{ and } \lambda > \omega$$

# Strain Softening: Force Component and Moment Arm at Each Stage of Normalized Top Compressive Strain ( $\lambda$ )



| Stage                                | $F'_{c2}$                                    | $F'_{c1}$                     | $F'_{t1}$            | $F'_{t2}$                               | $y'_{c2}$                              | $y'_{c1}$                    | $y'_{t1}$             | $y'_{t2}$                                     |
|--------------------------------------|----------------------------------------------|-------------------------------|----------------------|-----------------------------------------|----------------------------------------|------------------------------|-----------------------|-----------------------------------------------|
| $0 < \lambda \leq 1$                 | -                                            | $\frac{\lambda}{4}$           | $\frac{\lambda}{4}$  | -                                       | -                                      | $\frac{1}{3}$                | $\frac{1}{3}$         | -                                             |
| $1 < \lambda \leq \omega$            | -                                            | $\frac{\lambda k}{2}$         | $\frac{k}{2\lambda}$ | $\mu - \frac{\mu(\lambda+1)k}{\lambda}$ | -                                      | $\frac{2k}{3}$               | $\frac{2k}{3\lambda}$ | $\frac{1}{2} - \frac{(\lambda-1)k}{2\lambda}$ |
| $\omega < \lambda \leq \lambda_{cu}$ | $\frac{\omega k(\lambda - \omega)}{\lambda}$ | $\frac{\omega^2 k}{2\lambda}$ | $\frac{k}{2\lambda}$ | $\mu - \frac{\mu(\lambda+1)k}{\lambda}$ | $\frac{(\omega + \lambda)k}{2\lambda}$ | $\frac{2\omega k}{3\lambda}$ | $\frac{2k}{3\lambda}$ | $\frac{1}{2} - \frac{(\lambda-1)k}{2\lambda}$ |

$F_{xi}/(bdE\varepsilon_{cr})$  and  $y_{xi}/d$ ,  
respectively where  $xi$  refers to  $c1$ ,  $c2$ ,  $t1$  and  $t2$

# Normalized Force Terms and Moment Arms

Table 3

Normalized force component for each stage of normalized tensile strain at bottom fiber ( $\beta$ )

| Normalized force component | Stage 1 ( $0 \leq \beta \leq 1$ )<br>and ( $0 \leq \lambda \leq \omega$ ) | Stage 2.1 ( $1 < \beta \leq \alpha$ )<br>and ( $0 \leq \lambda \leq \omega$ ) | Stage 2.2 ( $1 < \beta \leq \alpha$ )<br>and ( $\omega < \lambda \leq \lambda_{cu}$ ) | Stage 3.1 ( $\beta > \alpha$ )<br>and ( $0 \leq \lambda \leq \omega$ ) | Stage 3.2 ( $\beta > \alpha$ )<br>and ( $\omega < \lambda \leq \lambda_{cu}$ ) |
|----------------------------|---------------------------------------------------------------------------|-------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|------------------------------------------------------------------------|--------------------------------------------------------------------------------|
| $\frac{F_{c2}}{bdE_{ct}}$  | —                                                                         | —                                                                             | $\frac{\omega\gamma}{\beta}(\beta k + \omega k - \omega)$                             | —                                                                      | $\frac{\omega\gamma}{\beta}(\beta k + \omega k - \omega)$                      |
| $\frac{F_{c1}}{bdE_{ct}}$  | $\frac{\beta\gamma k^2}{2(1-k)}$                                          | $\frac{\beta\gamma k^2}{2(1-k)}$                                              | $\frac{\omega^2\gamma}{2\beta}(1-k)$                                                  | $\frac{\beta\gamma k^2}{2(1-k)}$                                       | $\frac{\omega^2\gamma}{2\beta}(1-k)$                                           |
| $\frac{F_{t1}}{bdE_{ct}}$  | $\frac{\beta}{2}(1-k)$                                                    | $\frac{(1-k)}{2\beta}$                                                        | $\frac{(1-k)}{2\beta}$                                                                | $\frac{(1-k)}{2\beta}$                                                 | $\frac{(1-k)}{2\beta}$                                                         |
| $\frac{F_{t2}}{bdE_{ct}}$  | —                                                                         | $\frac{(1-k)(\beta-1)(\eta\beta-\eta+2)}{2\beta}$                             | $\frac{(1-k)(\beta-1)(\eta\beta-\eta+2)}{2\beta}$                                     | $\frac{(1-k)(\alpha-1)(\eta\alpha-\eta+2)}{2\beta}$                    | $\frac{(1-k)(\alpha-1)(\eta\alpha-\eta+2)}{2\beta}$                            |
| $\frac{F_{t3}}{bdE_{ct}}$  | —                                                                         | —                                                                             | —                                                                                     | $\frac{(1-k)(\beta-\alpha)\mu}{\beta}$                                 | $\frac{(1-k)(\beta-\alpha)\mu}{\beta}$                                         |

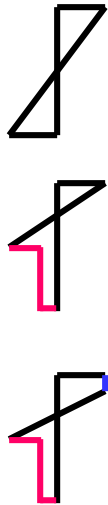
Table 4

Normalized moment arm of force component for each stage of normalized tensile strain at bottom fiber ( $\beta$ )

| Normalized moment arm | Stage 1 ( $0 \leq \beta \leq 1$ )<br>and ( $0 \leq \lambda \leq \omega$ ) | Stage 2.1 ( $1 < \beta \leq \alpha$ )<br>and ( $0 \leq \lambda \leq \omega$ )            | Stage 2.2 ( $1 < \beta \leq \alpha$ )<br>and ( $\omega < \lambda \leq \lambda_{cu}$ ) | Stage 3.1 ( $\beta > \alpha$ )<br>and ( $0 \leq \lambda \leq \omega$ )                       | Stage 3.2 ( $\beta > \alpha$ )<br>and ( $\omega < \lambda \leq \lambda_{cu}$ ) |
|-----------------------|---------------------------------------------------------------------------|------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------|
| $\frac{y_{c2}}{d}$    | —                                                                         | —                                                                                        | $\frac{\beta k + \omega(1-k)}{2\beta}$                                                | —                                                                                            | $\frac{\beta k + \omega(1-k)}{2\beta}$                                         |
| $\frac{y_{c1}}{d}$    | $\frac{2}{3}k$                                                            | $\frac{2}{3}k$                                                                           | $\frac{2}{3}\frac{\omega(1-k)}{\beta}$                                                | $\frac{2}{3}k$                                                                               | $\frac{2}{3}\frac{\omega(1-k)}{\beta}$                                         |
| $\frac{y_{t1}}{d}$    | $\frac{2}{3}(1-k)$                                                        | $\frac{2}{3}\frac{(1-k)}{\beta}$                                                         | —                                                                                     | $\frac{2}{3}\frac{(1-k)}{\beta}$                                                             | —                                                                              |
| $\frac{y_{t2}}{d}$    | —                                                                         | $\frac{2\eta\beta^2 - \eta\beta - \eta + 3\beta + 3}{3\beta(\eta\beta - \eta + 2)}(1-k)$ | —                                                                                     | $\frac{2\eta\alpha^2 - \eta\alpha - \eta + 3\alpha + 3}{3\beta(\eta\alpha - \eta + 2)}(1-k)$ | —                                                                              |
| $\frac{y_{t3}}{d}$    | —                                                                         | —                                                                                        | —                                                                                     | $\frac{(\alpha+\beta)}{2\beta}(1-k)$                                                         | —                                                                              |

Soranakom C, Mobasher B. “Correlation of tensile and flexural responses of strain softening and strain hardening cement composites”, Cem Concr Comp 2008;30:465–477.

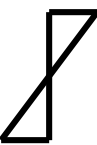
# Strain Softening: Net force, Moment and Curvature for Each Stage of Normalized Top Compressive Strain ( $\lambda$ )



| Stage                                | Net Force                            | Moment                                                      | Curvature                             |
|--------------------------------------|--------------------------------------|-------------------------------------------------------------|---------------------------------------|
| $0 < \lambda \leq 1$                 | $-F_{c1} + F_{t1}$                   | $F_{c1}y_{c1} + F_{t1}y_{t1}$                               | $\frac{\lambda \varepsilon_{cr}}{kd}$ |
| $1 < \lambda \leq \omega$            | $-F_{c1} + F_{t1} + F_{t2}$          | $F_{c1}y_{c1} + F_{t1}y_{t1} + F_{t2}y_{t2}$                |                                       |
| $\omega < \lambda \leq \lambda_{cu}$ | $-F_{c1} - F_{c2} + F_{t1} + F_{t2}$ | $F_{c1}y_{c1} + F_{c2}y_{c2} + F_{t1}y_{t1} + F_{t2}y_{t2}$ |                                       |



# Strain Softening: Simplified Solutions of Neutral Axis Depth and Moment-Curvature

|                                                                                  | Stage                                     | k                                                                   | M'=M/M <sub>cr</sub>                                                                         | $\phi'=\phi/\phi_{cr}$ |
|----------------------------------------------------------------------------------|-------------------------------------------|---------------------------------------------------------------------|----------------------------------------------------------------------------------------------|------------------------|
|  | 1<br>$0 < \lambda \leq 1$                 | $\phi_{cr} = \frac{2\varepsilon_{cr}}{d}$<br>$\frac{1}{2}$          | $\frac{\lambda}{2k}$                                                                         |                        |
|  | 2<br>$1 < \lambda \leq \omega$            | $\frac{2\mu\lambda}{\lambda^2 + 2\mu(\lambda+1)-1}$                 | $\frac{(2\lambda^3 + 3\mu\lambda^2 - 3\mu + 2)k^2}{\lambda^2} - 3\mu(2k-1)$                  | $\frac{\lambda}{2k}$   |
|  | 3<br>$\omega < \lambda \leq \lambda_{cu}$ | $\frac{2\mu\lambda}{-\omega^2 + 2\lambda(\omega + \mu) + 2\mu - 1}$ | $\frac{(3\omega\lambda^2 - \omega^3 + 3\mu\lambda^2 - 3\mu + 2)k^2}{\lambda^2} - 3\mu(2k-1)$ |                        |

$$M_{cr} = \frac{1}{6} bd^2 E \varepsilon_{cr}$$

# Closed Form Solutions for Strain Hardening/Softening material

Table 6

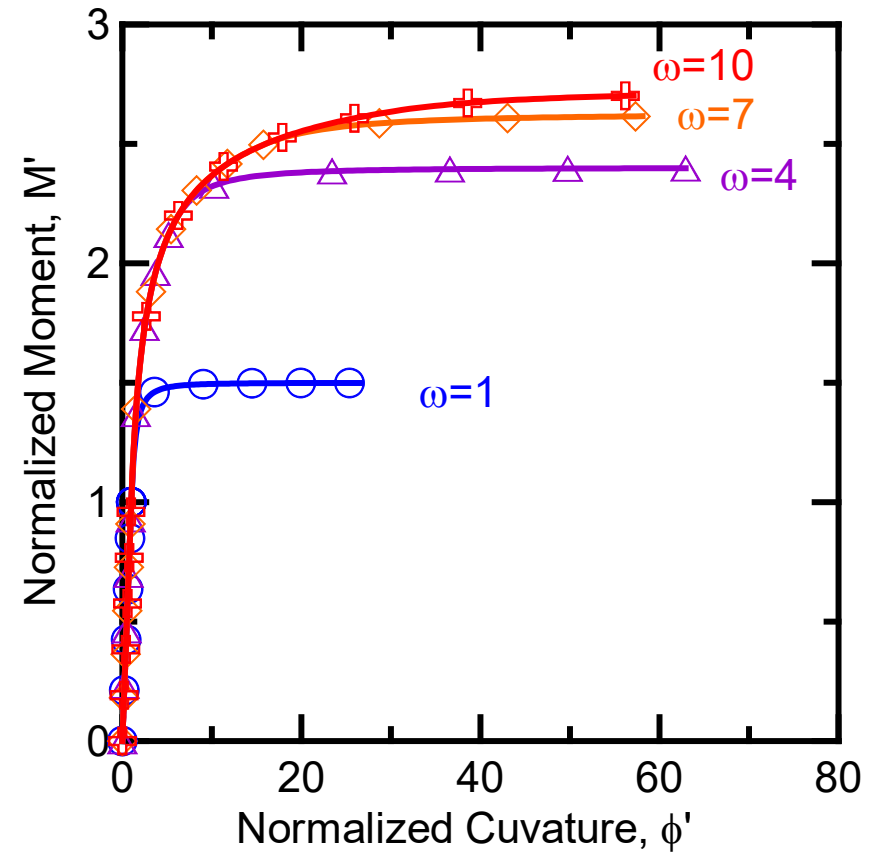
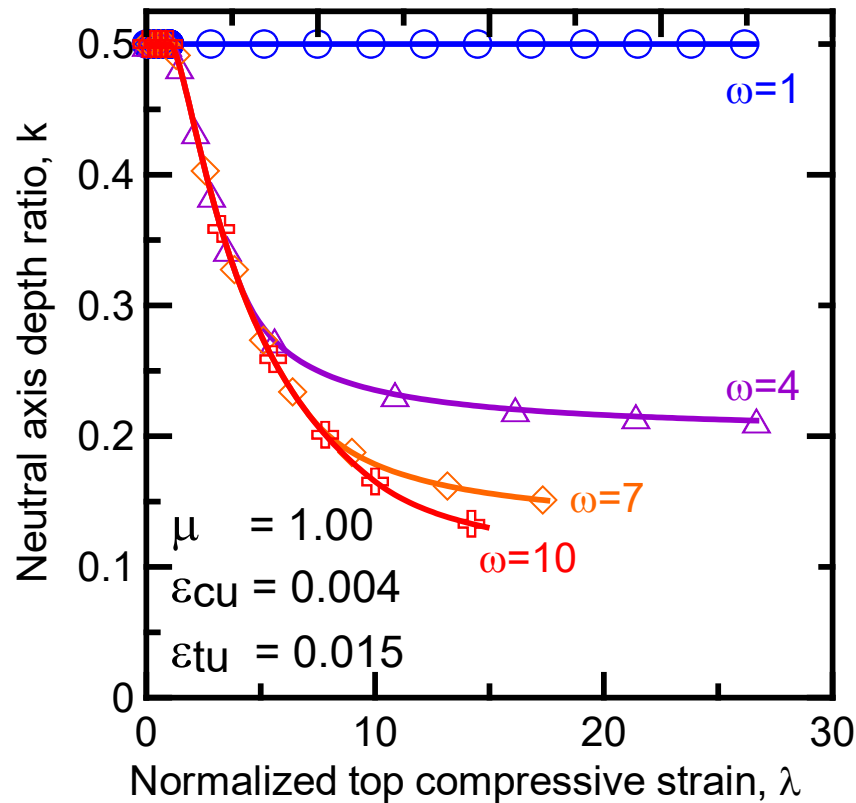
Neutral axis depth ratio, normalized moment and curvature for each stage of normalized tensile strain at bottom fiber ( $\beta$ )

| Stage | $k$                                                                                                                                                                   | $M'$                                                                                                                                                                                                                     | $\phi'$                                  |
|-------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------|
| 1     | $k_1 = \begin{cases} \frac{1}{2} & \text{for } \gamma = 1 \\ \frac{-1+\sqrt{\gamma}}{-1+\gamma} & \text{for } \gamma < 1 \text{ or } \gamma > 1 \end{cases}$          | $M'_1 = \frac{2\beta[(\gamma-1)k_1^3 + 3k_1^2 - 3k_1 + 1]}{1-k_1}$                                                                                                                                                       | $\phi'_1 = \frac{\beta}{2(1-k_1)}$       |
| 2.1   | $k_{21} = \frac{\beta^2\gamma + D_{21} - \sqrt{\gamma^2\beta^4 + D_{21}\gamma\beta^2}}{D_{21}}$<br>$D_{21} = \eta(\beta^2 - 2\beta + 1) + 2\beta - \beta^2\gamma - 1$ | $M'_{21} = \frac{(2\beta\gamma + C_{21})k_{21}^3 - 3C_{21}k_{21}^2 + 3C_{21}k_{21} - C_{21}}{1-k_{21}}$<br>$C_{21} = \frac{-2\eta\beta^3 + 3\eta\beta^2 - 3\beta^2 - \eta + 1}{\beta^2}$                                 | $\phi'_{21} = \frac{\beta}{2(1-k_{21})}$ |
| 2.2   | $k_{22} = \frac{D_{22}}{D_{22} + 2\omega\eta\beta}$<br>$D_{22} = \eta(\beta^2 - 2\beta + 1) + 2\beta + \omega^2\gamma - 1$                                            | $M'_{22} = (3\omega\gamma + C_{22})k_{22}^2 - 2C_{22}k_{22} + C_{22}$<br>$C_{22} = \frac{2\eta\beta^3 - 3\eta\beta^2 + 3\beta^2 - \omega^3\gamma + \eta - 1}{\beta^2}$                                                   | $\phi'_{22} = \frac{\beta}{2(1-k_{22})}$ |
| 3.1   | $k_{31} = \frac{D_{31} - \sqrt{\gamma\beta^2 D_{31}}}{D_{31} - \beta^2\gamma}$<br>$D_{31} = \eta(\alpha^2 - 2\alpha + 1) + 2\mu(\beta - \alpha) + 2\alpha - 1$        | $M'_{31} = \frac{(C_{31} - 2\beta\gamma)k_{31}^3 - 3C_{31}k_{31}^2 + 3C_{31}k_{31} - C_{31}}{k_{31} - 1}$<br>$C_{31} = \frac{3(\mu\beta^2 - \mu\alpha^2 - \eta\alpha^2 + \alpha^2) + 2\eta\alpha^3 + \eta - 1}{\beta^2}$ | $\phi'_{31} = \frac{\beta}{2(1-k_{31})}$ |
| 3.2   | $k_{32} = \frac{D_{32}}{D_{32} + 2\omega\eta\beta}$<br>$D_{32} = \omega^2\gamma + \eta\alpha^2 + 2(\mu\beta - \eta\alpha - \mu\alpha + \alpha) + \eta - 1$            | $M'_{32} = (C_{32} + 3\omega\gamma)k_{32}^2 - 2C_{32}k_{32} + C_{32}$<br>$C_{32} = \frac{3(\mu\beta^2 - \mu\alpha^2 - \eta\alpha^2 + \alpha^2) + 2\eta\alpha^3 - \omega^3\gamma + \eta - 1}{\beta^2}$                    | $\phi'_{32} = \frac{\beta}{2(1-k_{32})}$ |

$$M = M' M_{cr} \quad \phi = \phi' \phi_{cr}$$

$$M_{cr} = \frac{1}{6} b d^2 E \varepsilon_{cr} \quad \phi_{cr} = \frac{2\varepsilon_{cr}}{d}$$

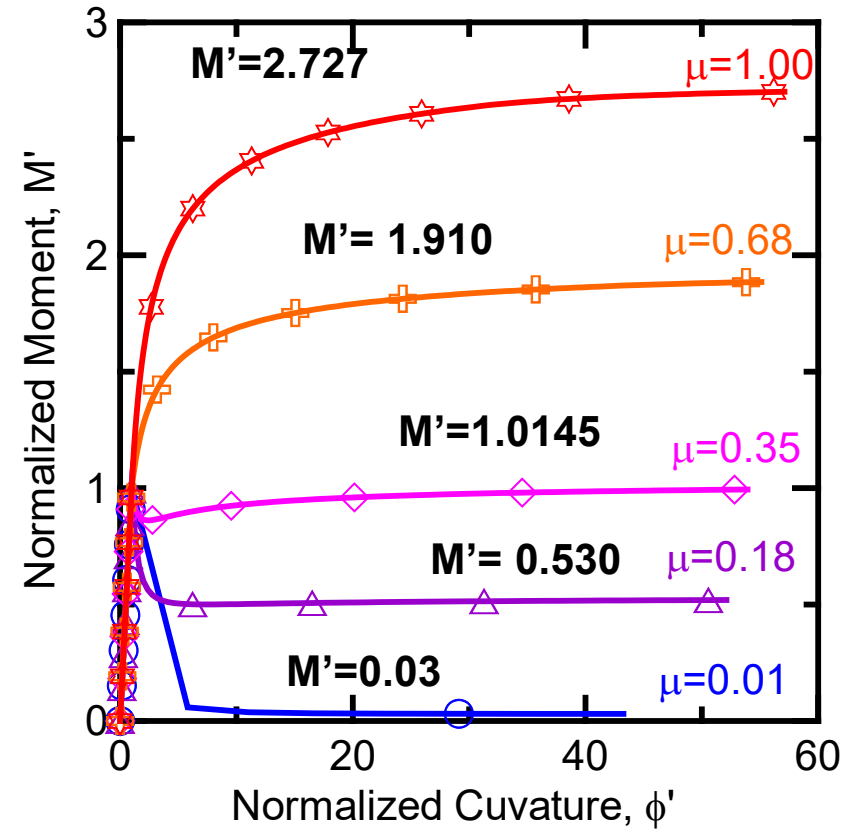
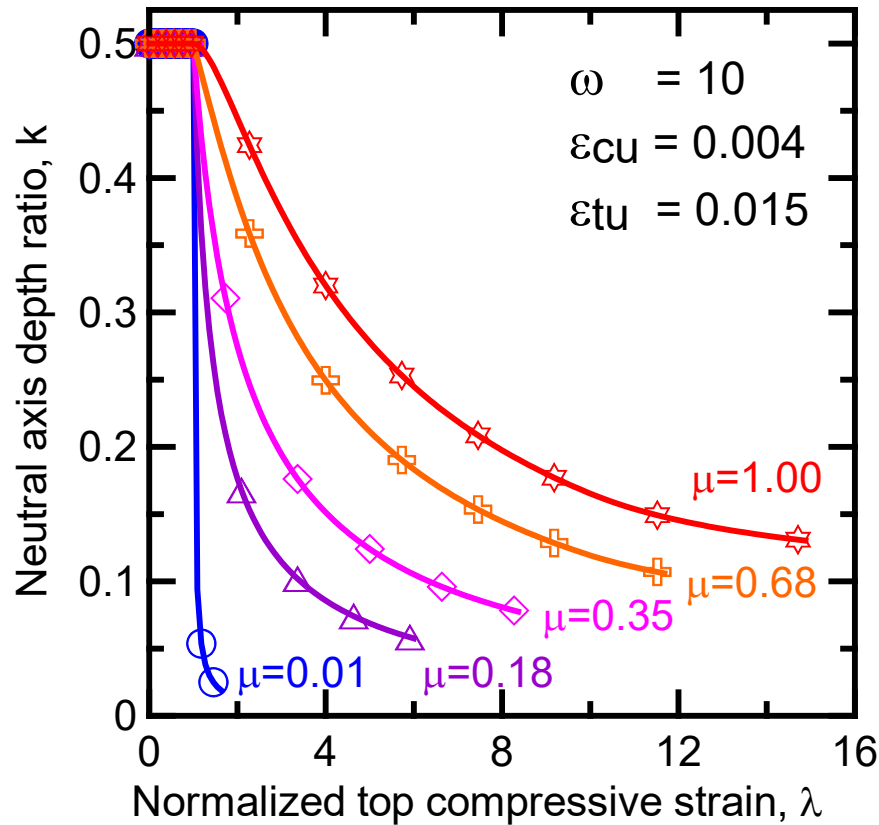
# Effect of compression to Tension strength ratio, $\omega$ (Elastic Plastic Material, $\mu=1$ )



$$M_{cr} = \frac{1}{6} b d^2 E \varepsilon_{cr} \quad \phi_{cr} = \frac{2 \varepsilon_{cr}}{d}$$

# Strain Softening Materials ( $\eta=0$ , $\alpha=1$ )

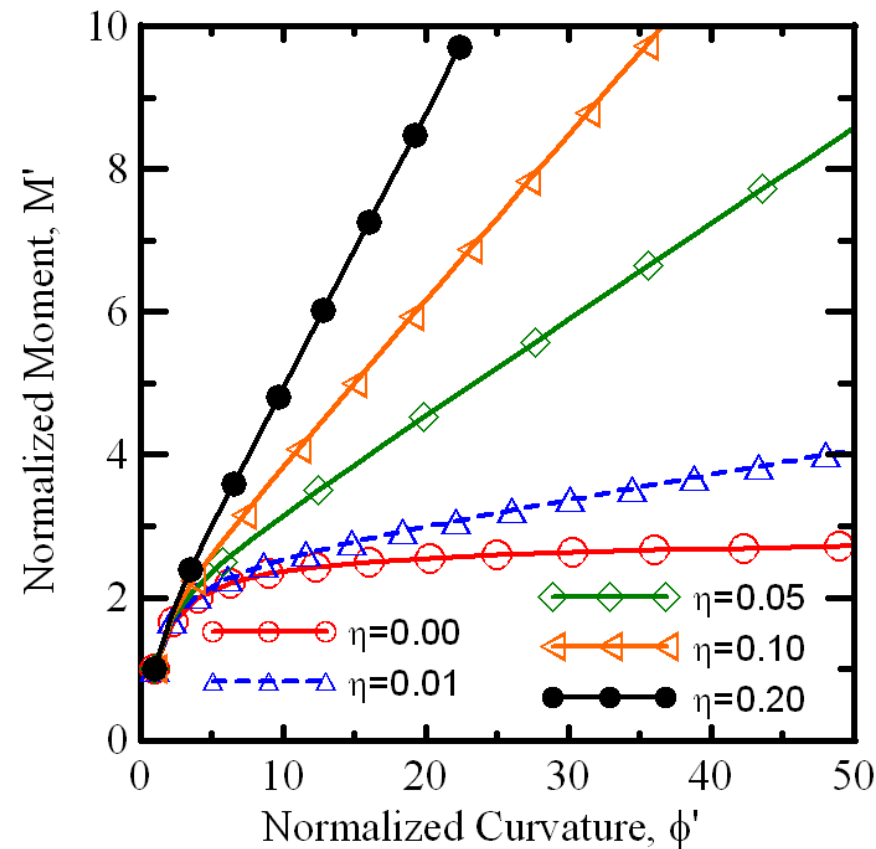
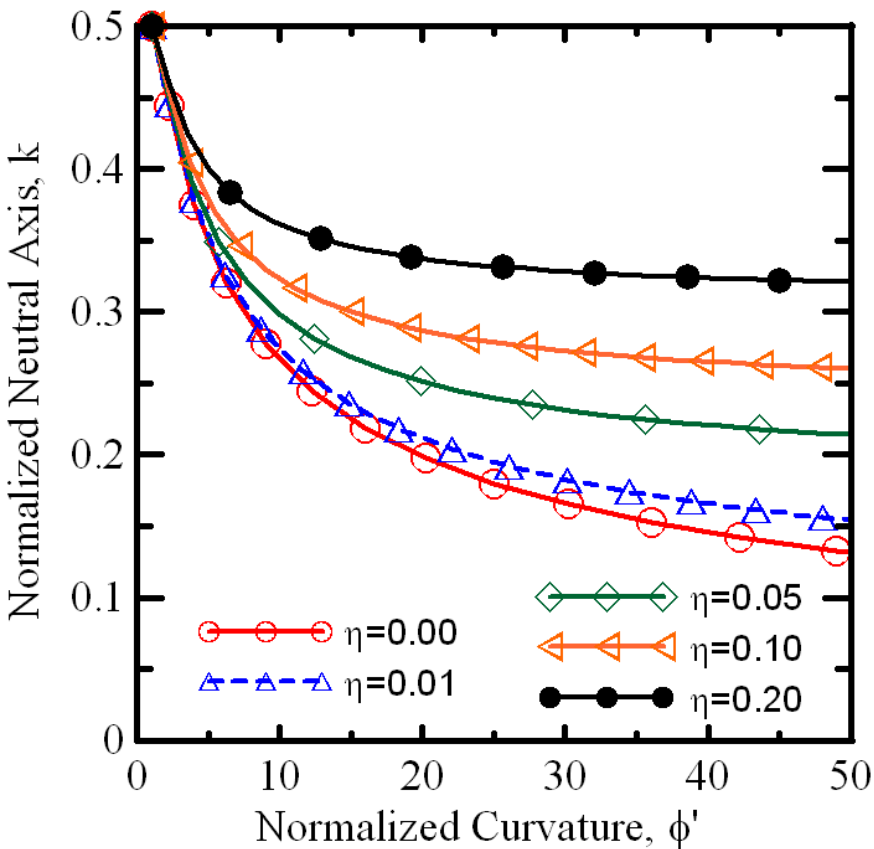
## Effect of post-peak tensile strength, $\mu$



$$M'(\phi \rightarrow \infty) = 3 \frac{\mu\omega}{\mu + \omega}$$

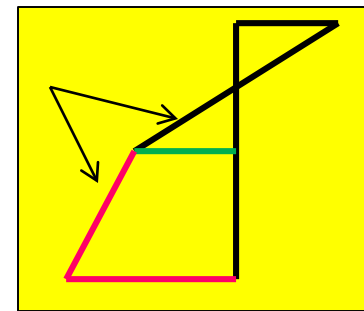
$$M_{cr} = \frac{1}{6} b d^2 E \varepsilon_{cr} \quad \phi_{cr} = \frac{2\varepsilon_{cr}}{d}$$

## Range 2.1 - Compression Elastic - Tension Hardening



Elastic perfectly plastic tension ( $\eta=0$ )  
Elastic ( $\eta=1$ )

$\eta$ =ratio of Slopes



# Development of Design Guides

$$M_{cr} = \frac{1}{6} b d^2 E \varepsilon_{cr}$$

- Deflection Softening-Deflection Hardening Transition

$$\mu_{crit} = \frac{\omega}{3\omega - 1}$$

– For  $\omega = 10$ ,  $\mu_{crit} = 0.355$

- Ultimate Moment Capacity

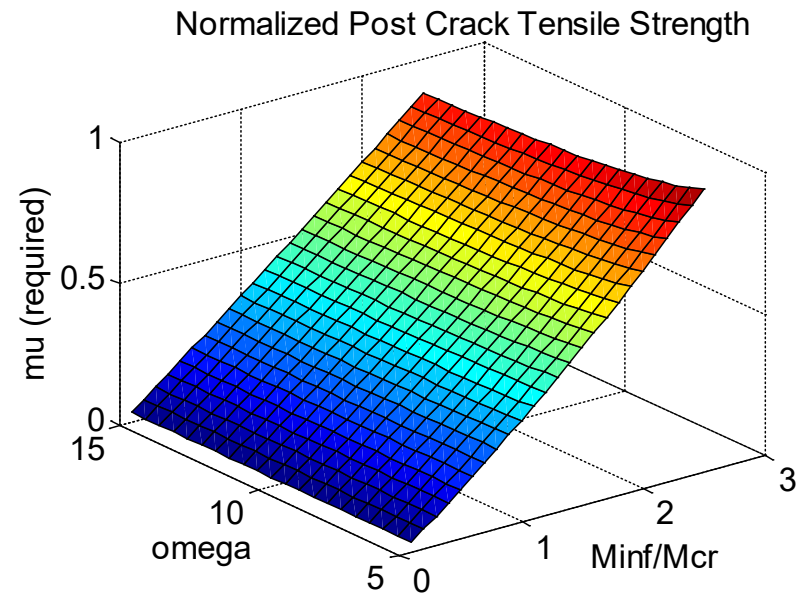
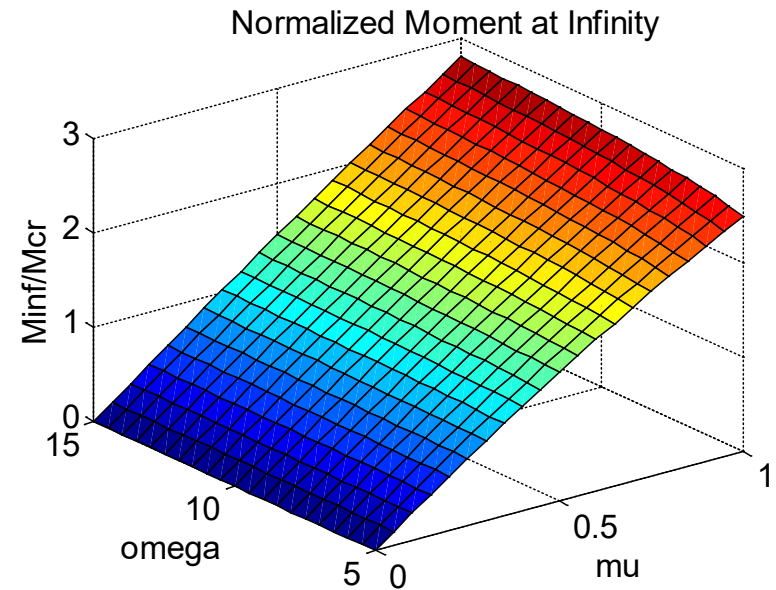
–  $\mu = 0 - 1$  and  $\omega = 5 - 15$

$$M'_{\infty} = \frac{3\mu\omega}{\omega + \mu}$$

- Required Post crack strength for a given moment

–  $M'_{\infty} = 0.1 - 2.5$  and  $\omega = 5 - 15$

$$\mu = \frac{\omega M'_{\infty}}{3\omega - M'_{\infty}}$$



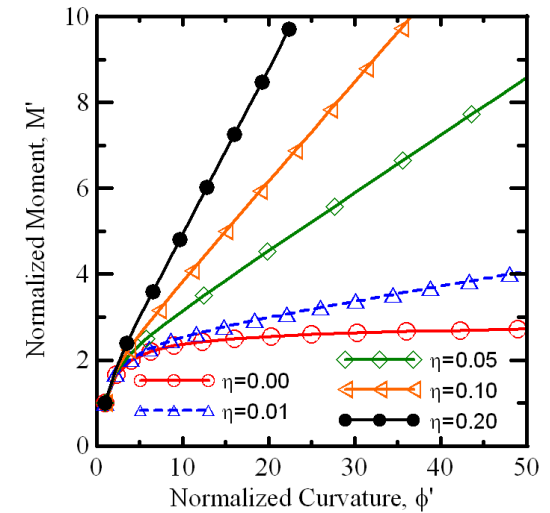
## Range 2.1 -Compression Elastic - Tension Hardening

$$k = \frac{\sqrt{A}}{\sqrt{A} + \beta\sqrt{\gamma}}$$

$$A = \eta(\beta^2 + 1 - 2\beta) + 2\beta - 1$$

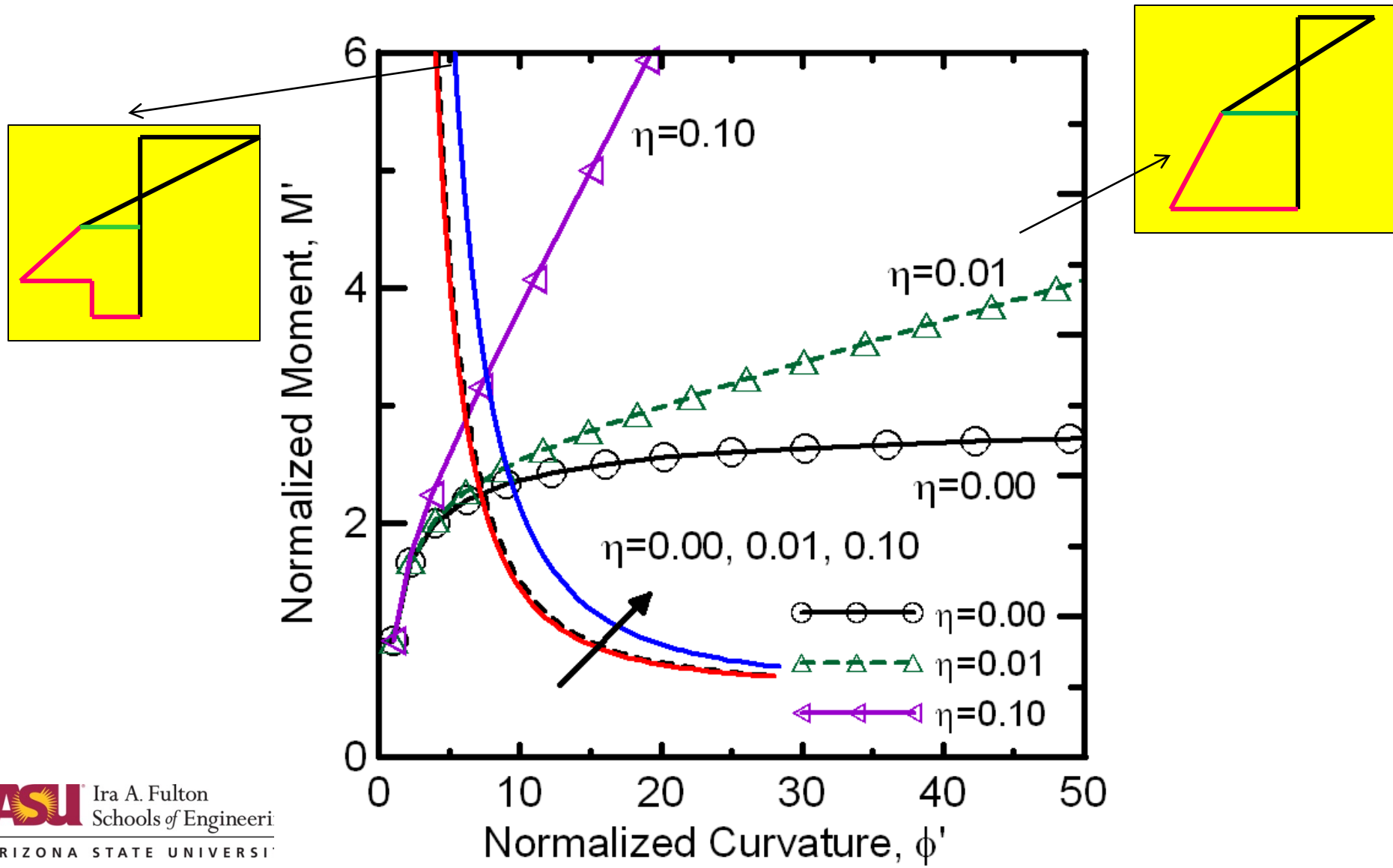
Special Case:  $E_c = E_t$  ( $\gamma=1$ ),  
Elastic perfectly plastic tension ( $\eta=0$ )

$$k = \frac{\sqrt{2\beta - 1}}{\sqrt{2\beta - 1} + \beta}$$



| $\eta$ | $A, (k = \frac{\sqrt{A}}{\sqrt{A} + \beta})$ | $M'(k)$                                | $M'(\phi)$                  |
|--------|----------------------------------------------|----------------------------------------|-----------------------------|
| 0.5    | $0.5(\beta^2 + 1 - 2\beta) + 2\beta - 1$     | $-0.773 + 0.108 \times 10^{-1} k^{-6}$ | $0.507 + 0.686 \phi$        |
| 0.2    | $0.2(\beta^2 + 1 - 2\beta) + 2\beta - 1$     | $0.654 + 0.516 \times 10^{-2} k^{-6}$  | $1.105 + 0.383 \phi$        |
| 0.1    | $0.1(\beta^2 + 1 - 2\beta) + 2\beta - 1$     | $1.276 + 0.289 \times 10^{-2} k^{-6}$  | $1.461 + .234 \phi$         |
| 0.05   | $0.05(\beta^2 + 1 - 2\beta) + 2\beta - 1$    | $1.645 + .1632 \times 10^{-2} k^{-6}$  | $1.720 + .1401 \phi$        |
| 0.01   | $0.01(\beta^2 + 1 - 2\beta) + 2\beta - 1$    | $0.852 + 0.456 k^{-1}$                 | $1.342 + 0.371 \sqrt{\phi}$ |

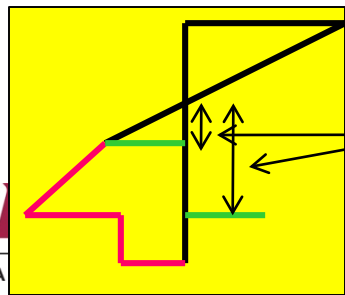
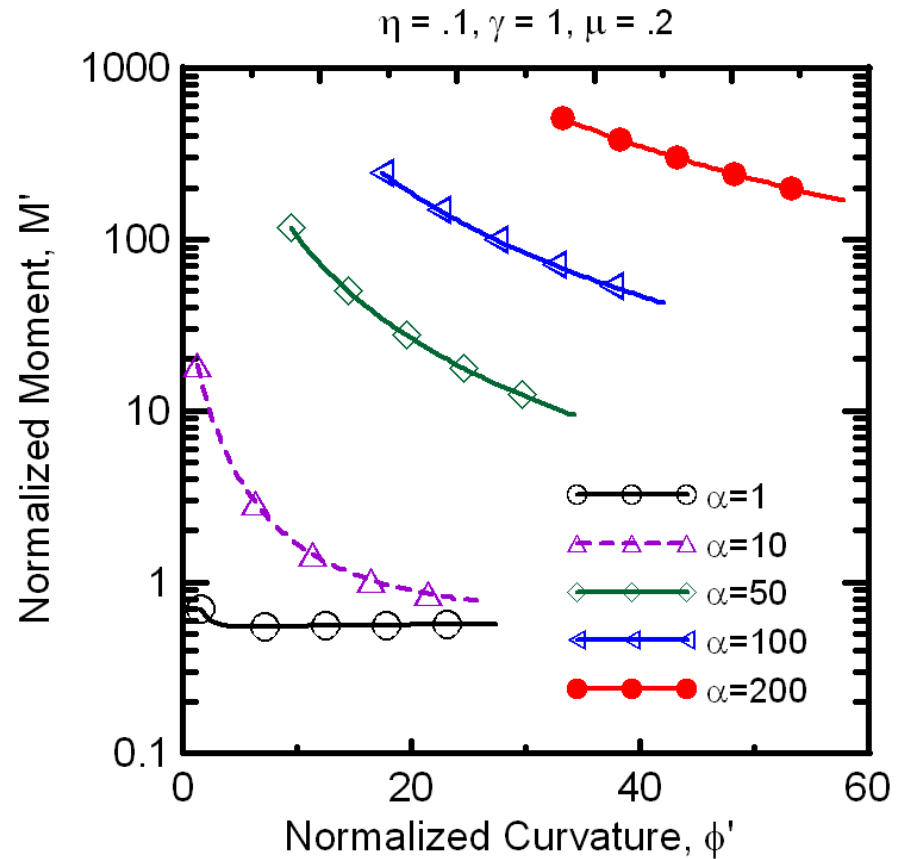
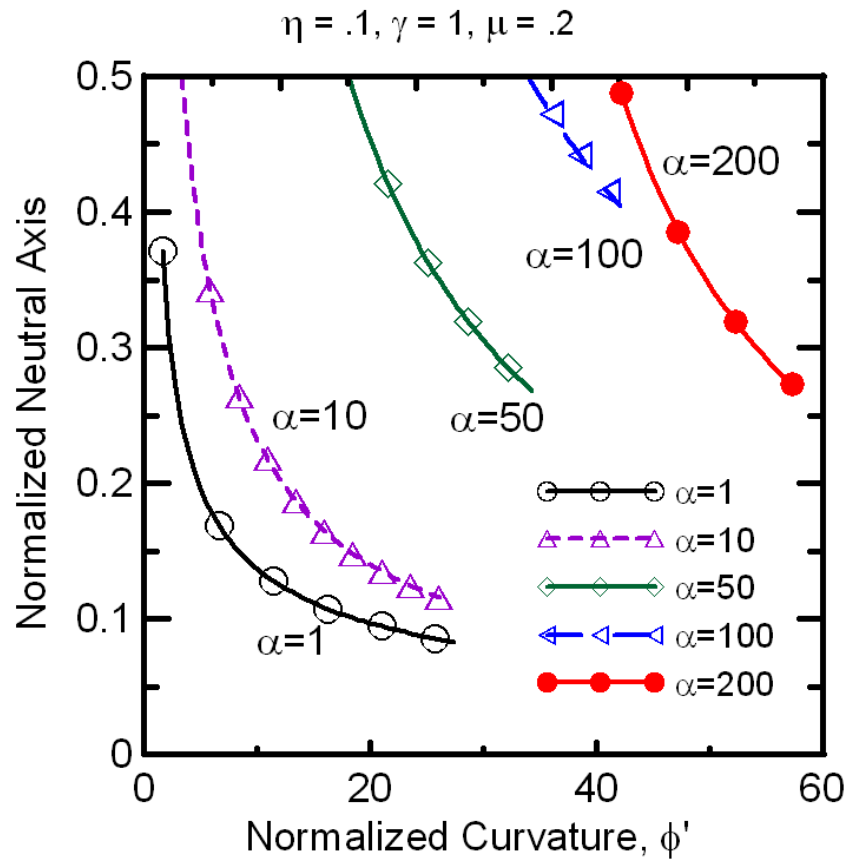
# Interaction of Failure Zones (three tension modes, one compression)





# Range 3.1 Compression Elastic Tension-Softening

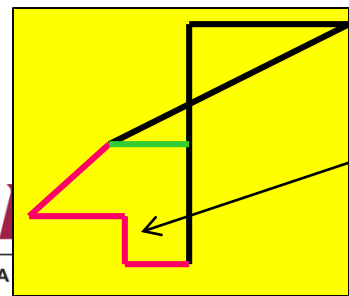
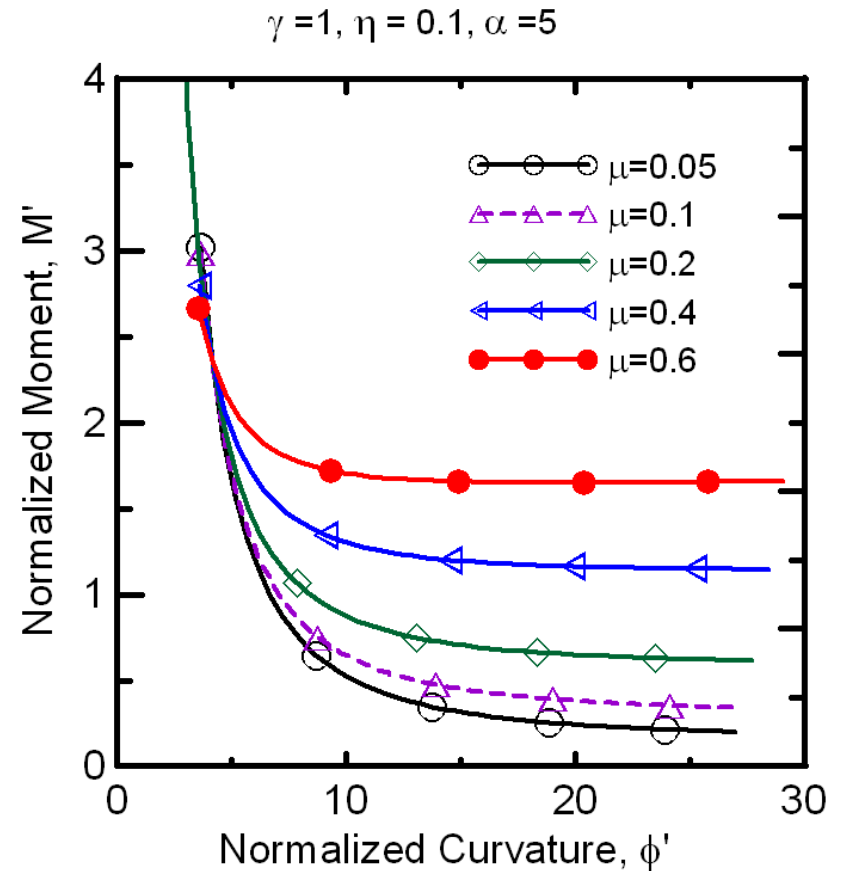
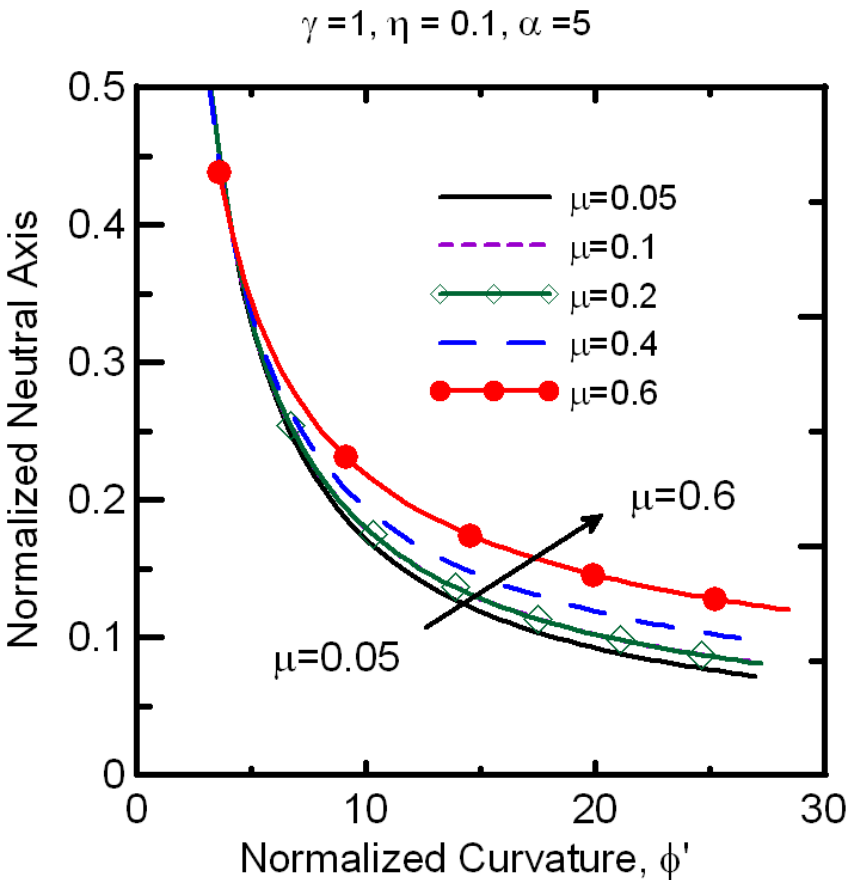
## Effect of Ultimate Strain Capacity, $\alpha$



$\alpha$ =ratio of ultimate strain capacity to first crack strain

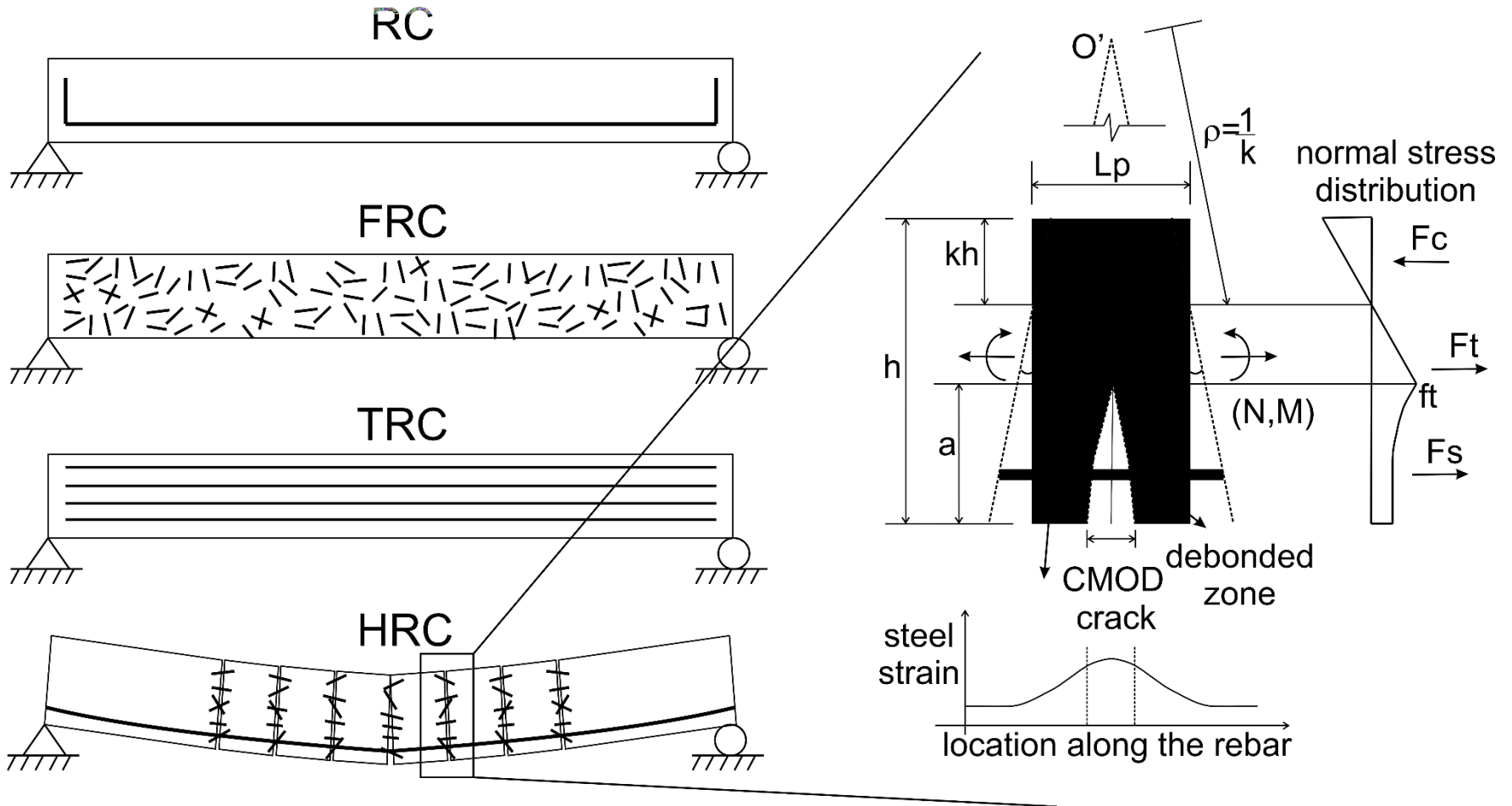
# Range 3.1 Compression Elastic Tension-Softening

## Effect of Post peak Stress level, $\mu$



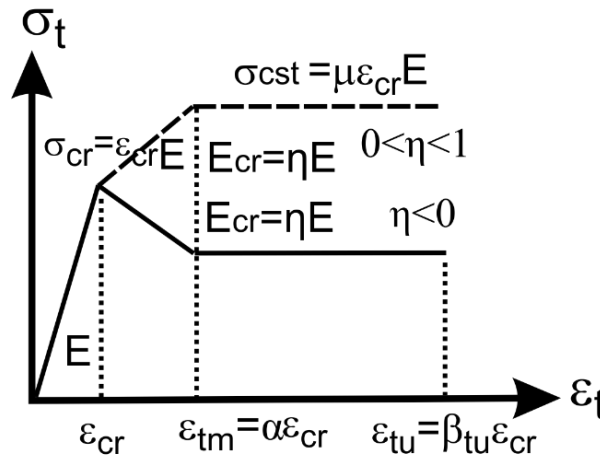
Post peak Stress level,  $\mu$

# ASU Localized hinge Model

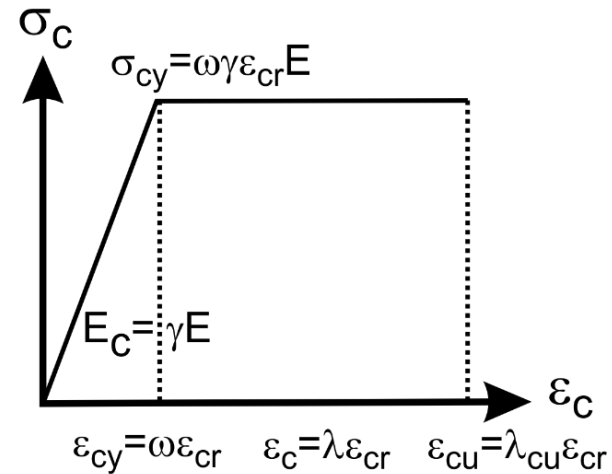


# Parameterized Material Models

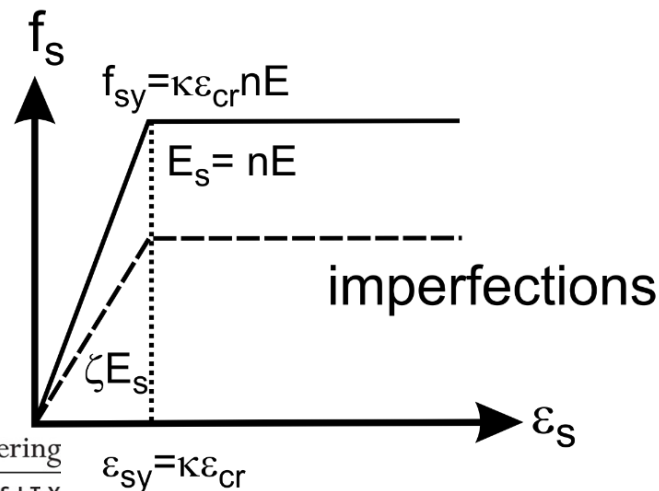
## Tension



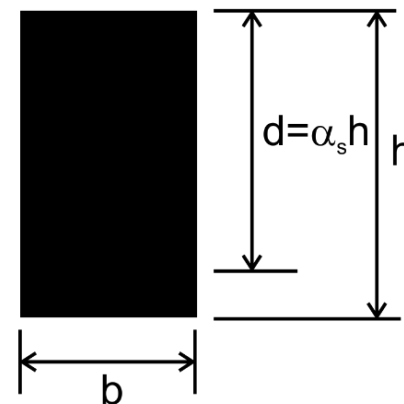
## Compression



## Steel rebar

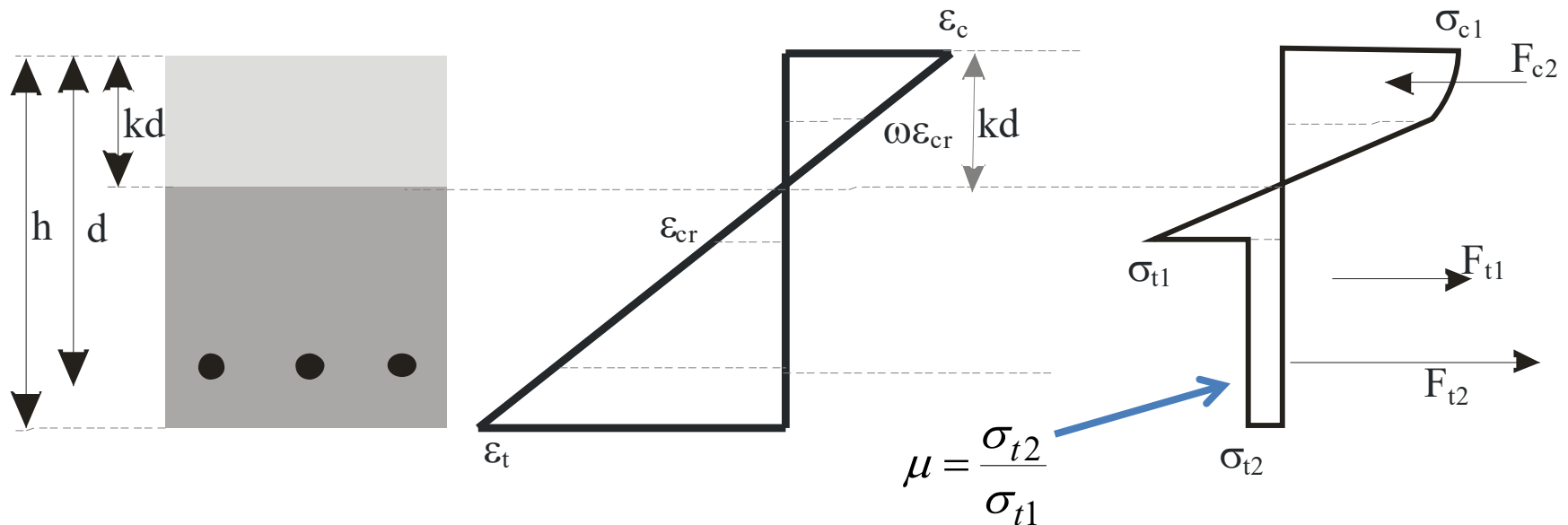


## Cross section



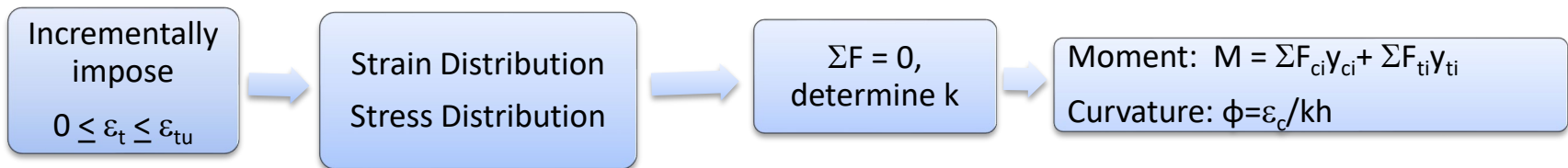
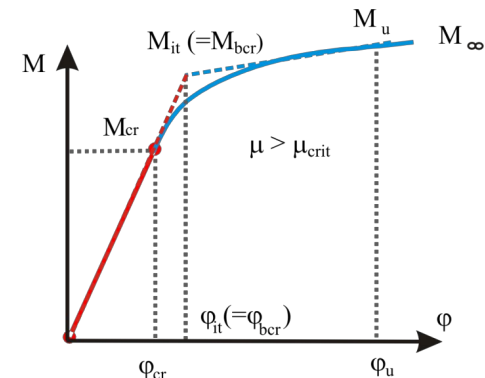
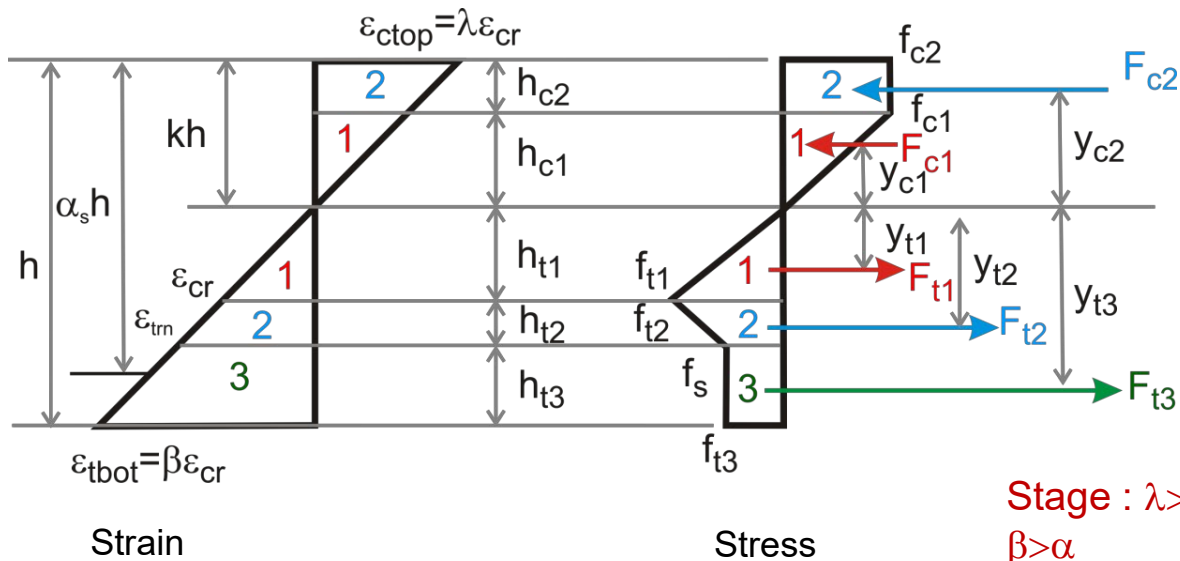
# Simplified Design Approach for FRC

- materials model, testing techniques, and design codes
- Simplifies FRC Flexural design assumption for a beam with traditional and fiber reinforcement.
- Use a stress distribution analogous to Whitney's stress block diagram
- Tensile stresses in accordance with stress-strain response applied over the tensile domain
- Integrate to compute the moment capacity



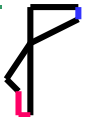
tensile stress transmitted by the fibers across the flexural crack. A measure of residual tensile stress-strain response.

# Derivation of Moment-Curvature Relationship

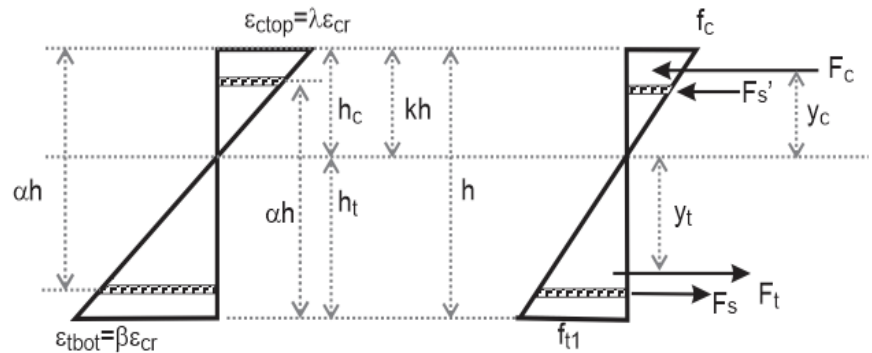
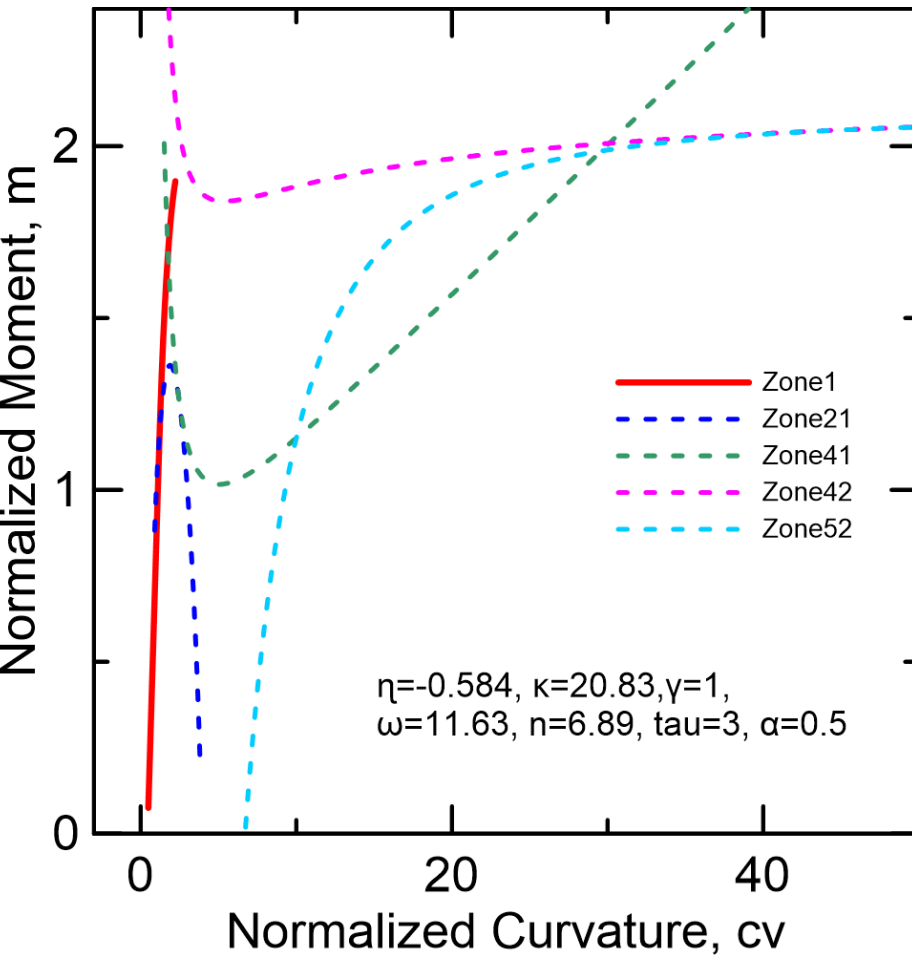


# Moment Curvature Diagram for hardening –softening materials

$$\phi' = \phi / \phi_{cr} \quad \phi'_i = \frac{\beta}{2(1 - k_i)}$$

| Stage | Range                                            | k                                                                                                                                                                     | M' = M/M <sub>cr</sub>                                                                                                                                                                                                   |                                                                                       |
|-------|--------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|
| 1     | (0 < β ≤ 1)<br>and<br>(λ ≤ ω)                    | $k_1 = \begin{cases} \frac{-1 + \sqrt{\gamma}}{-1 + \gamma} & \text{for } \gamma > 1 \\ \frac{1}{2} & \text{for } \gamma = 1 \end{cases}$                             | $M'_1 = \frac{2\beta[(\gamma - 1)k_1^3 + 3k_1^2 - 3k_1 + 1]}{1 - k_1}$                                                                                                                                                   |    |
| 2.1   | (1 < β ≤ α)<br>and<br>(λ ≤ ω)                    | $k_{21} = \frac{\beta^2\gamma + D_{21} - \sqrt{\gamma^2\beta^4 + D_{21}\gamma\beta^2}}{D_{21}}$<br>$D_{21} = \eta(\beta^2 - 2\beta + 1) + 2\beta - \beta^2\gamma - 1$ | $M'_{21} = \frac{(2\beta\gamma + C_{21})k_1^3 - 3C_{21}k_1^2 + 3C_{21}k_1 - C_{21}}{1 - k_1}$<br>$C_{21} = \frac{-2\eta\beta^3 + 3\eta\beta^2 - 3\beta^2 - \eta + 1}{\beta^2}$                                           |    |
| 2.2   | (1 < β ≤ α)<br>and<br>(ω < λ ≤ λ <sub>cu</sub> ) | $k_{22} = \frac{D_{22}}{D_{22} + 2\omega\gamma\beta}$<br>$D_{22} = \eta(\beta^2 - 2\beta + 1) + 2\beta + \omega^2\gamma - 1$                                          | $M'_{22} = (3\omega\gamma + C_{22})k_{22}^2 - 2C_{22}k_{22} + C_{22}$<br>$C_{22} = \frac{2\eta\beta^3 - 3\eta\beta^2 + 3\beta^2 - \omega^3\gamma + \eta - 1}{\beta^2}$                                                   |    |
| 3.1   | (β > α)<br>and<br>(λ ≤ ω)                        | $k_{31} = \frac{D_{31} - \sqrt{\gamma\beta^2 D_{31}}}{D_{31} - \beta^2\gamma}$<br>$D_{31} = \eta(\alpha^2 - 2\alpha + 1) + 2\mu(\beta - \alpha) + 2\alpha - 1$        | $M'_{31} = \frac{(C_{31} - 2\beta\gamma)k_{31}^3 - 3C_{31}k_{31}^2 + 3C_{31}k_{31} - C_{31}}{k_{31} - 1}$<br>$C_{31} = \frac{3(\mu\beta^2 - \mu\alpha^2 - \eta\alpha^2 + \alpha^2) + 2\eta\alpha^3 + \eta - 1}{\beta^2}$ |  |
| 3.2   | (β > α)<br>and<br>(ω < λ ≤ λ <sub>cu</sub> )     | $k_{32} = \frac{D_{32}}{D_{32} + 2\omega\gamma\beta}$<br>$D_{32} = \omega^2\gamma + \eta\alpha^2 + 2(\mu\beta - \eta\alpha - \mu\alpha + \alpha) + \eta - 1$          | $M'_{32} = (C_{32} + 3\omega\gamma)k_{32}^2 - 2C_{32}k_{32} + C_{32}$<br>$C_{32} = \frac{3(\mu\beta^2 - \mu\alpha^2 - \eta\alpha^2 + \alpha^2) + 2\eta\alpha^3 - \omega^3\gamma + \eta - 1}{\beta^2}$                    |  |

# Interaction of modes of strain and stress distribution at all stages

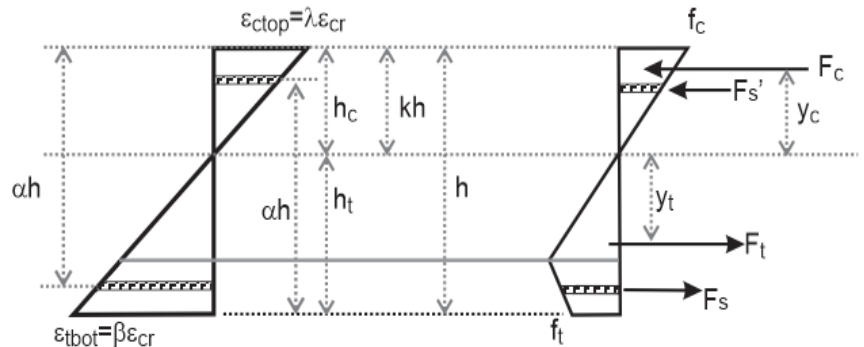
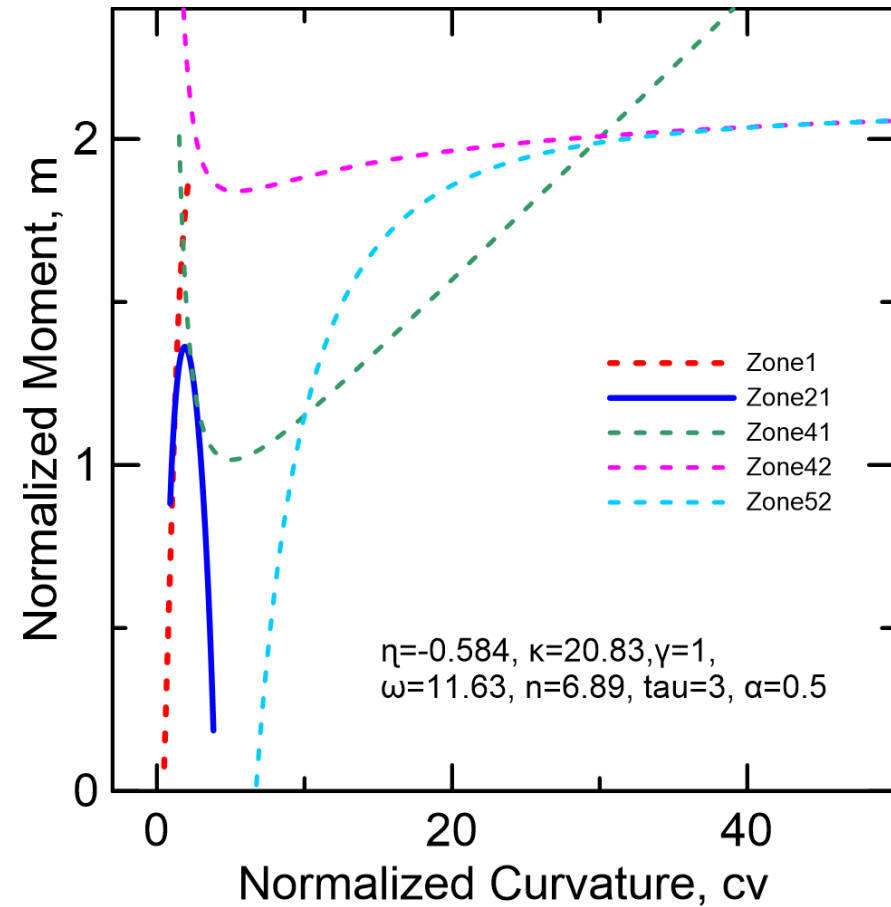


Zone1 :

- compression and tension are in elastic zone
- Steel is elastic



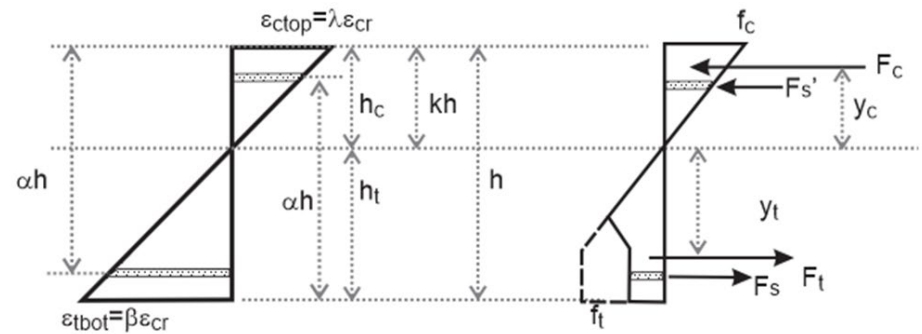
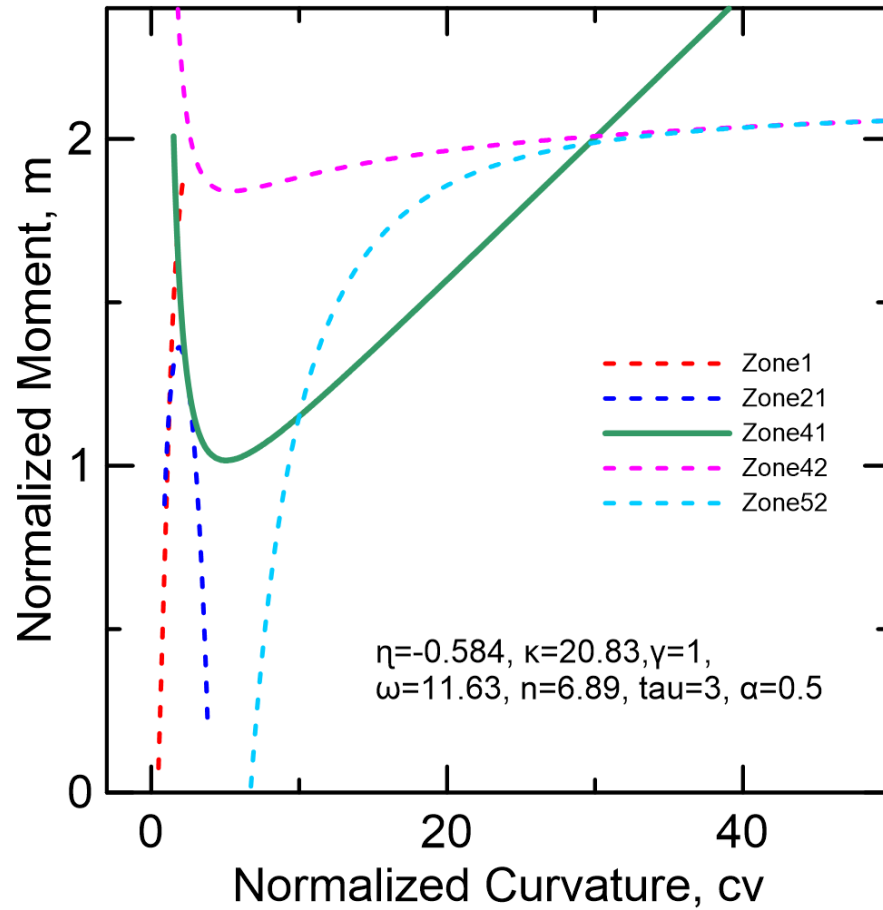
## Zone2.1



### Zone21

- compression is in elastic zone
- tension is second slope zone, first crack has formed
- Steel is elastic

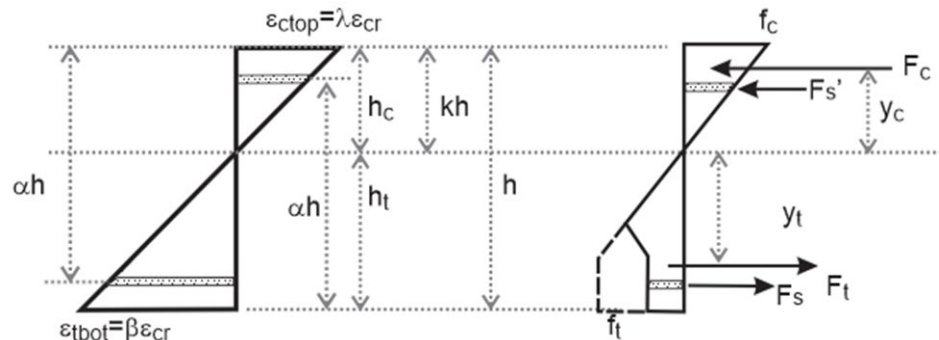
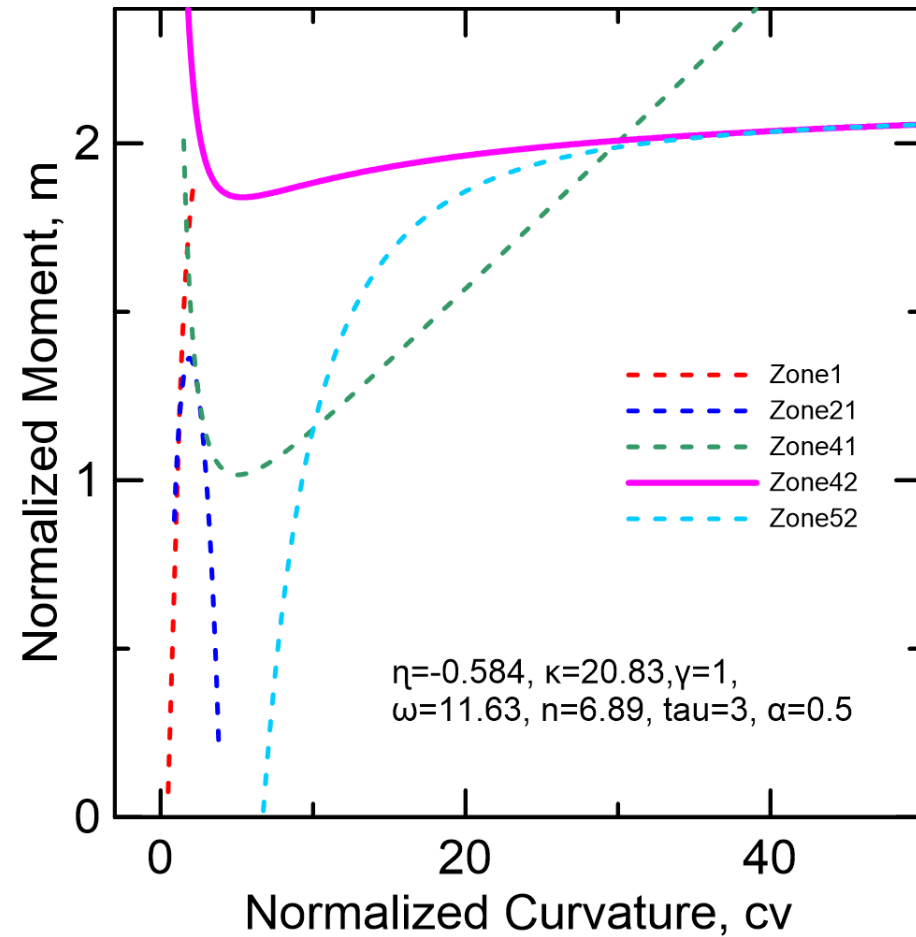
# Zone4.1



## Stage41 :

- compression is in elastic zone
- tension is in residual third range
- Steel is elastic

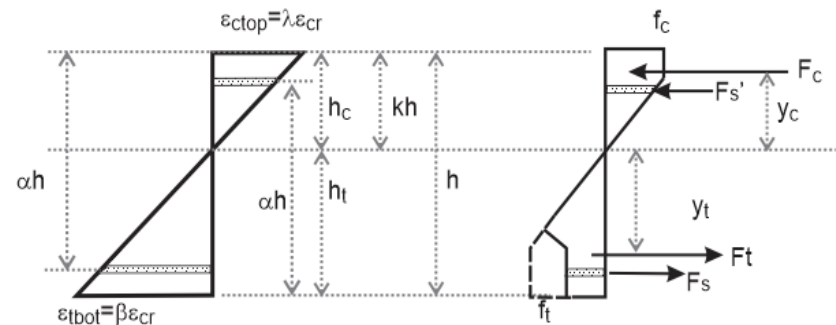
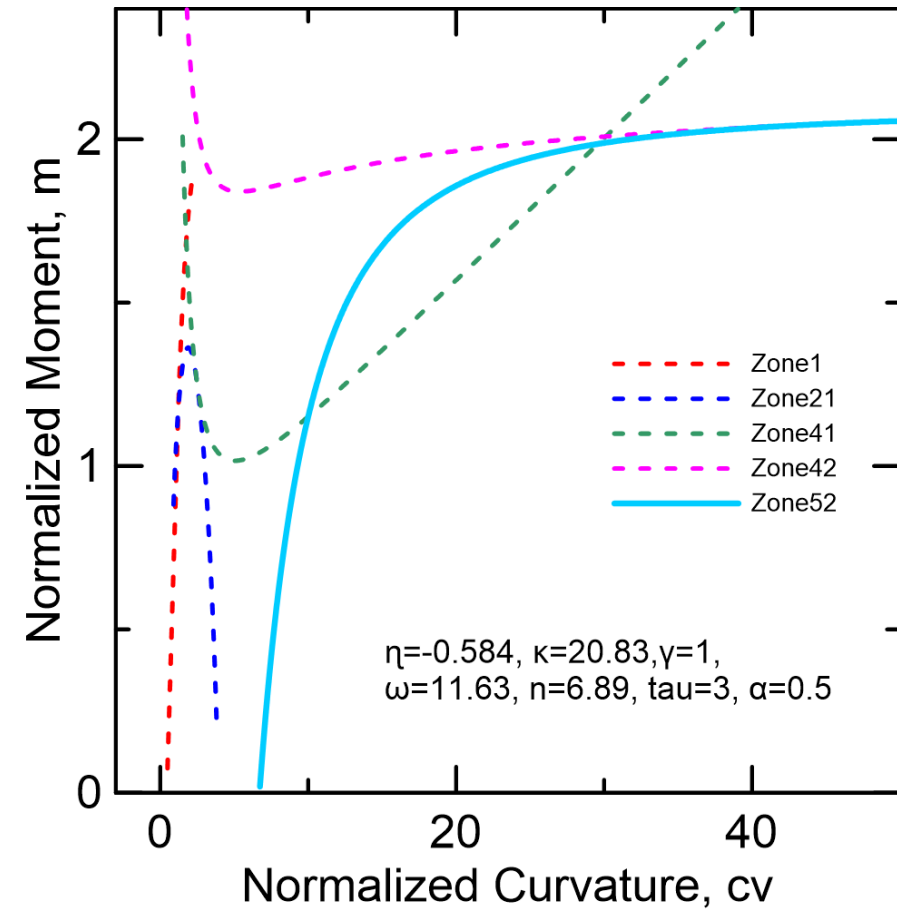
## Zone4.2



### Stage42 :

- compression is in elastic zone
- tension is in residual range
- Steel has yielded

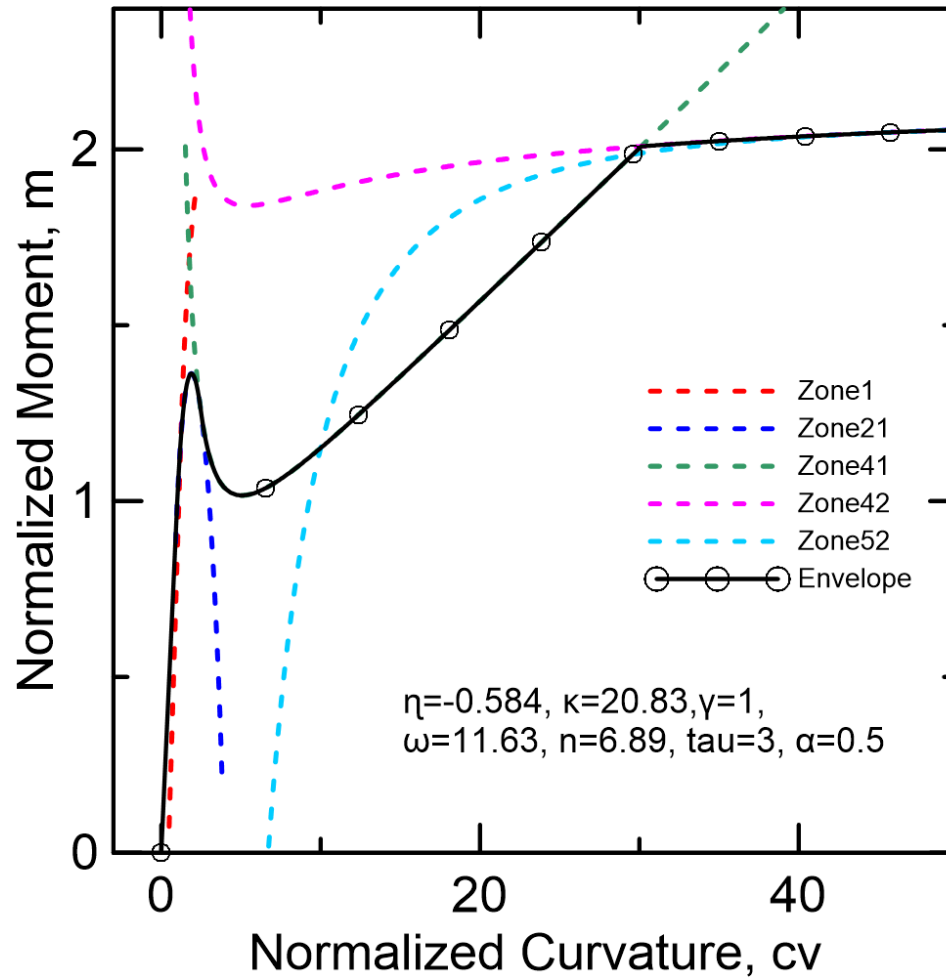
## Zone5.2



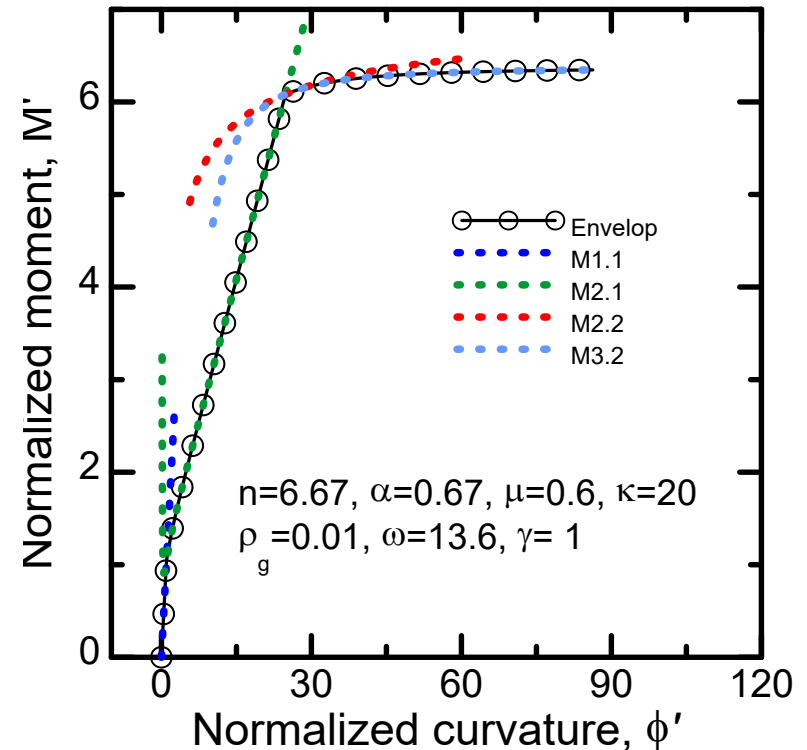
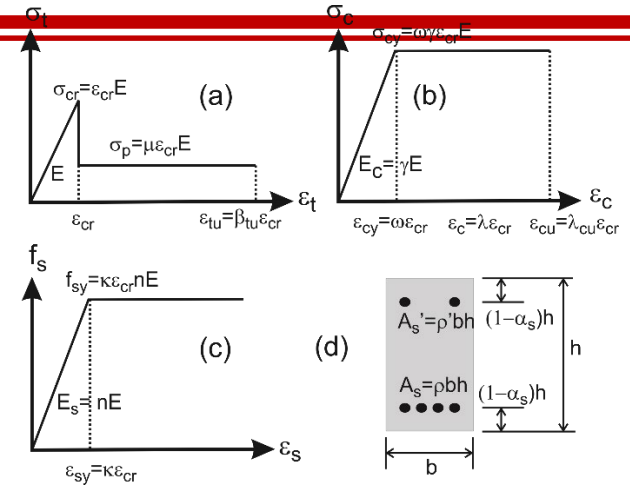
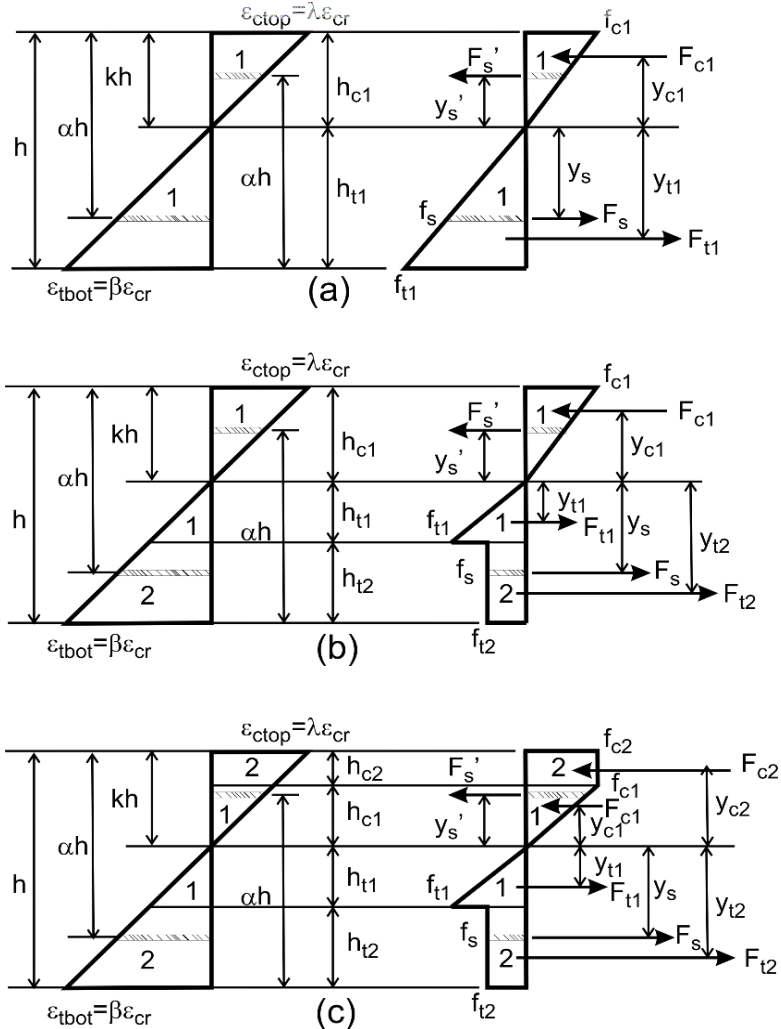
Stage52 :

- compression is in plastic zone
- tension is in residual range
- Steel has yielded

# Envelope curve



# Development of Envelope Moment-Curvature for Hybrid Response



# Moment-Curvature Response in Flexure

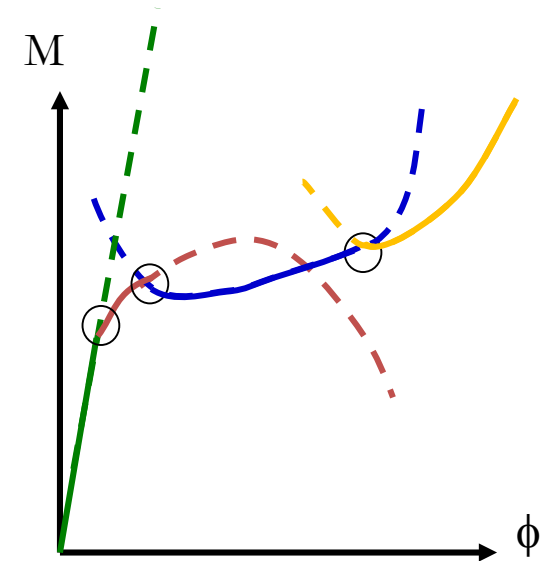
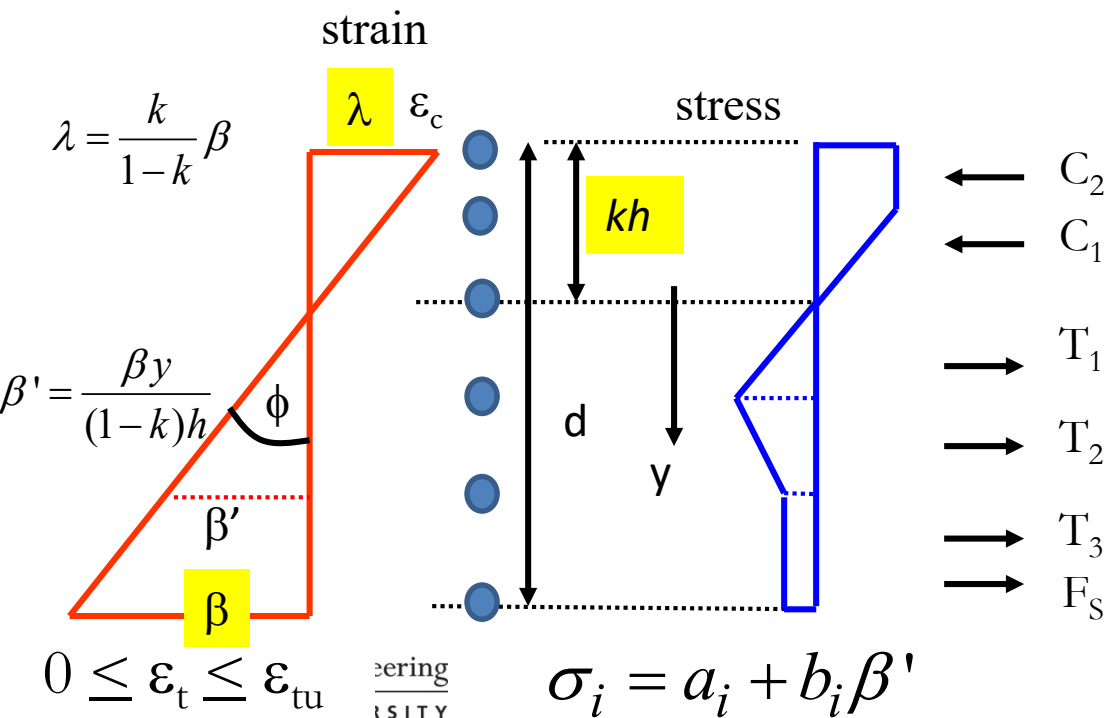
Change of variables, location of Neutral axis,  $[\beta, \lambda]$  to  $[\beta, k]$

- Incrementally impose  $0 \leq \varepsilon_t \leq \varepsilon_{tu}$
- Strain Distribution, linear,  $\beta, \beta'$
- Stress Distribution defined by linear segments
- $\Sigma F = 0$ , determine  $k$  (Neutral axis)
- $M = \Sigma C_i y_{ci} + \Sigma T_i y_{ti}$  and  $\phi = \varepsilon_c / kd$
- Normalization  $M' = M / M_0$  and  $\phi' = \phi / \phi_{cr}$

$$\sigma_i = a_i + b_i \beta'$$

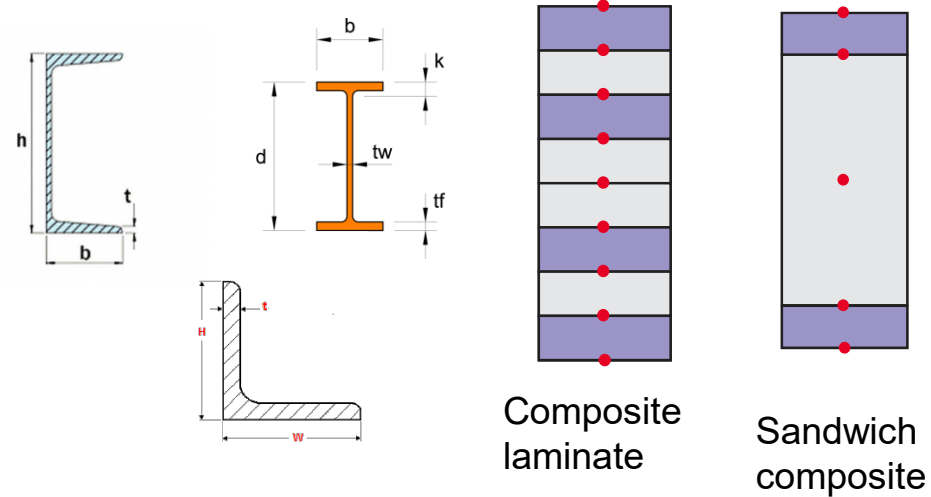
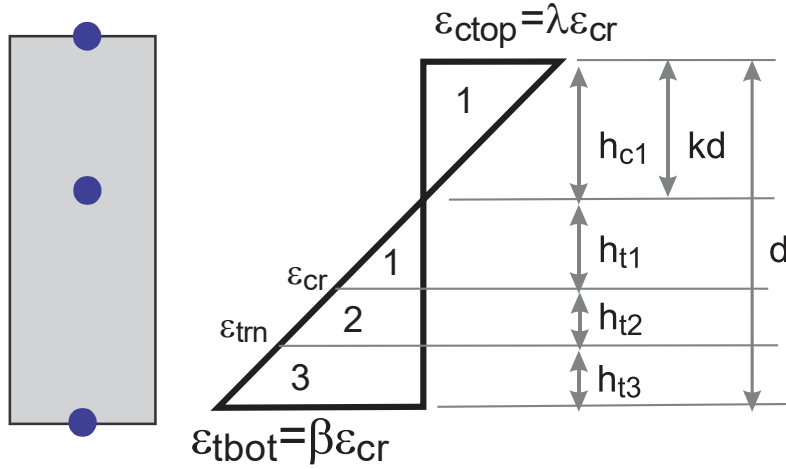
$$F_{c1} = b \int_0^{kd} \sigma_i(y) dy$$

$$y_{c1} = \frac{b}{F_{c1}} \int_0^{kd} \sigma_i(y) y dy$$

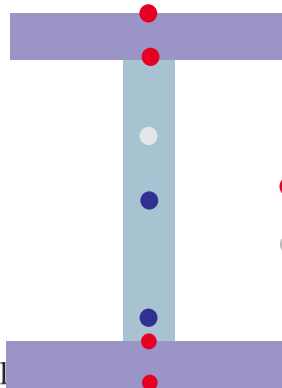


Moment curvature diagram

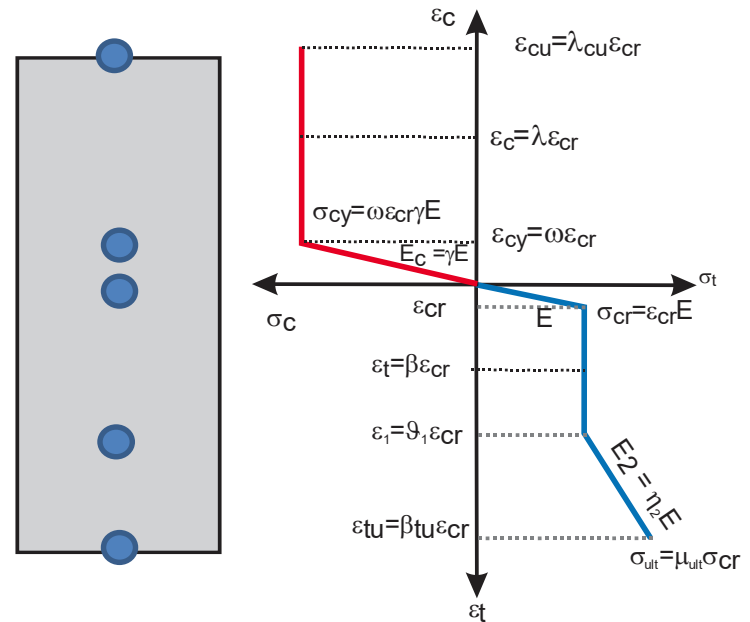
# Control Points, Geometrical and Material



Rectangular section



- stress-strain control point
- geometrical control point
- neutral axis control point





# Example for Control points for geometry and sections for a doubly Reinforced Concrete section

Define the strain distribution variable along the depth of the section

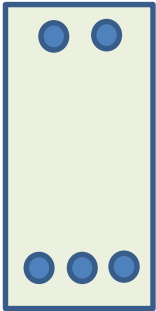
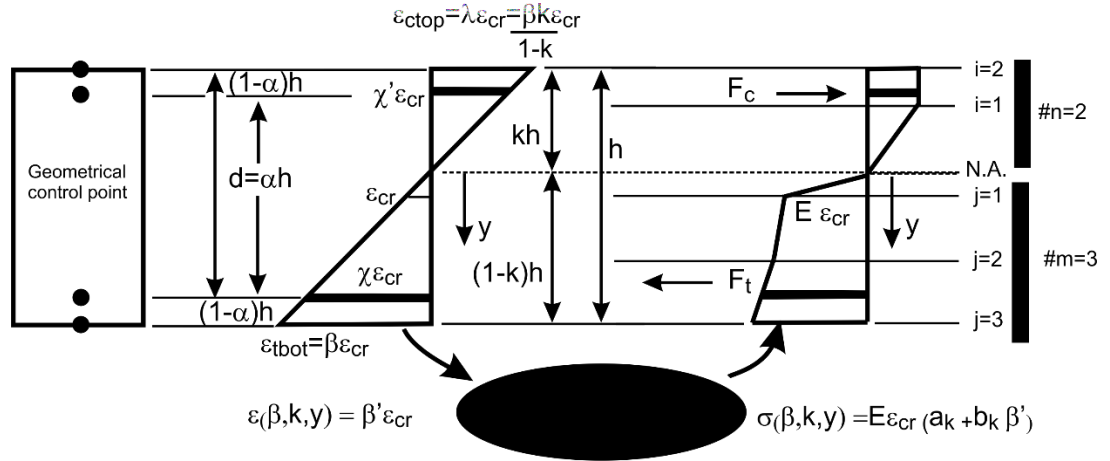
$$\beta' = \frac{\varepsilon(y)}{\varepsilon_{cr}} = \beta'(\beta, k, \frac{y}{h}) = \frac{\beta y}{(1-k)h}$$

$\lambda$  = Compressive strain

$\beta'$  = Tensile strain across depth

$k$  = Neutral axis location

$\beta$  = Max. Tensile strain, indep. variable



| Geometrical control point | Depth, $\frac{y}{h}$ | Strain Magnitude                                      |
|---------------------------|----------------------|-------------------------------------------------------|
| Extreme compression fiber | $-kh$                | $\lambda = \beta' = \frac{-\beta k}{1-k}$             |
| Compression reinforcement | $-(\alpha + k - 1)$  | $\chi' = \beta' = \frac{-(\alpha + k - 1)\beta}{1-k}$ |
| Neutral axis              | 0                    | $\beta' = 0$                                          |
| Tension reinforcement     | $(\alpha - k)h$      | $\chi = \beta' = \frac{(\alpha - k)\beta}{1-k}$       |
| Extreme tensile fiber     | $(1-k)h$             | $\beta' = \beta$                                      |

# Derivation of Moment-Curvature Relationship

Incrementally impose

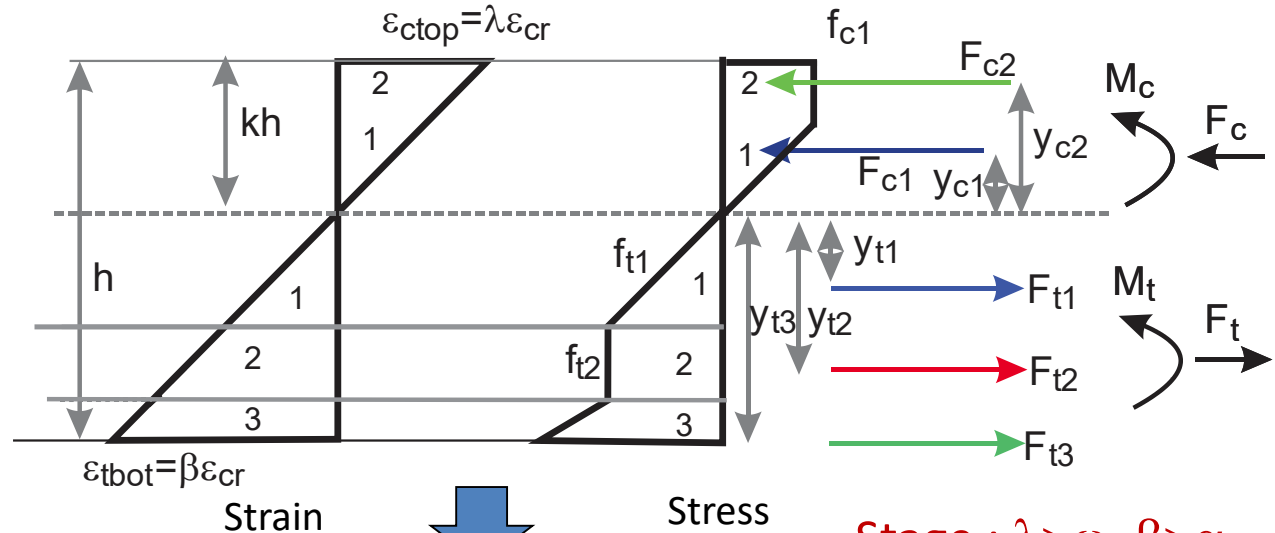
$$0 \leq \varepsilon_t \leq \varepsilon_{tu} \quad \beta$$

Strain Distribution  
Stress Distribution

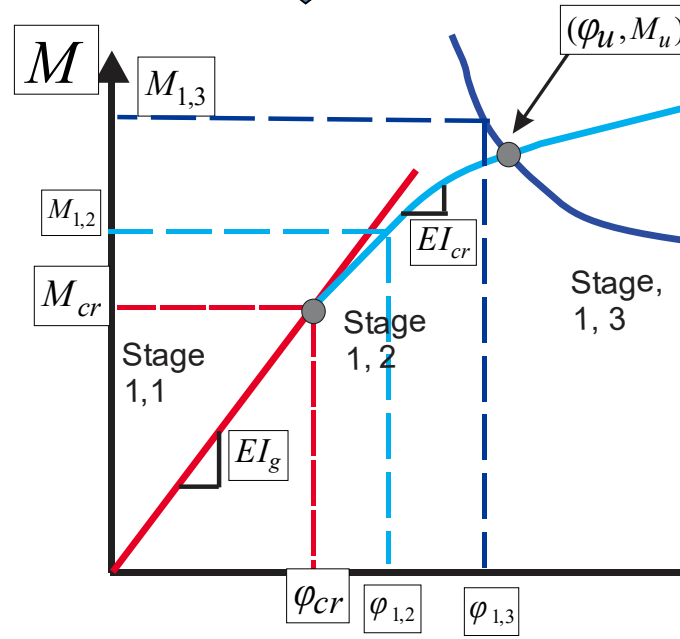
$\Sigma F = 0$ , determine  $k$

Moment:  $M = \Sigma F_{ci} y_{ci} + \Sigma F_{ti} y_{ti}$

Curvature:  $\phi = \varepsilon_c / kh$



Stage :  $\lambda > \omega$ ,  $\beta > \alpha$



# Evaluation of Moment-Curvature using Piecewise Functions

| Stage    | Tension              | TRC tensile strain                    | Compression       | TRC compressive strain               |
|----------|----------------------|---------------------------------------|-------------------|--------------------------------------|
| Zone 1.1 | Elastic              | $0 \leq \beta < 1$                    | Elastic           | $0 \leq \lambda < \omega$            |
| Zone 1.2 | Distributed Cracking | $1 \leq \beta < \vartheta_1$          | Elastic           | $0 \leq \lambda < \omega$            |
| Zone 1.3 | Hardening            | $\vartheta_1 \leq \beta < \beta_{tu}$ | Elastic           | $0 \leq \lambda < \omega$            |
| Zone 2.2 | Distributed Cracking | $1 \leq \beta < \vartheta_1$          | Perfectly plastic | $\omega \leq \lambda < \lambda_{cu}$ |
| Zone 2.3 | Hardening            | $\vartheta_1 \leq \beta < \beta_{tu}$ | Perfectly plastic | $\omega \leq \lambda < \lambda_{cu}$ |

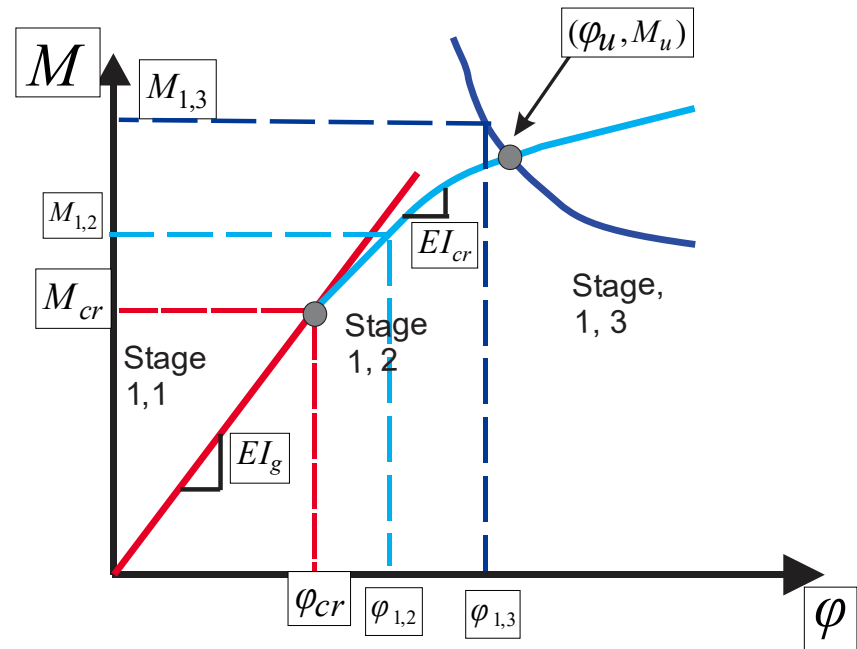
$$\phi = \frac{\beta \varepsilon_{cr}}{(1-k)h}$$

$$\phi_{cr} = \frac{2\varepsilon_{cr}}{h}$$

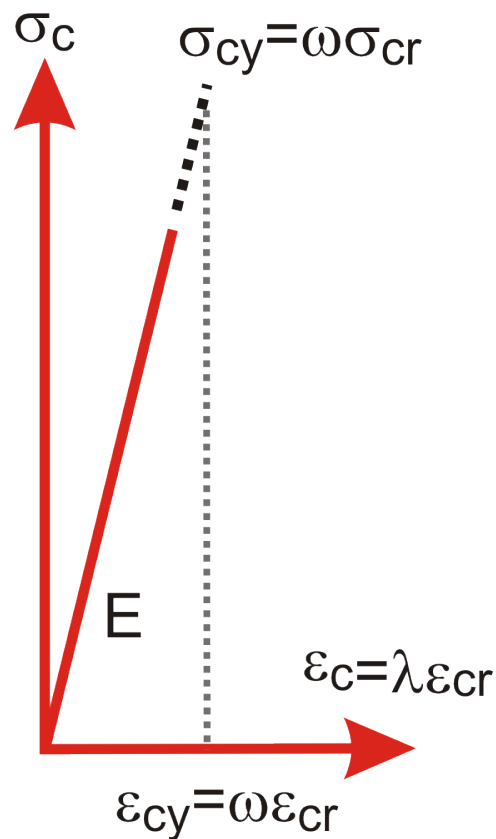
$$M_{cr} = \frac{1}{6}bd^2E\varepsilon_{cr}$$

$$\phi = \phi'_{i,j} \phi_{cr}$$

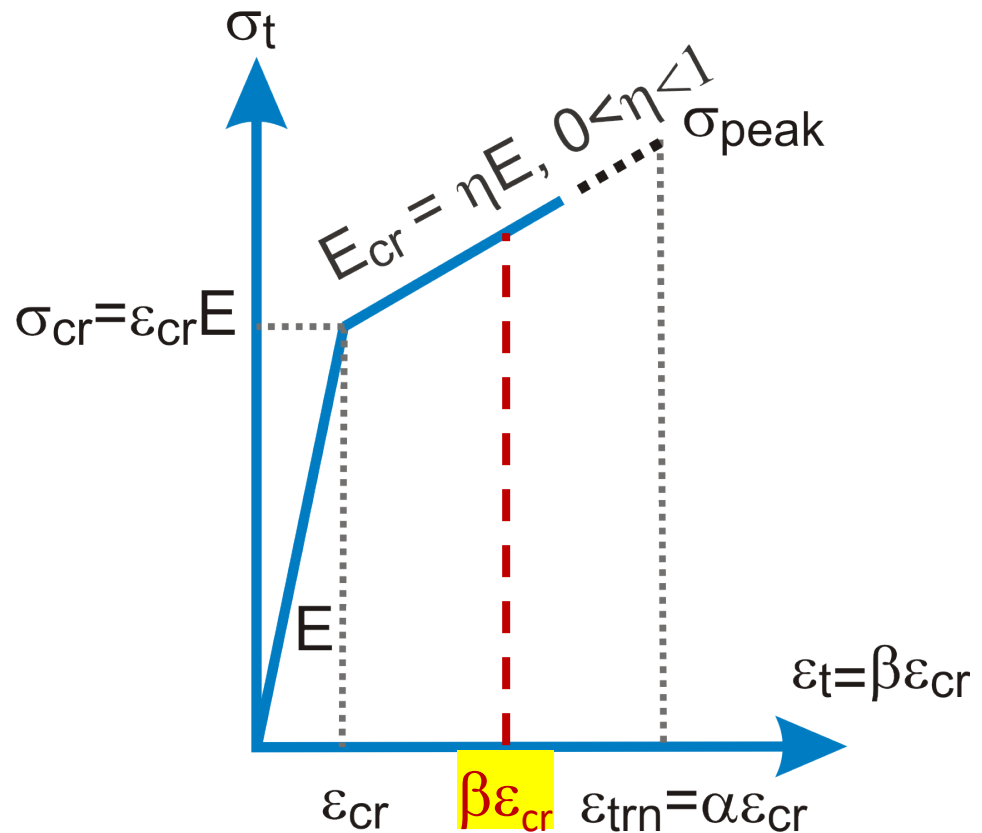
$$M_n = m_n M_{cr}$$



Serviceability based design, limit the moment and curvature to a specific limit of strain, Variable  $\beta$



Compression model



Tension model

Tension design Valid if  $\alpha > \beta$

# Design guide for strain hardening Systems

- 1st Crack tensile strain, (material property)  $\varepsilon_{cr} = \frac{\sigma_{cr}}{E}$
- Allowable tensile strain, (tensile limit,  $\beta_{max}$ )  $\beta_{max} = \frac{\varepsilon_t}{\varepsilon_{cr}} \leq \alpha = \frac{\varepsilon_{trn}}{\varepsilon_{cr}}$
- post crack tensile modulus  $\eta = \frac{E_{cr}}{E} = \frac{\sigma_{peak} - \sigma_{cr}}{E} (\varepsilon_{trn} - \varepsilon_{cr})$
- compressive yield strain, Reduction factor  $\omega = \frac{\phi f'_c}{\sigma_{cr}} \quad \phi=0.8$
- Neutral axis depth and Normalized moment and Curvature at any given strain:

$$k = \frac{C_1 - \sqrt{\beta^2 C_1}}{C_1 - \beta^2} \quad M' = C_2 \frac{k^2 - 2k + 1}{\beta^2} + \frac{2\beta k^3}{1 - k} \quad \phi' = \frac{\beta}{2(1 - k)}$$

$$C_1 = \eta (\beta^2 - 2\beta + 1) + 2\beta - 1 \quad M_{cr} = \frac{\sigma_{cr} b h^2}{6} \quad \phi_{cr} = \frac{2\varepsilon_{cr}}{d} \quad \phi = \phi'(\beta, \eta) \phi_{cr}$$

$$C_2 = C_1 + 2C_1\beta - \beta^2$$

$$M_n = M'(\beta, \eta) M_{cr} \quad \phi = \phi'(\beta, \eta) \phi_{cr}$$

# Design guide for strain hardening FRC Composites

2. Generate the moment curvature for the fixed value of post crack stiffness,  $\eta$  and all values of  $\beta$  as an array. Plot the moment and curvature response.

3. Check the normalized compressive strain developed at top fiber must be less than the normalized compressive yield strain

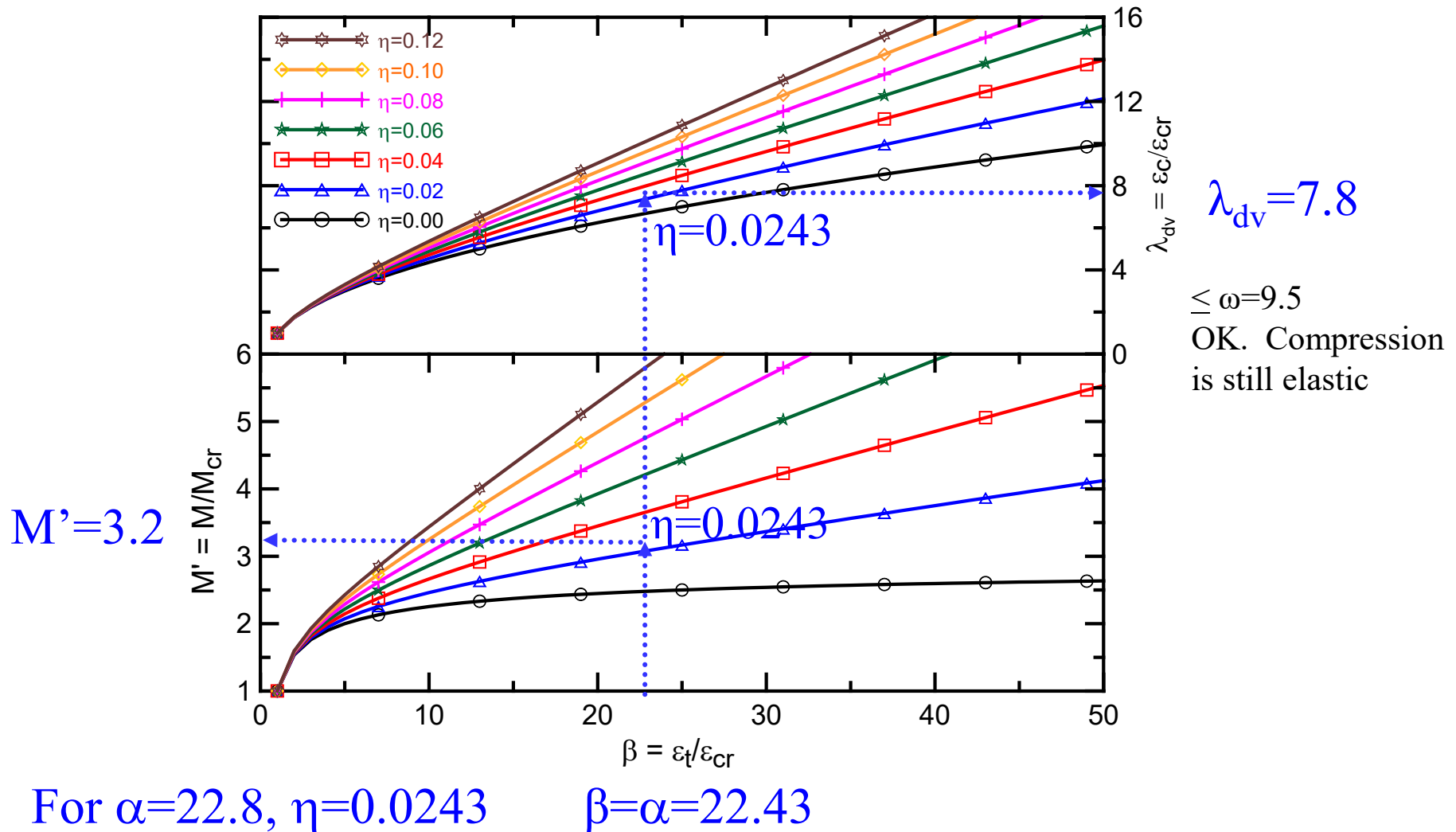
$$\lambda_{dv} = \frac{k}{1-k} \beta_{max} \leq \omega$$

4. Since  $M'$  and  $\lambda_{dv}$  are functions of  $\beta$  and  $\eta$ , it is convenient to create design charts

4.1  $M' - \beta \& \eta$

4.2  $\lambda_{dv} - \beta \& \eta$

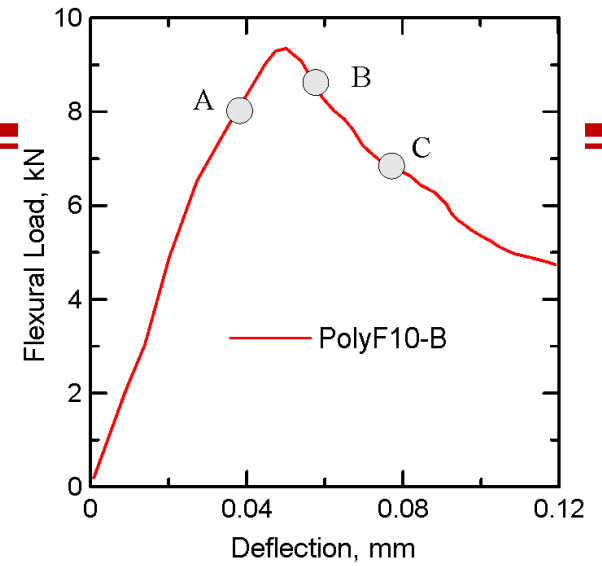
# Design example of strain hardening TRC composites



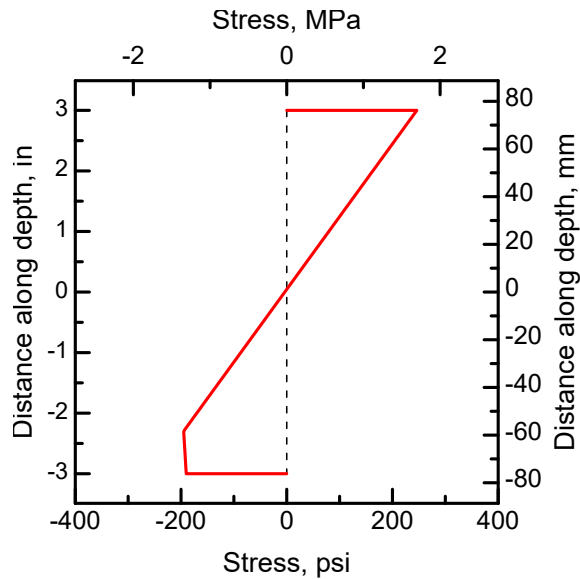
For  $\alpha = 22.8$ ,  $\eta = 0.0243$

$\beta = \alpha = 22.43$

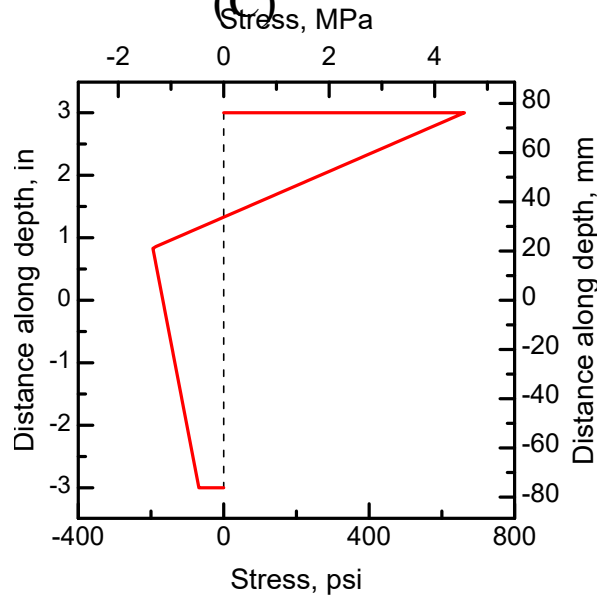
# Evolution of Stress Distribution Profile



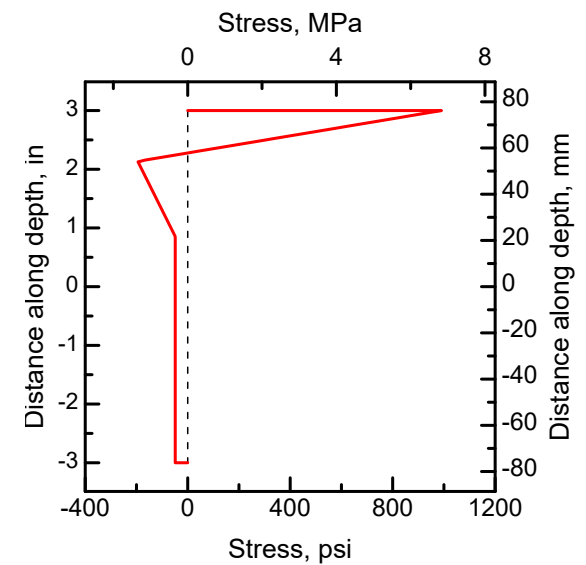
(A)



(C)

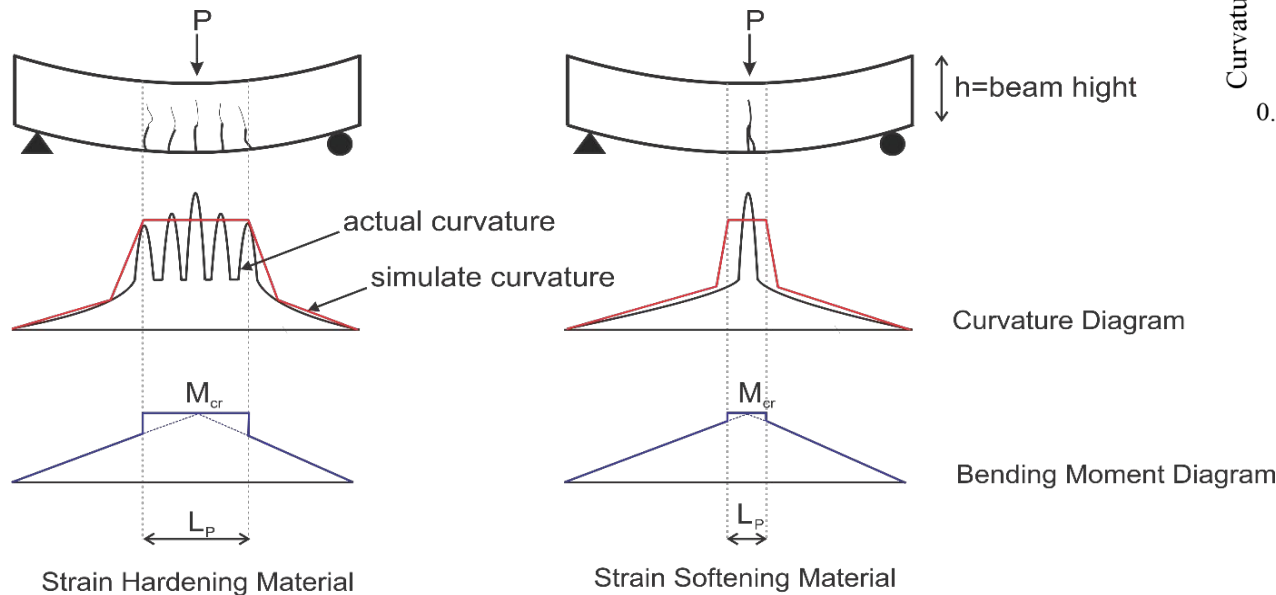


(B)

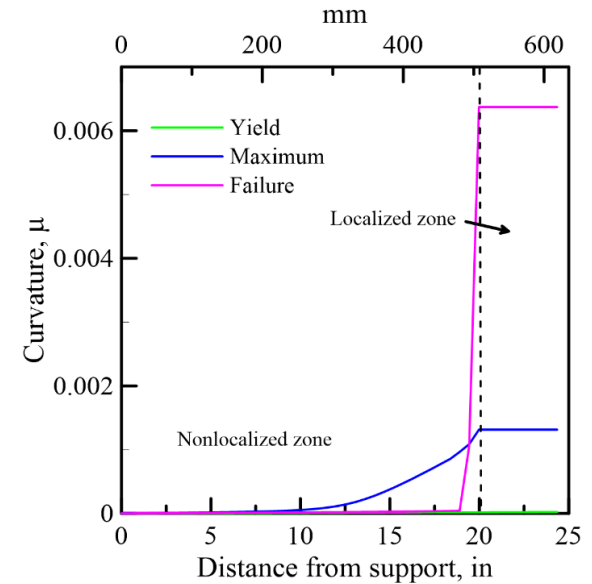




# Modeling the localization effect and effect of residual tensile strength under flexure using deflection criteria

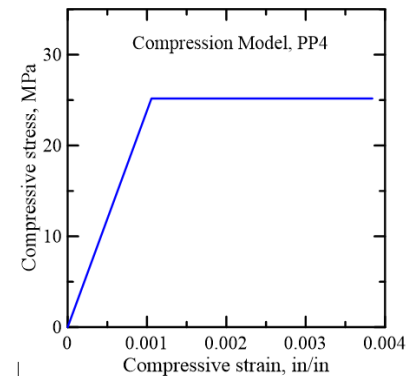
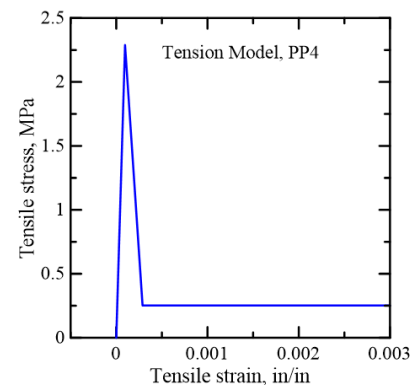
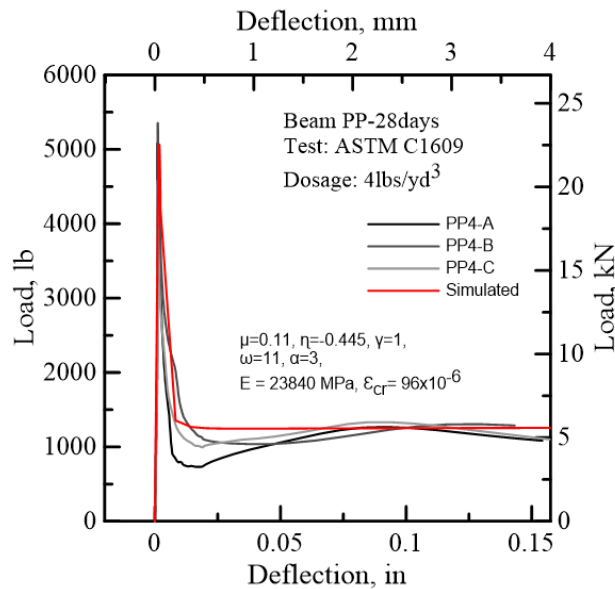
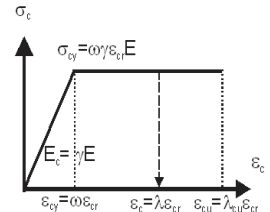
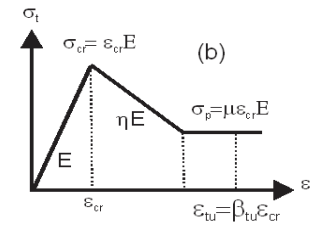
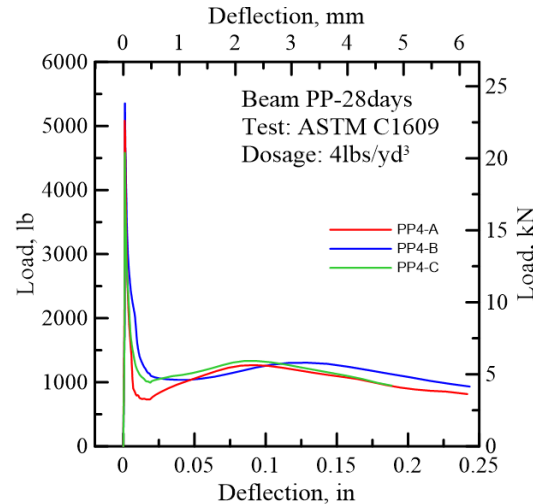
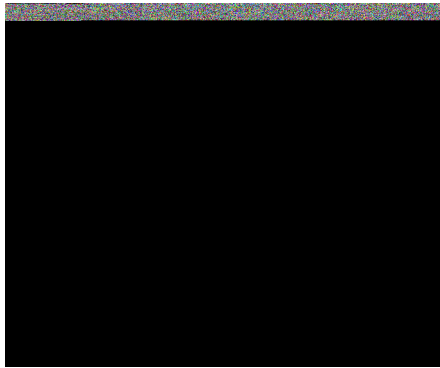


(a) schematic figure comparing in strain hardening and softening materials.

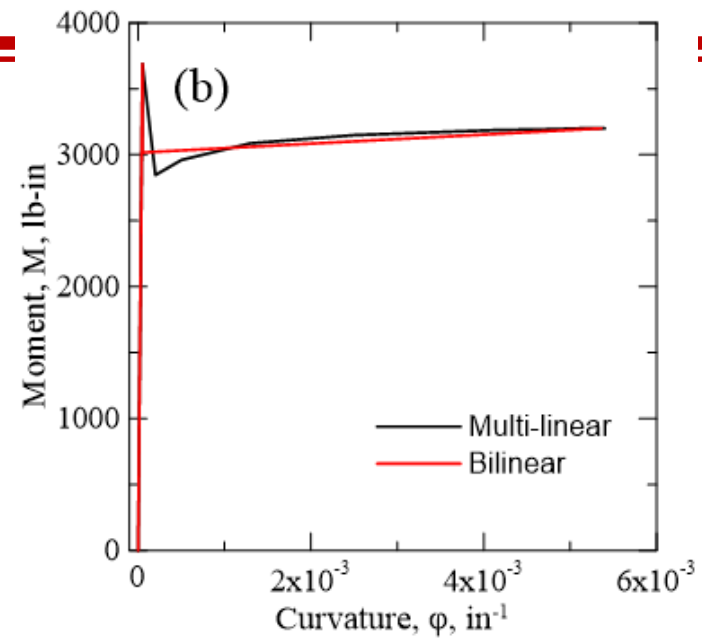
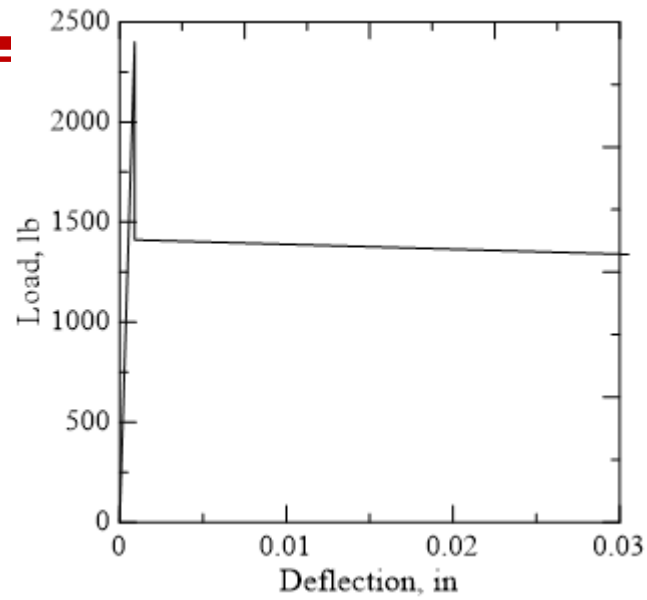
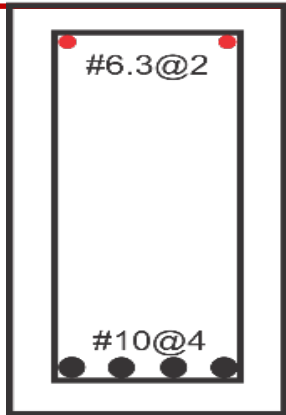


(b) curvature in different stages of loading

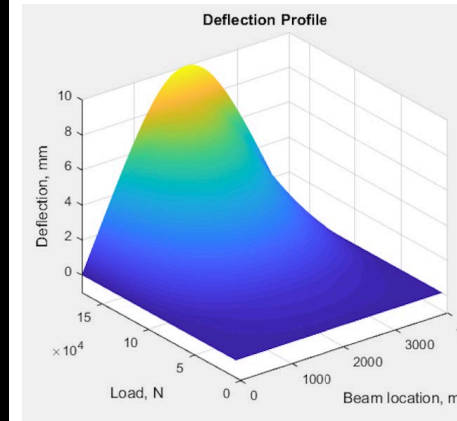
# Standard Design Procedure using ACI 544.4R method (Parametric-based design)



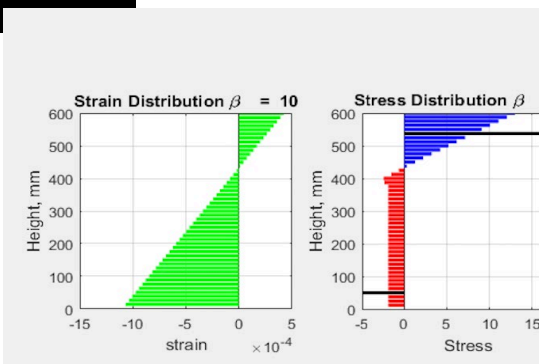
Using these parameters, the location of neutral axis, moment capacity, and curvature of the section can be obtained at any imposed strain



# Hybrid Reinforced Concrete Design & Back-Calculation Software (Compiled Matlab GUI)

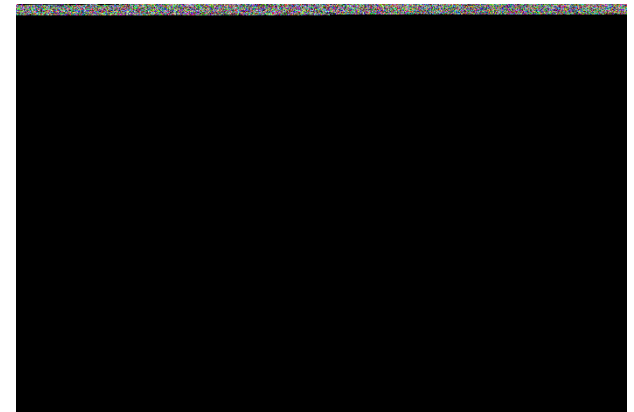
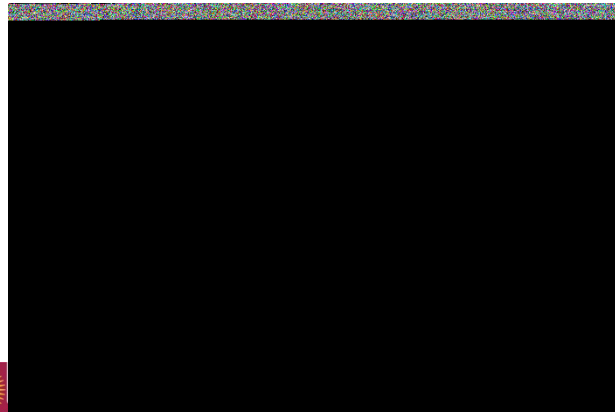
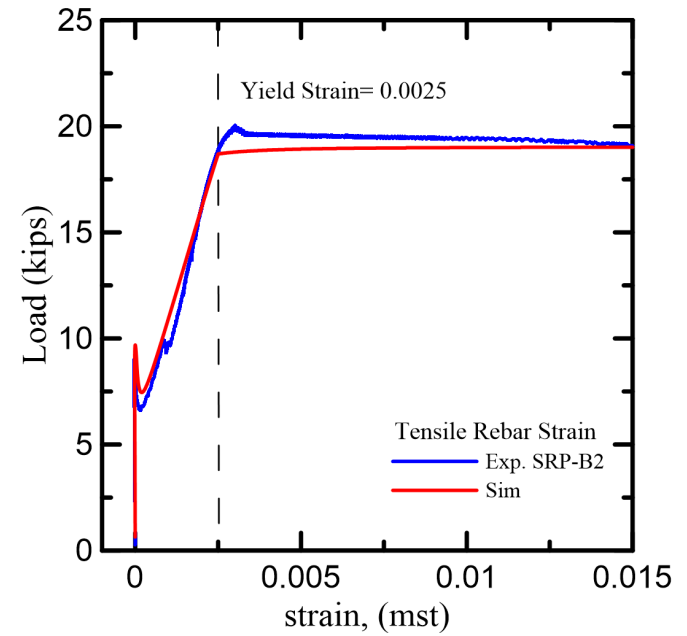
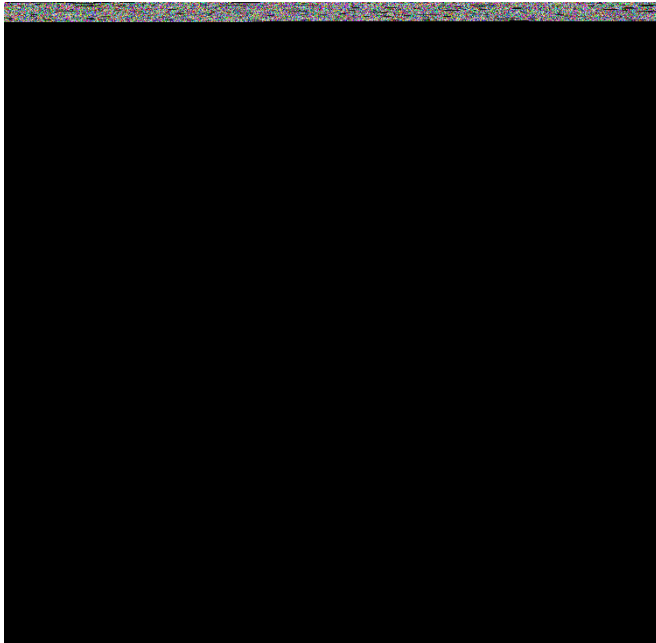


- Design parameters can be specified based on specifications on maximum allowed crack width, specific deflection, maximum curvature, or stiffness criteria.
- Output will produce 12 plots of various responses



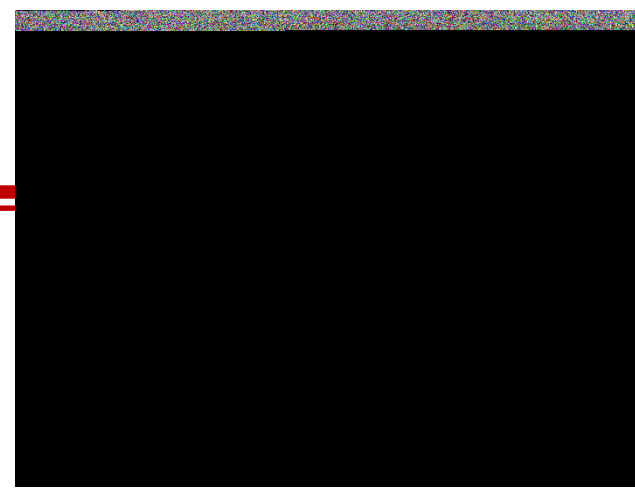
# Panel flexural tests Simulation

Variables Include: support Conditions, rebar configuration, and fiber content



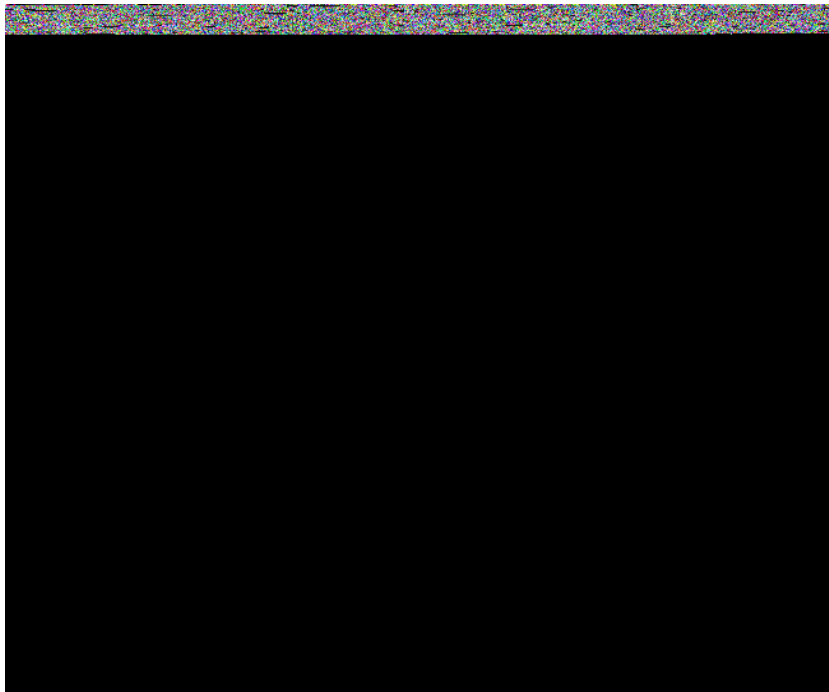
# Force distribution in conventional RC and HRC

---



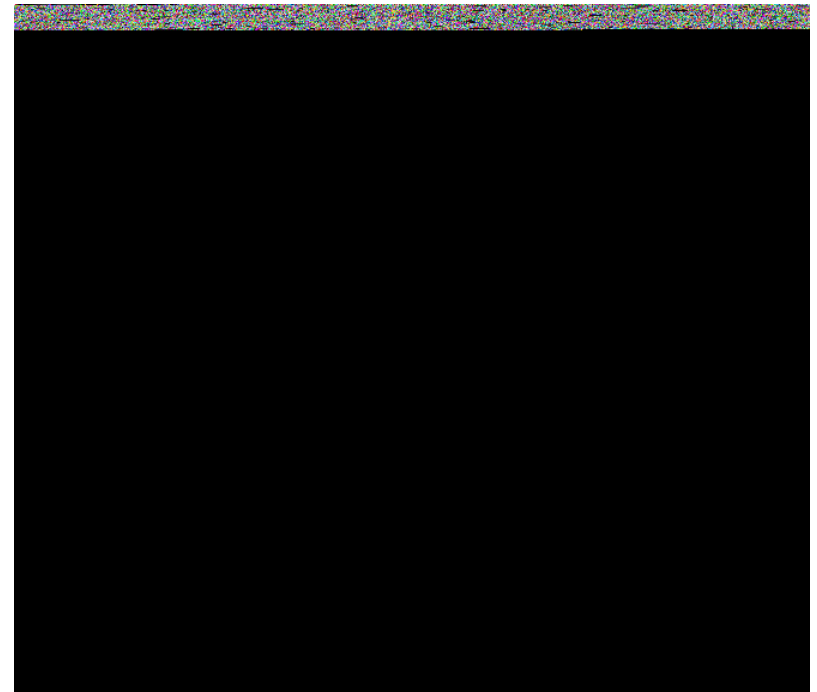
## conventional RC

Sudden change as soon as cracking occurs with rebar tension taking over 100% of the tension load within the serviceability range



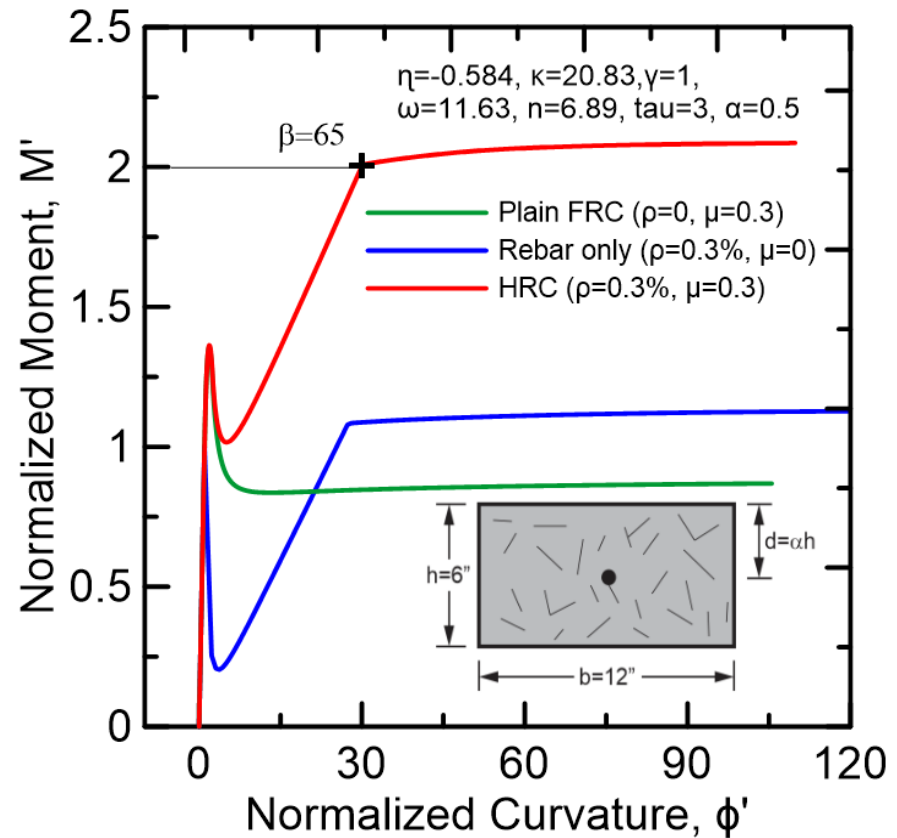
## Hybrid RC

With 1% fiber, there is gradual change with cracking with rebar tension taking over only 20% of the tension load at limit state

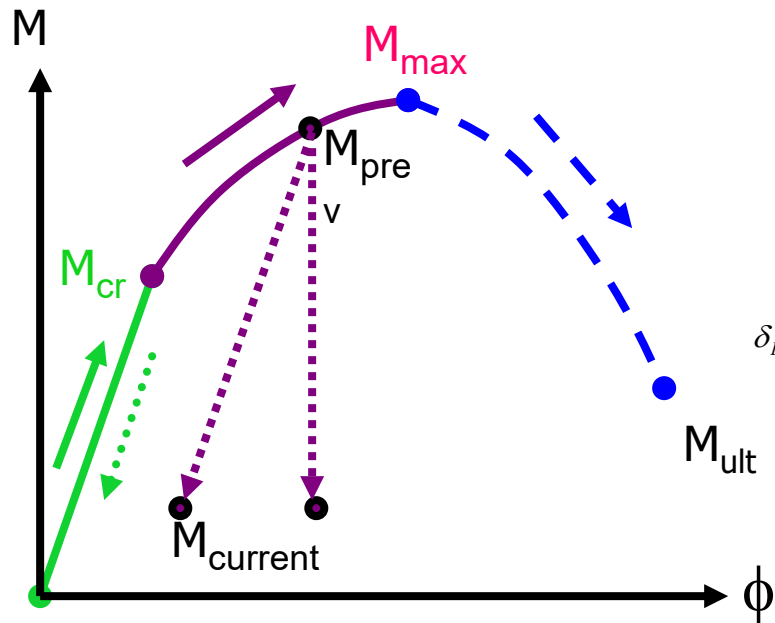


# Three main contributions of the Model Representing the Synergy of fiber and rebar combination

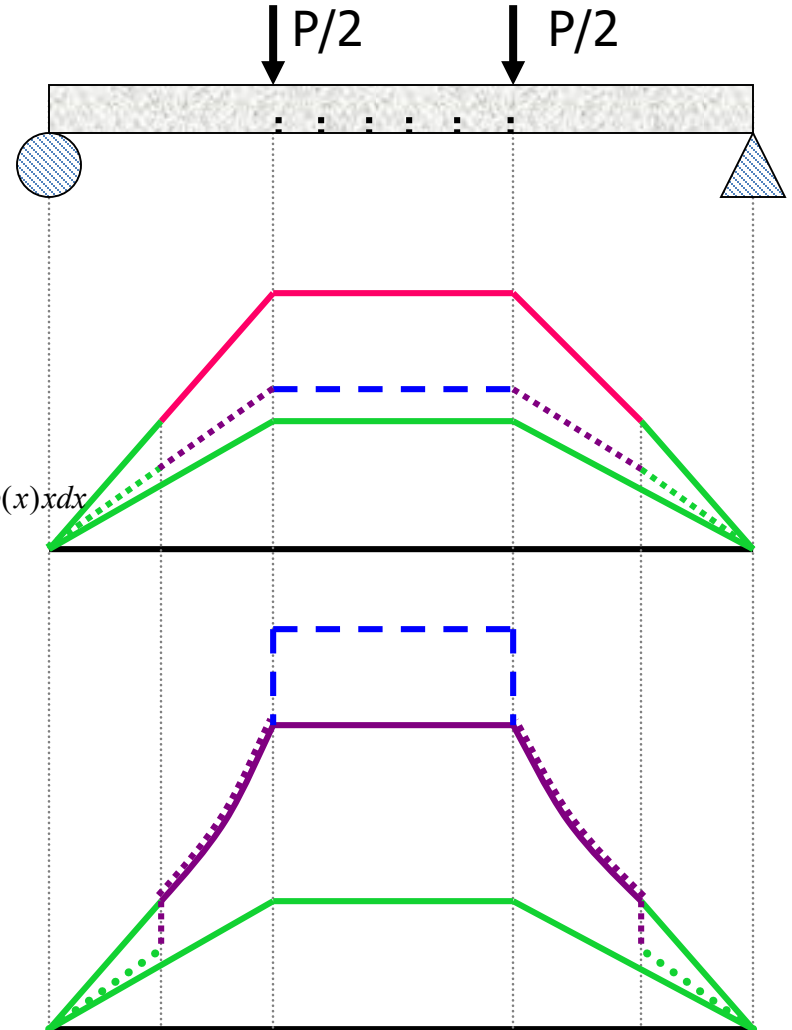
1. Example Slab on Ground with a single rebar at the mid-height
2. Simulated Moment-Curvature Response comparing HRC with plain FRC and conventional rebar
3. Additive contribution of fiber and steel based on compatibility of strain analysis
4. Capability to use a serviceability-based limit strain or curvature analysis



# Load Deflection by Moment Area Method



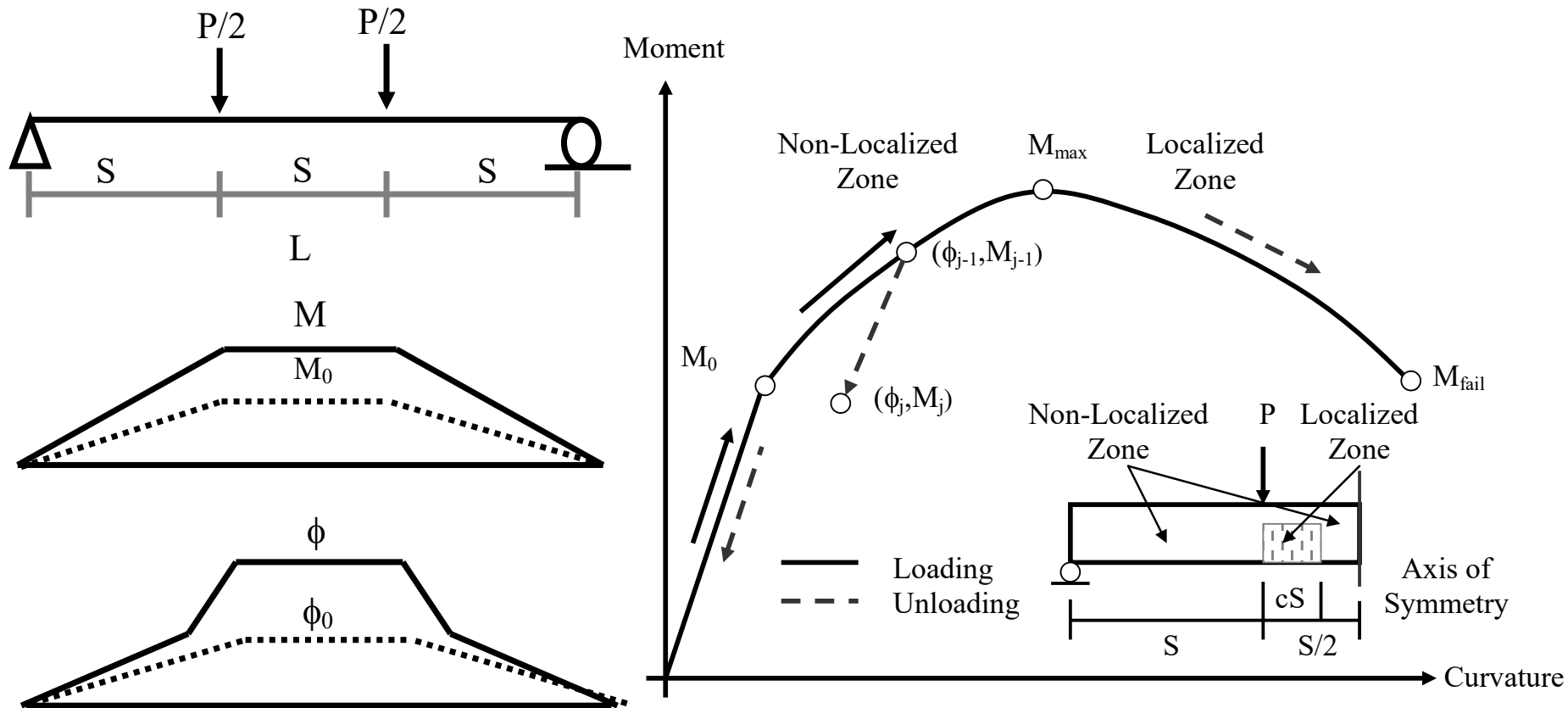
$$\delta_{L/2} = \int_0^{L/2} \phi(x) x dx$$



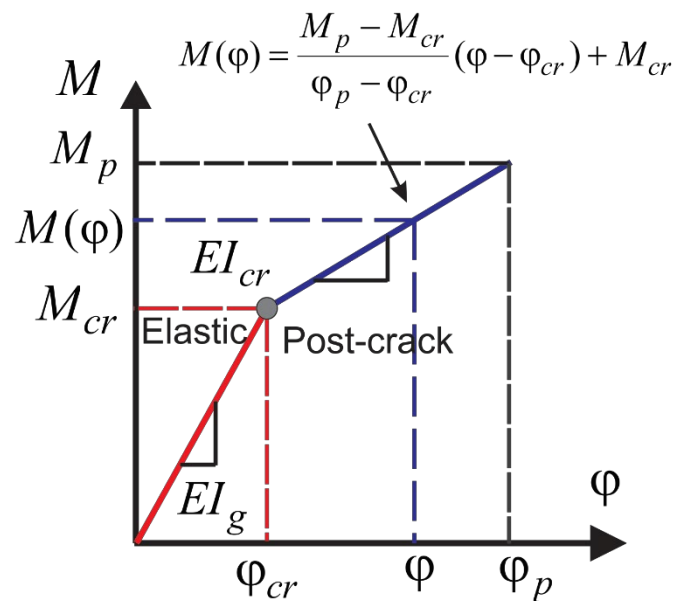


# Load Deflection Response

- Moment area method
- Crack localization rules



# Bilinear Moment-Curvature Model



$$M_{cr} = \frac{1}{6}bd^2E\varepsilon_{cr} \quad \phi_{cr} = \frac{2\varepsilon_{cr}}{d}$$

$$\begin{aligned} M(\phi) &= EI_g \phi & 0 < M \leq M_{cr} & \quad 0 < \phi \leq \phi_{cr} \\ M(\phi) &= \frac{M_p - M_{cr}}{\phi_p - \phi_{cr}}(\phi - \phi_{cr}) + M_{cr} & M_{cr} \leq M \leq M_p & \quad \phi_{cr} \leq \phi \leq \phi_p \end{aligned}$$

Normalized moment-curvature

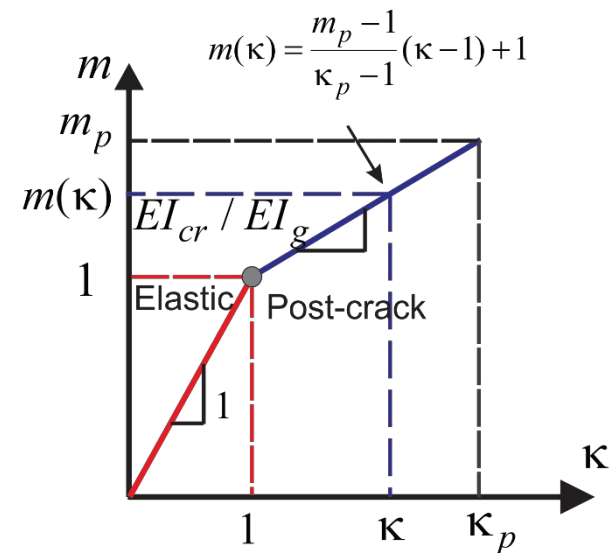
$$m(\kappa) = \kappa$$

$$m(\kappa) = \frac{m_p - 1}{\kappa_p - 1}(\kappa - 1) + 1$$

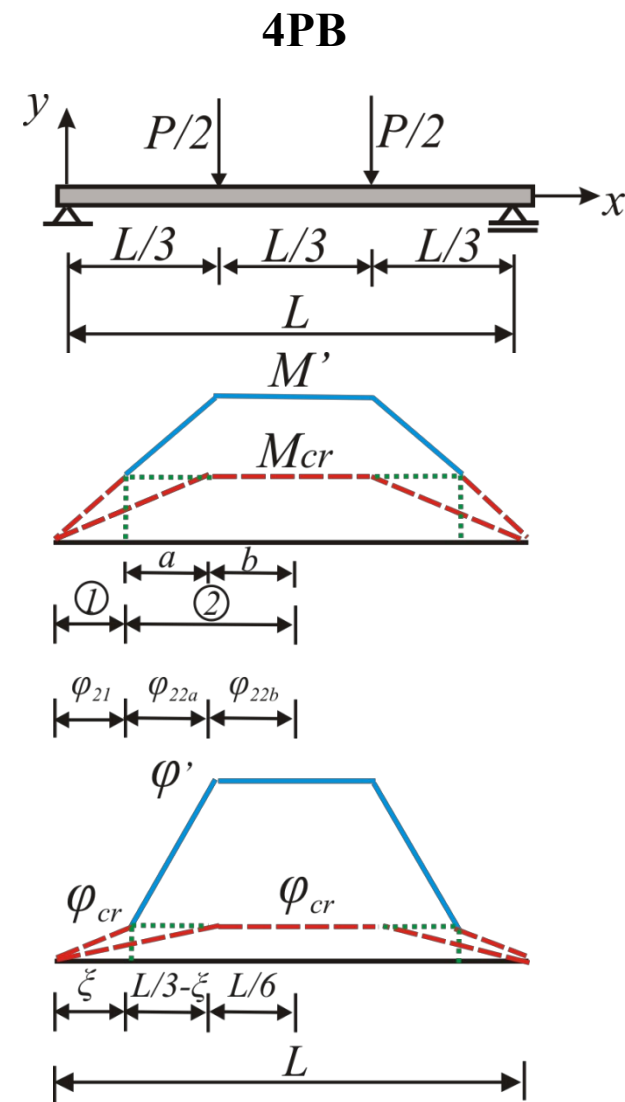
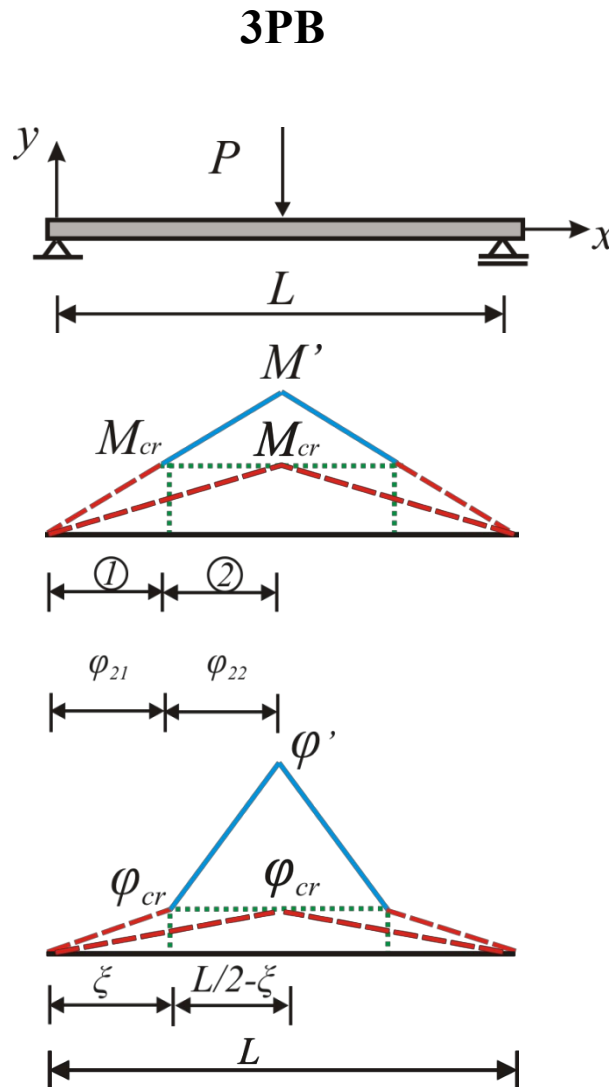
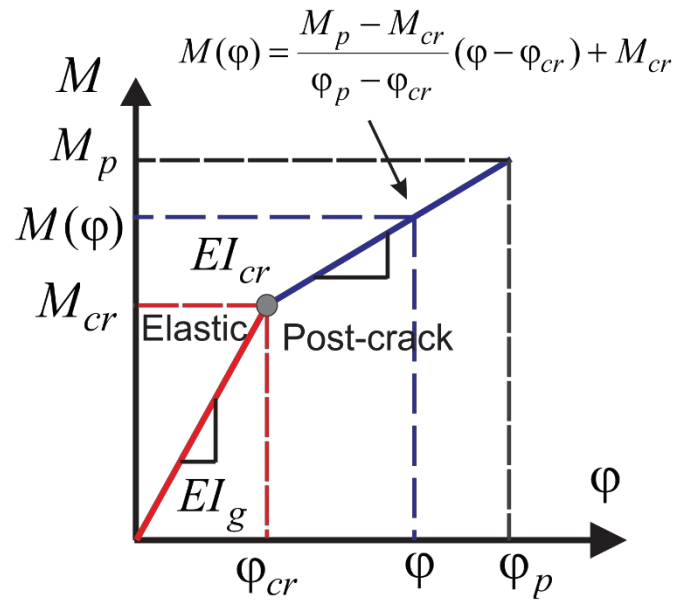
$$m(\kappa) = \frac{M(\phi)}{M_{cr}} \quad \kappa = \frac{\phi}{\phi_{cr}} \quad m = \frac{M_u}{M_{cr}} \quad \kappa_p = \frac{\phi_p}{\phi_{cr}} \quad \eta = \frac{EI_{cr}}{EI_g}$$

$$0 < m \leq 1 \quad 0 < \kappa \leq 1$$

$$1 \leq m \quad 1 \leq \kappa$$



# 1-D Member: Moment-Curvature Distributions



■ Use static equilibrium to get moment distributions

■ Moment distributions

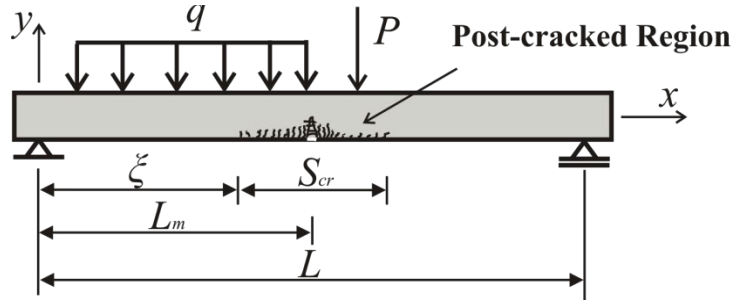


Bilinear moment-curvature



Curvature distributions

# Analytical Deflection Solutions



## ➤ Rotation

$$\text{Region ① : } \theta_{21} = \int_0^x \varphi_{21}(x) dx + C_1 \quad 0 \leq x \leq \xi$$

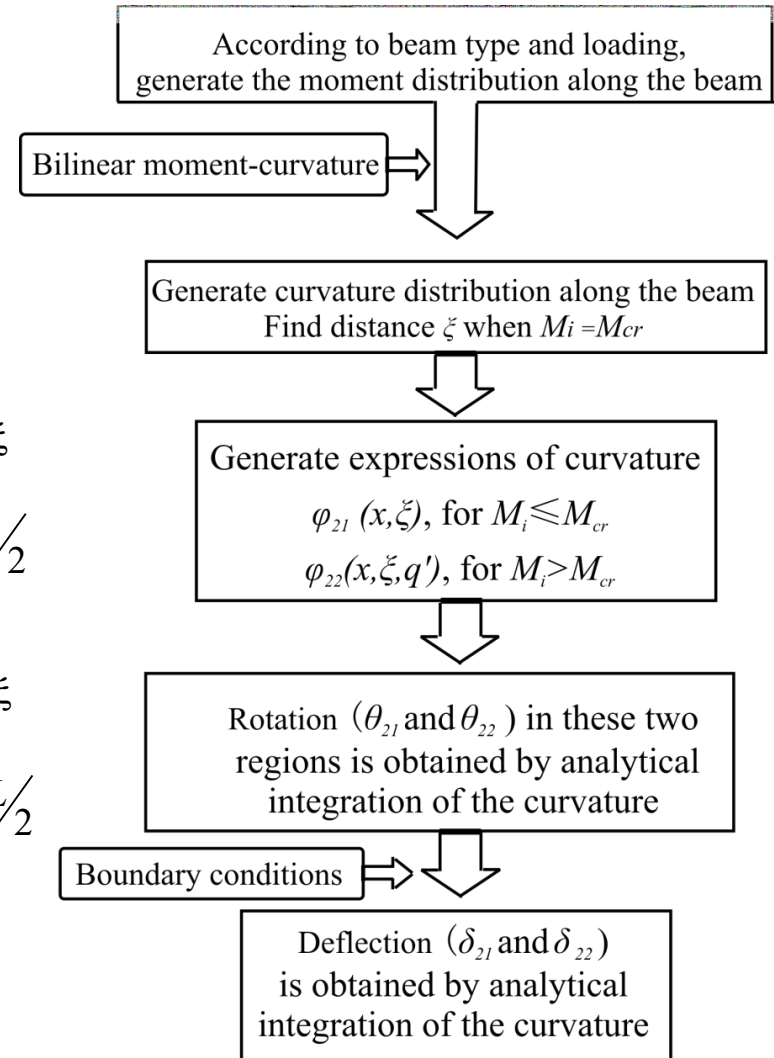
$$\text{Region ② : } \theta_{22} = \int_0^\xi \varphi_{21}(x) dx + \int_\xi^x \varphi_{21}(x) dx + C_2 \quad \xi \leq x \leq L/2$$

## ➤ Deflection

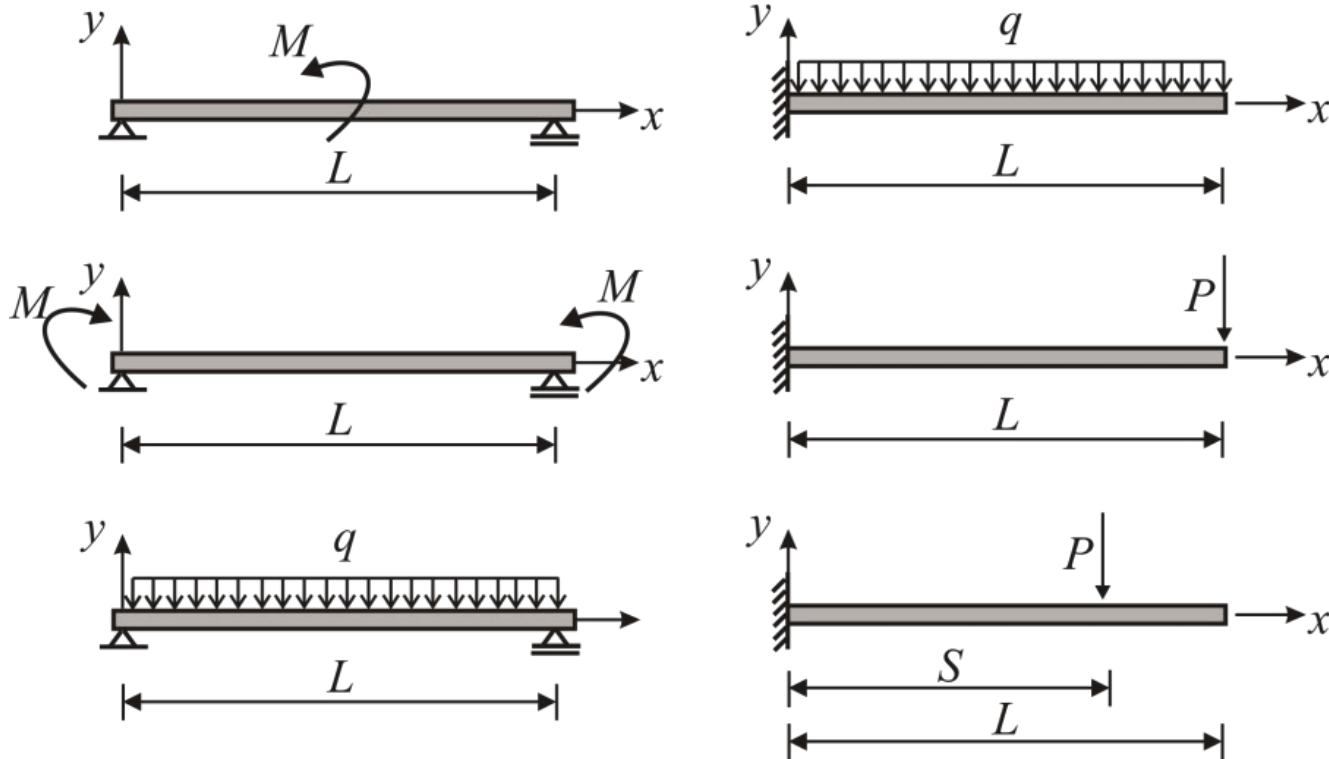
$$\text{Region ① : } \delta_{21} = \int_0^x \theta_{21}(x) dx + C_3 \quad 0 \leq x \leq \xi$$

$$\text{Region ② : } \delta_{22} = \int_0^\xi \theta_{21}(x) dx + \int_\xi^x \theta_{22}(x) dx + C_4 \quad \xi \leq x \leq L/2$$

The constants of integration defined in terms of  $C_i$  are numerical values which are determined based on the boundary conditions.



# Solutions are available for other loading types



- Different equilibrium equations affecting Moment distribution
- Simply supported beam and cantilever beam.
- 8 different loading types

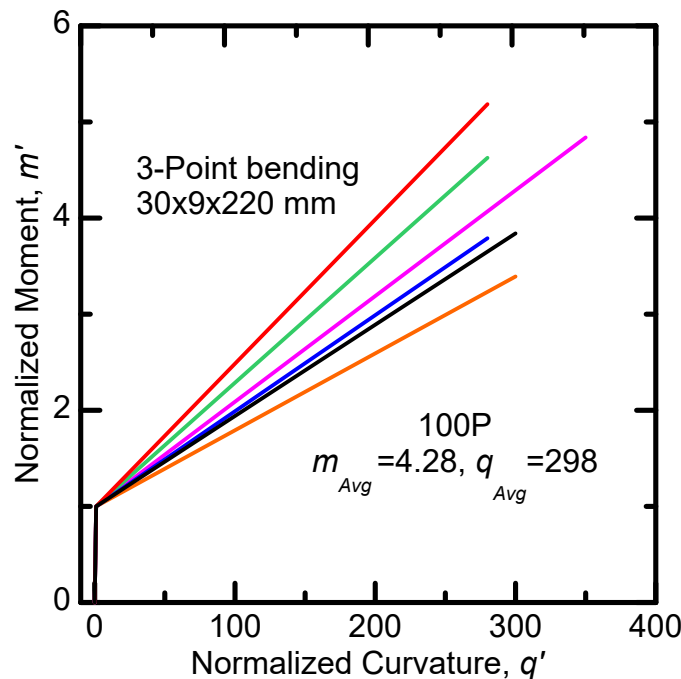
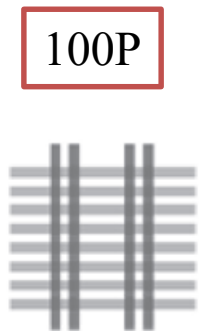
# Experimental Verification of bilinear moment-curvature

## Data Set 1 – TRC PP and Aramid Fibers

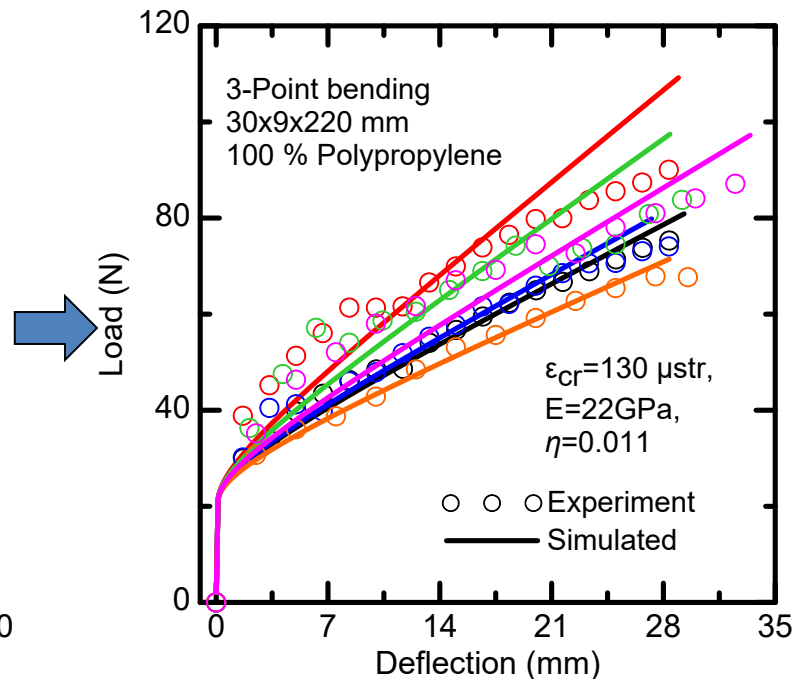
- Three-point bending test
- 30 (b) × 9(d) × 220 (L) mm
- Textile fabrics:
  - Polypropylene
  - Aramid

Simulated Parameters (Avg.)

| ID     | $\epsilon_{cr}, \mu\text{str}$ | E, Gpa | $\eta$ | m     | q   | EI, $10^7$ | EI <sub>cr</sub> , $10^7$ |
|--------|--------------------------------|--------|--------|-------|-----|------------|---------------------------|
| 100P   | 130                            | 22     | 0.01   | 4.28  | 298 | 4.0        | 0.044                     |
| 100A   | 130                            | 22     | 0.1    | 17    | 157 | 4.0        | 0.41                      |
| 25A75P | 130                            | 22     | 0.05   | 10.48 | 198 | 4.0        | 0.202                     |



Normalized curvature vs. Normalized moment



Deflection at mid-span vs. Applied load

# Simulations

| Composite       | Classification                                     | Model used                |
|-----------------|----------------------------------------------------|---------------------------|
| (1) ECC         | Strain hardening                                   | Full                      |
| (2) GFRC        | Strain hardening                                   | Full **                   |
| (3) SFRC        | Strain Softening, (deflection hardening)           | Simplified *              |
| (4) SFRC        | Strain Softening, (deflection softening)           | Simplified *              |
| (5) Ferrocement | Strain Softening, deflection softening & hardening | Bilinear/<br>Softening ** |

\* Soranakom, C., and Mobasher, B., "Closed Form Solutions for Flexural Response of Fiber Reinforced Concrete Beams," in press, ASCE, *Journal of Engineering Mechanics*, August- 2007

\*\* Soranakom, C., and Mobasher, B., "Closed-Form Moment-Curvature Expressions For Homogenized Fiber Reinforced Concrete" ACI Materials Journal, in press, July-August, 2007

# Engineered Cementitious Composite (ECC)

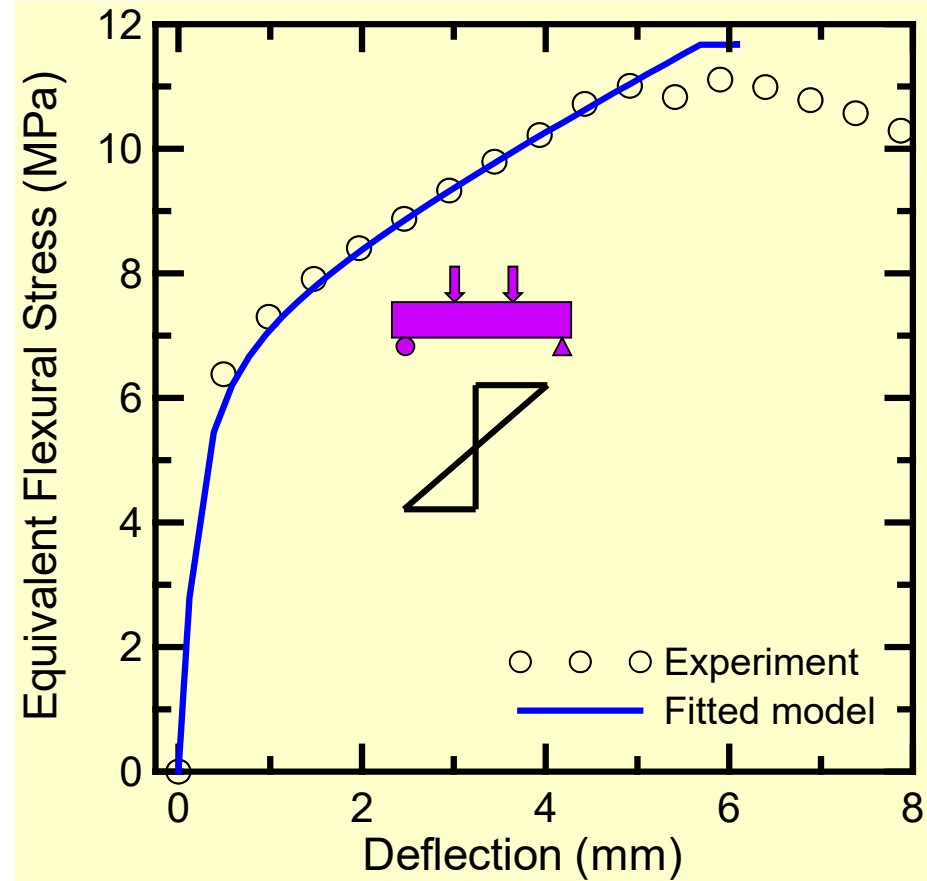
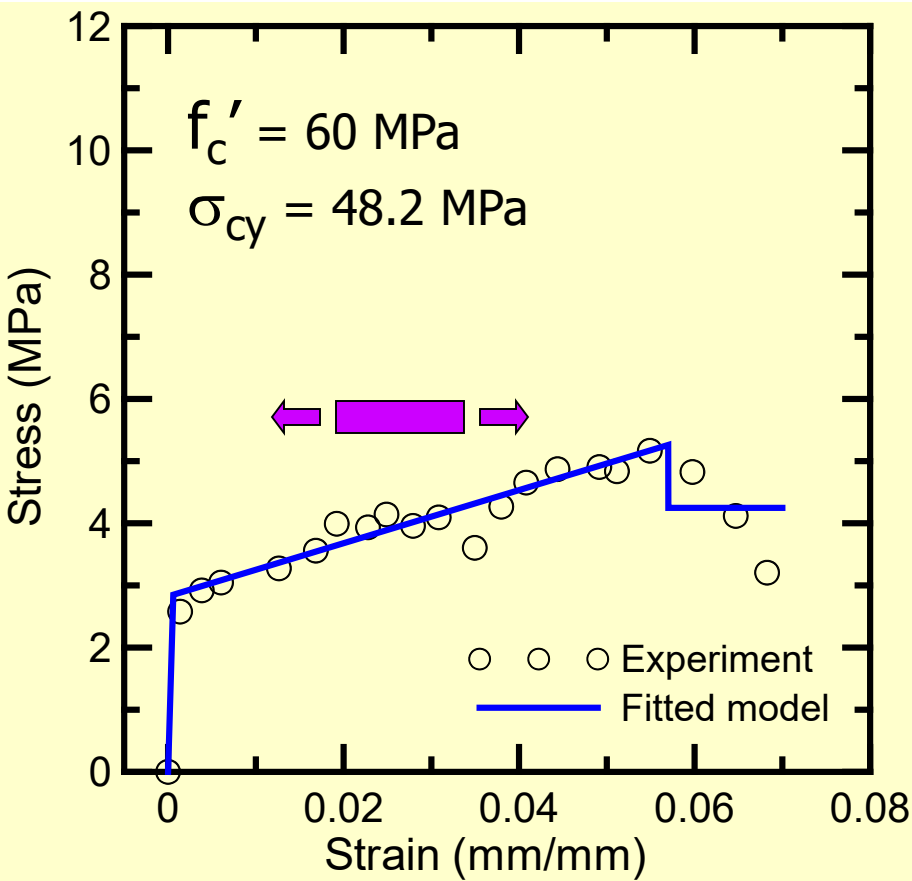
## Simulation of Strain hardening

- 2% Polyethylene (PE), Maalej, M. (1999)
- Parameters:  $\varepsilon_{cr} = 0.006$ ,  $E = 4750$  MPa,  
 $\alpha = 95$ ,  $\beta_{tu} = 117$ ,  $\mu = 1.49$ ,  $\eta = 0.009$ ,  
 $\gamma = 18.4$ ,  $\lambda_{cu} = 20$

$$M_{cr} = 37363 \text{ N-mm}, \phi_{cr} = 1.181 \times 10^{-5}$$

$$M'_{max} = 4.1, \phi'_{max} = 67.9$$

$$M'_{ult} = 4.1, \phi'_{ult} = 67.9$$





# Glass Fiber Reinforced Concrete (GFRC)

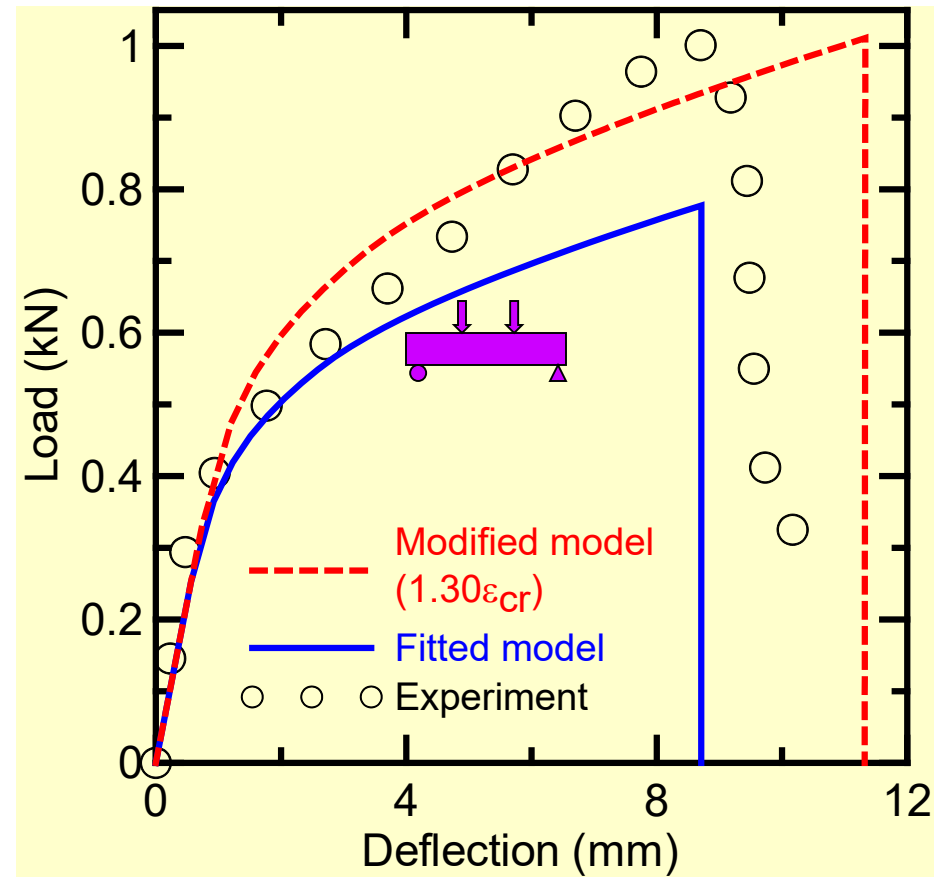
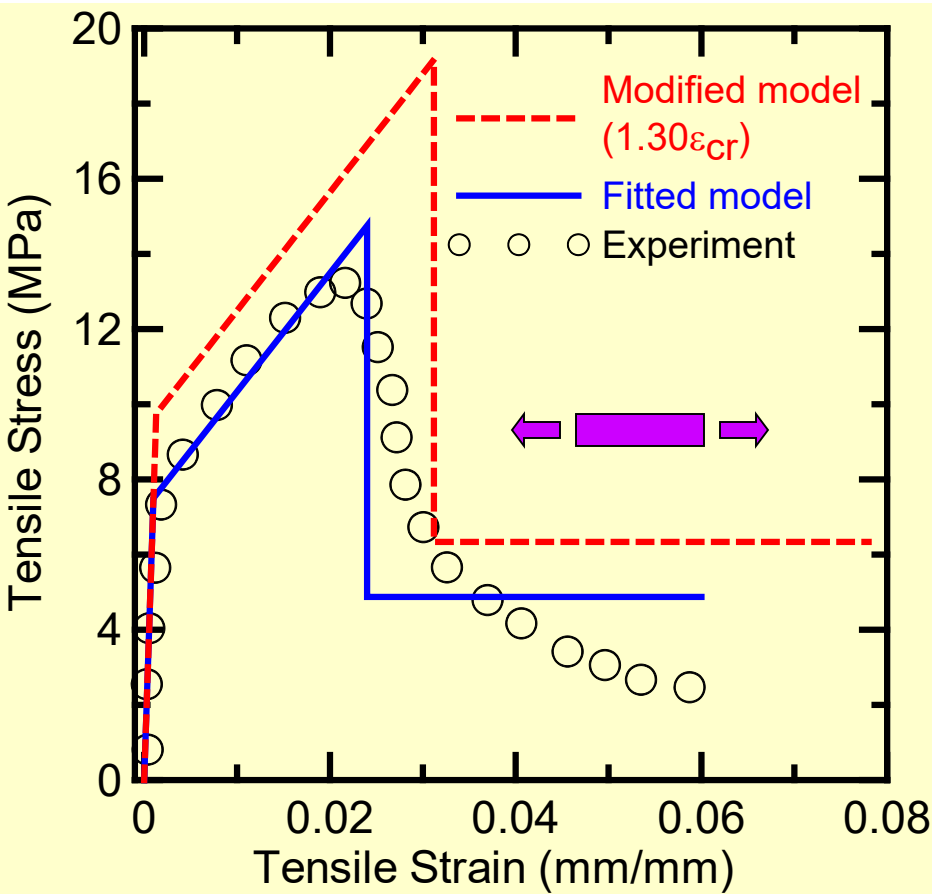
## Simulation of Strain hardening

- 5% by weight AR-glass Marikunte, S. (1997)
- Parameters:  $\varepsilon_{cr} = 0.001$ ,  $E = 7500$  MPa,  $\alpha = 24$ ,
- $\beta_{tu} = 60$ ,  $\mu = 0.65$ ,  $\eta = 0.042$ ,  $\gamma = 2.02$ ,  $\lambda_{cu} = 8$

$$M_{cr} = 6250 \text{ N-mm}, \phi_{cr} = 2.0 \times 10^{-4}$$

$$M'_{max} = 3.63, \phi'_{max} = 15.5$$

$$M'_{ult} = 3.63, \phi'_{ult} = 15.5$$



# Steel Fiber Reinforced Concrete (SFRC)

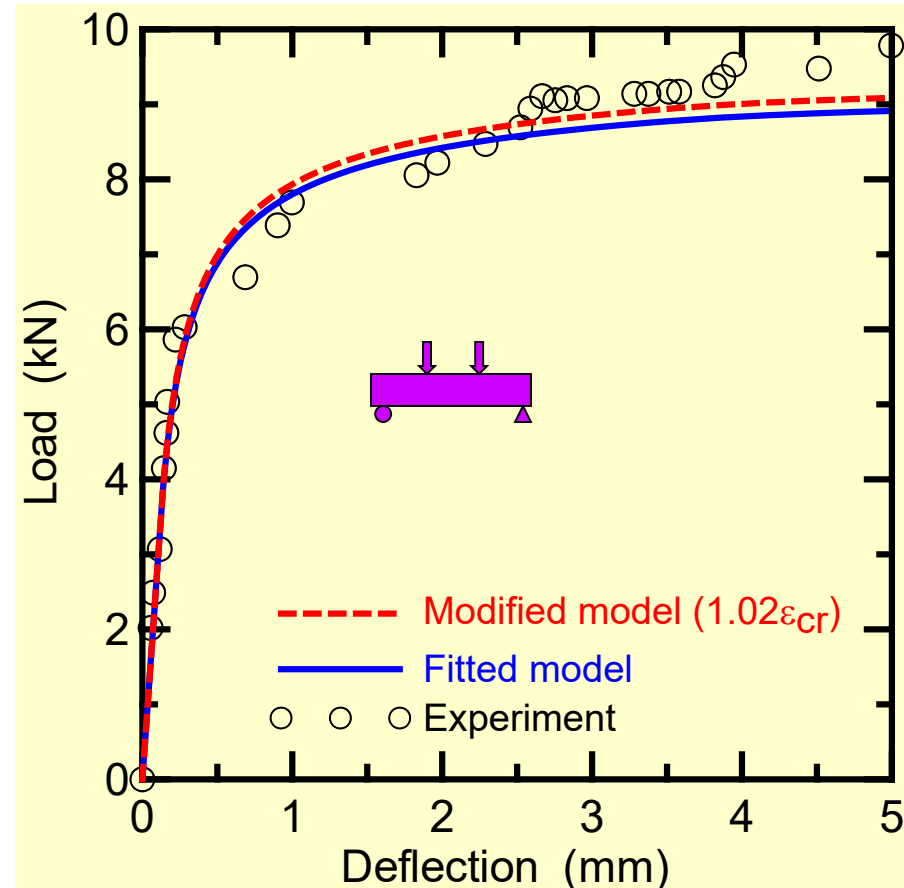
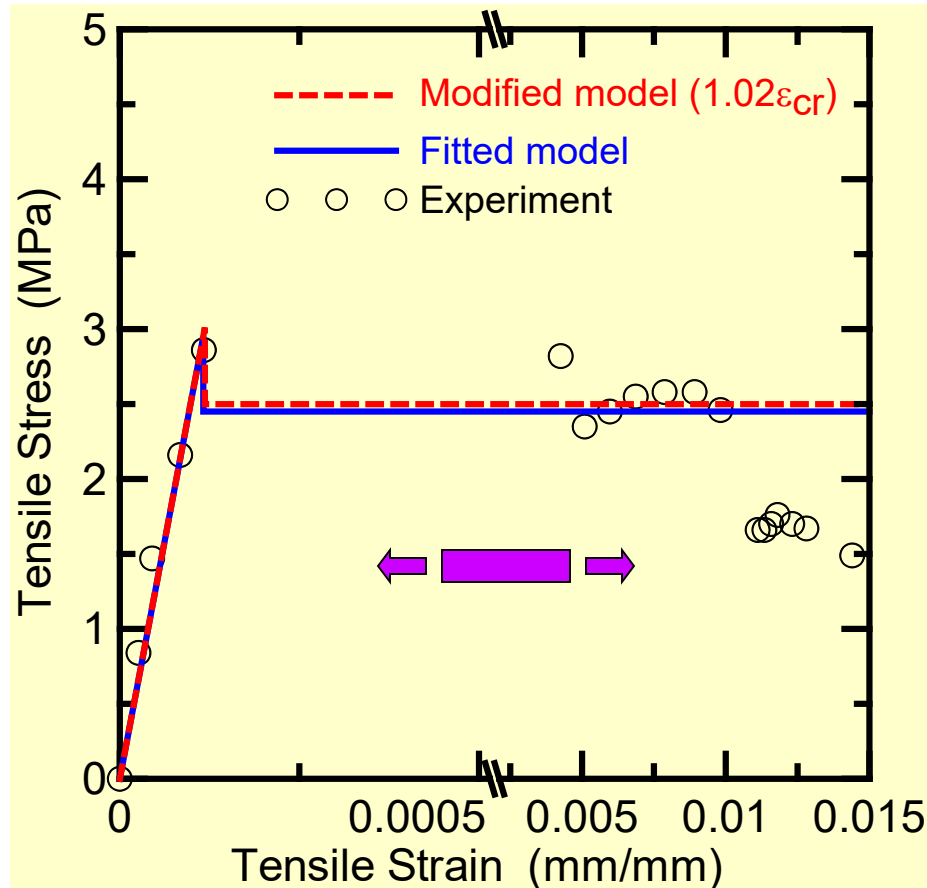
## Strain Softening, deflection hardening

- 1.0% hook end steel fiber, Lim, T.Y. (1987)
- Parameters:  $\varepsilon_{cr} = 0.000116$ ,  $E = 25400$  MPa,  $\mu = 0.83$ ,  $\beta_{tu} = 129$ ,  $\lambda_{cu} = 34.4$

$$M_{cr} = 49191 \text{ N-mm}, \phi_{cr} = 2.32 \times 10^{-6}$$

$$M'_{max} = 2.28, \phi'_{max} = 66.2$$

$$M'_{ult} = 2.28, \phi'_{ult} = 66.2$$



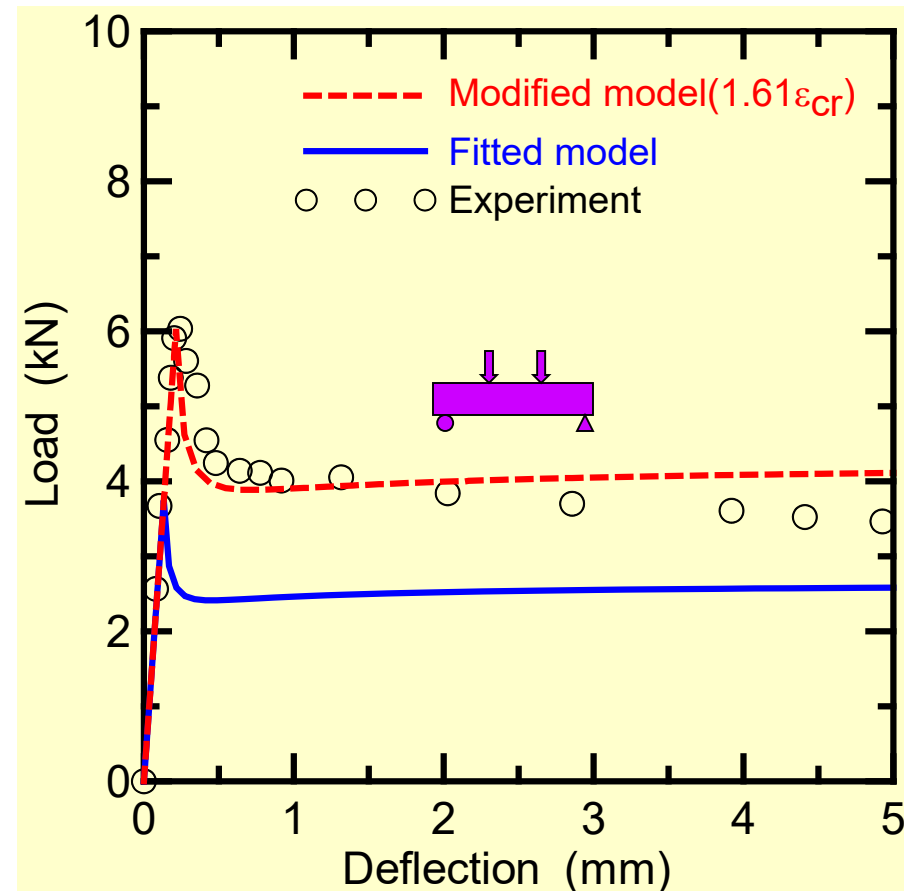
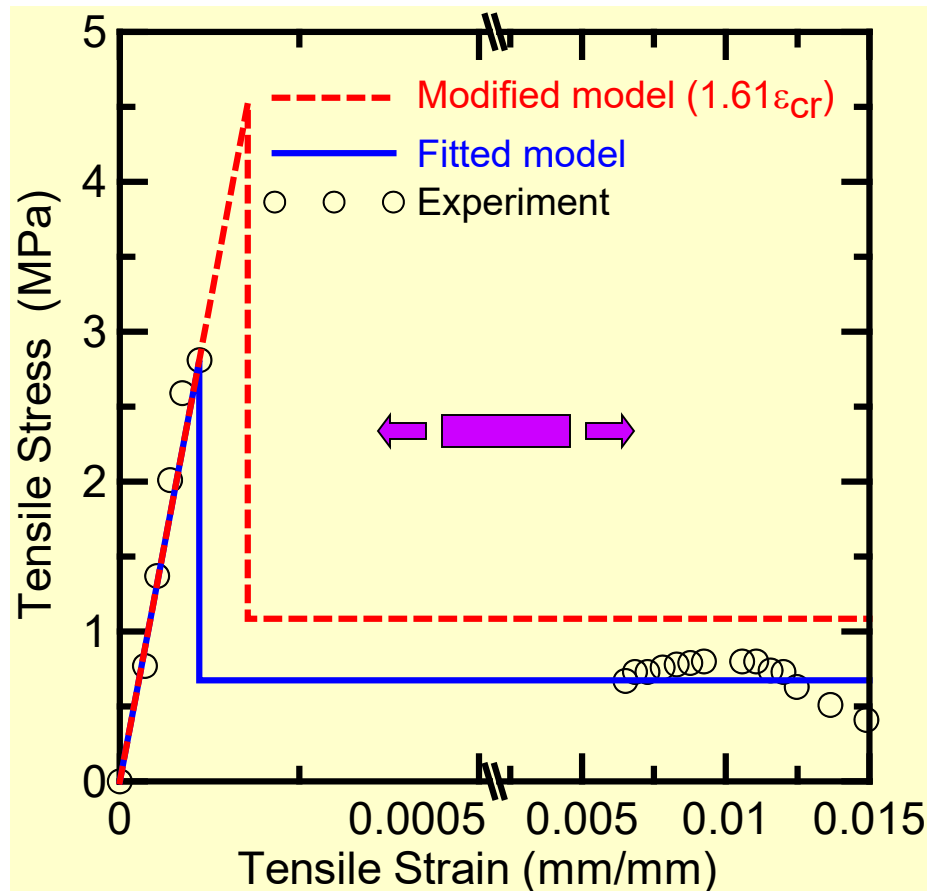
# Steel Fiber Reinforced Concrete (SFRC) Strain Softening, deflection softening

- 0.5% hook end steel fiber, Lim, T.Y. (1987)
- Parameters:  $\varepsilon_{cr} = 0.000111$ ,  $E = 25400$  MPa,  $\mu = 0.24$ ,  $\beta_{tu} = 136$ ,  $\lambda_{cu} = 36.2$

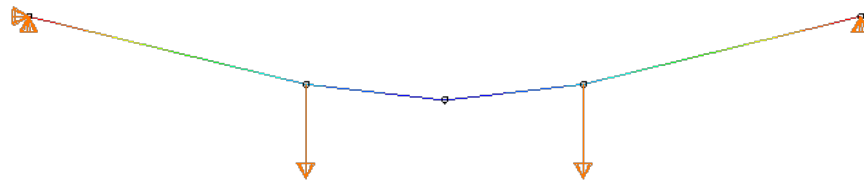
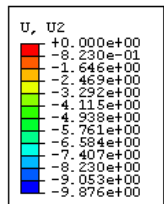
$$M_{cr} = 46821 \text{ N-mm}, \phi_{cr} = 2.21 \times 10^{-6}$$

$$M'_{max} = 1.00, \phi'_{max} = 1.00$$

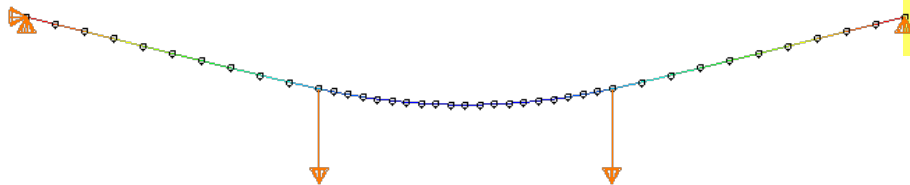
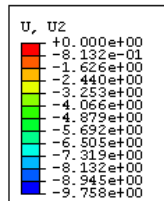
$$M'_{ult} = 0.69, \phi'_{ult} = 70.5$$



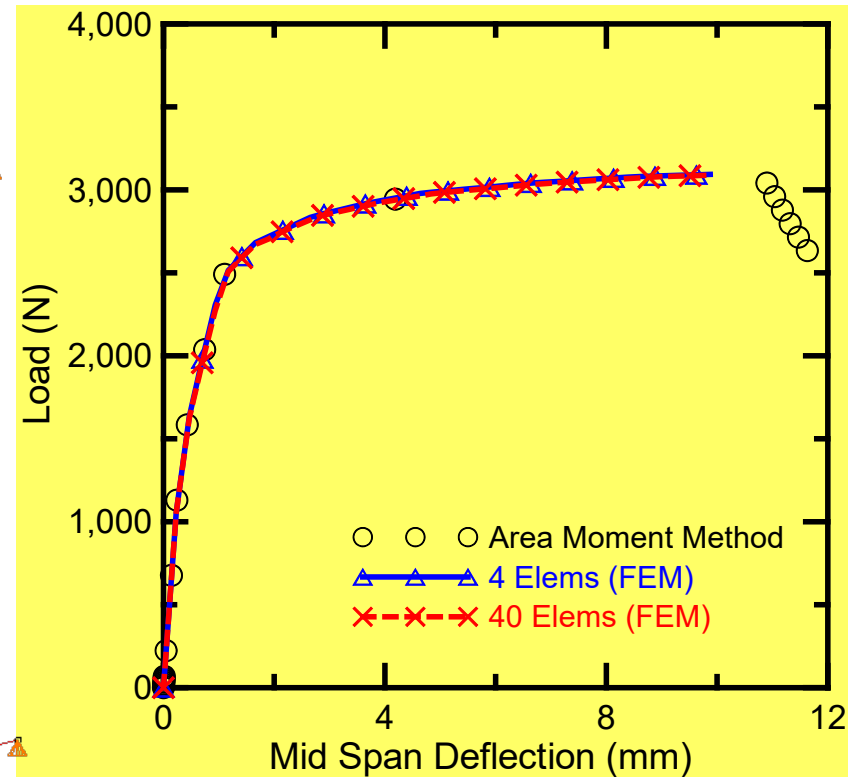
# Use of Moment-Curvature response as material property of 2 node, 3- DOF per node beam elements



Ferrocement,  $V_f = 1.51\%$ , 4- B23 bending elements,

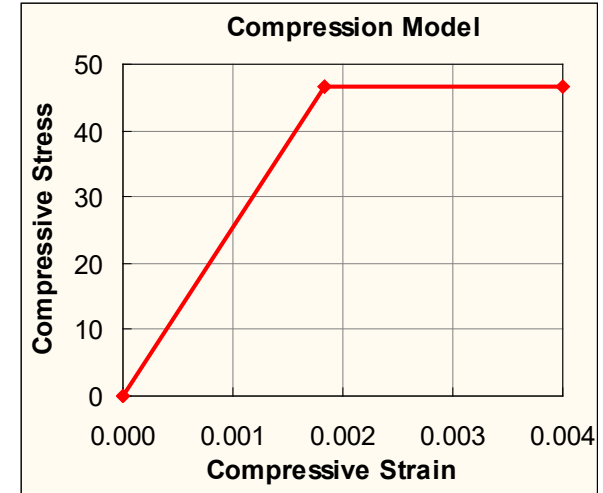
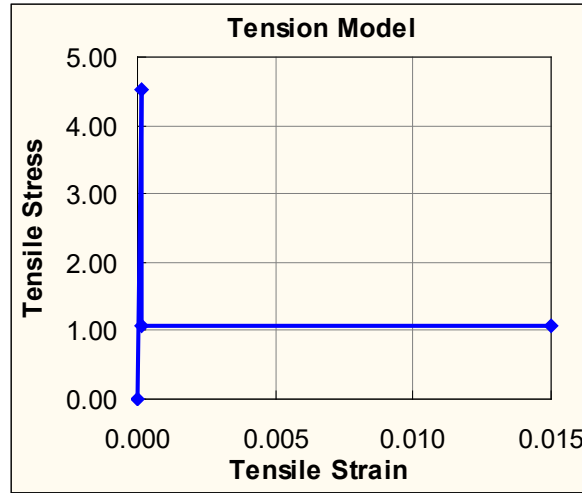
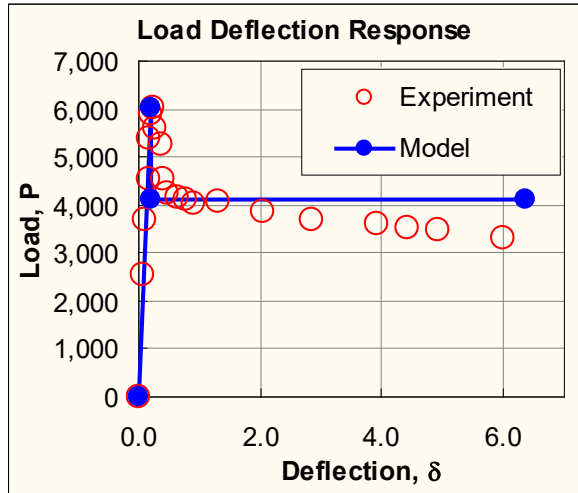


Ferrocement,  $V_f = 1.51\%$ , 40- B23 bending elements,



Soranakom, C., and Mobasher, B., "Correlation Of Tensile And Flexural Behavior Of Fiber Reinforced Cement Composites," Ferro8, proceedings of the 8th International Ferrocement and thin reinforced cement composites Conference, Bangkok, Thailand, 2006.

# Back Calculation of Material Properties



## Beam Size

test method **4** point bending  
Don't need  $L_p$  **0**

$b =$  **100**  
 $d =$  **100**

$L =$  **750**  
 $S =$  **250**

## Material Model

$E =$  **25400**  
 $\epsilon_{cr} =$  **0.0001781**

### tension

$\mu =$  **0.24**  
 $\beta_{tu} =$  **84.24**

### strain

0.0000000  
 $\epsilon_{cr} =$  0.0001781  
 $\epsilon_{cr} =$  0.0001781  
 $\epsilon_{tu} =$  0.0150000

### stress

0.00  
 $\sigma_{cr} =$  4.52  
 $\sigma_p =$  1.09  
 $\sigma_{tu} =$  1.09

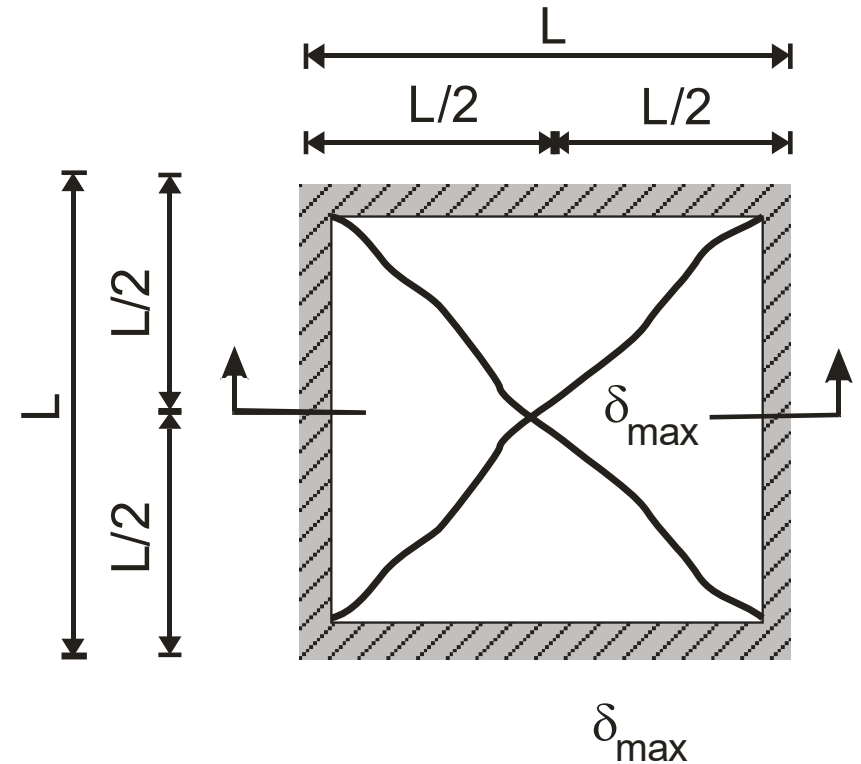
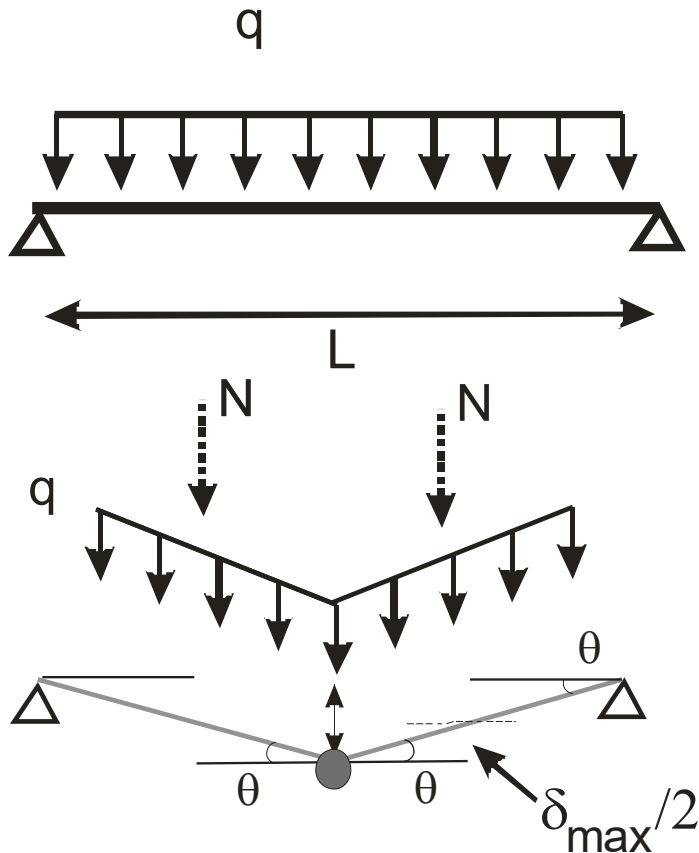
### compression

$\omega =$  **10.30**  
 $\lambda_{cu} =$  **22.46**

0.0000000  
 $\epsilon_{cy} =$  0.0018341  
 $\epsilon_{cu} =$  0.0040000

0.00  
 $\sigma_{cy} =$  46.59  
 $\sigma_{cu} =$  46.59

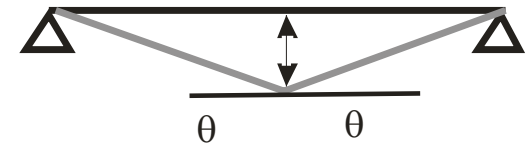
# Plastic Analysis Beam and Panel Flexure



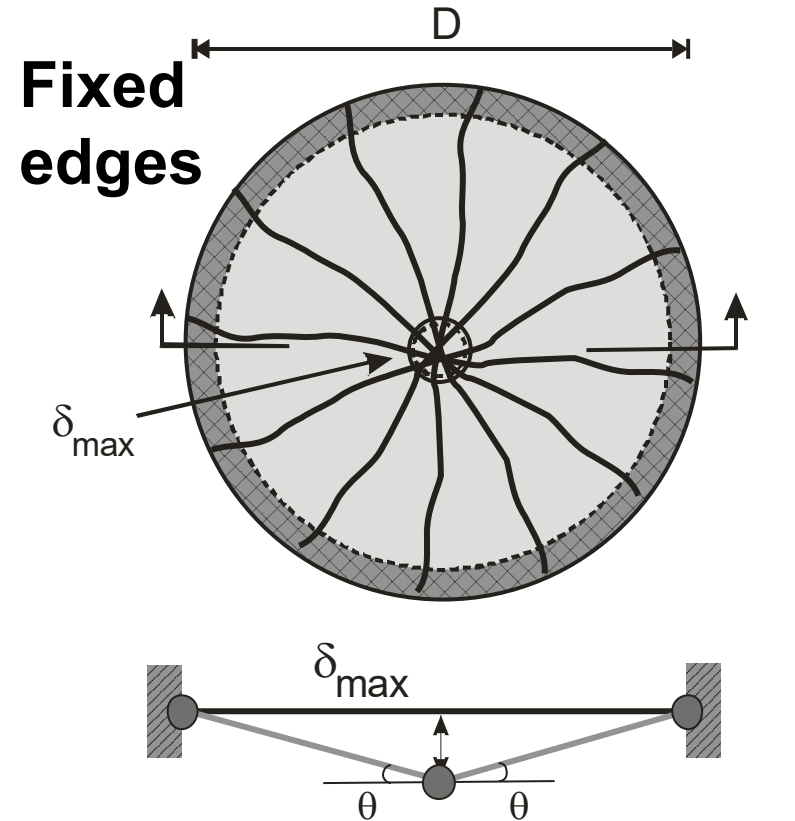
$$W_{int} = W_{ext}$$

$$2M_P\theta = 2M_P\left(\frac{2\delta}{L}\right) = 2\left[q\left(\frac{L}{2}\right)\left(\frac{\delta}{2}\right)\right]$$

$$M_P = \frac{q_{ult}L^2}{8}$$

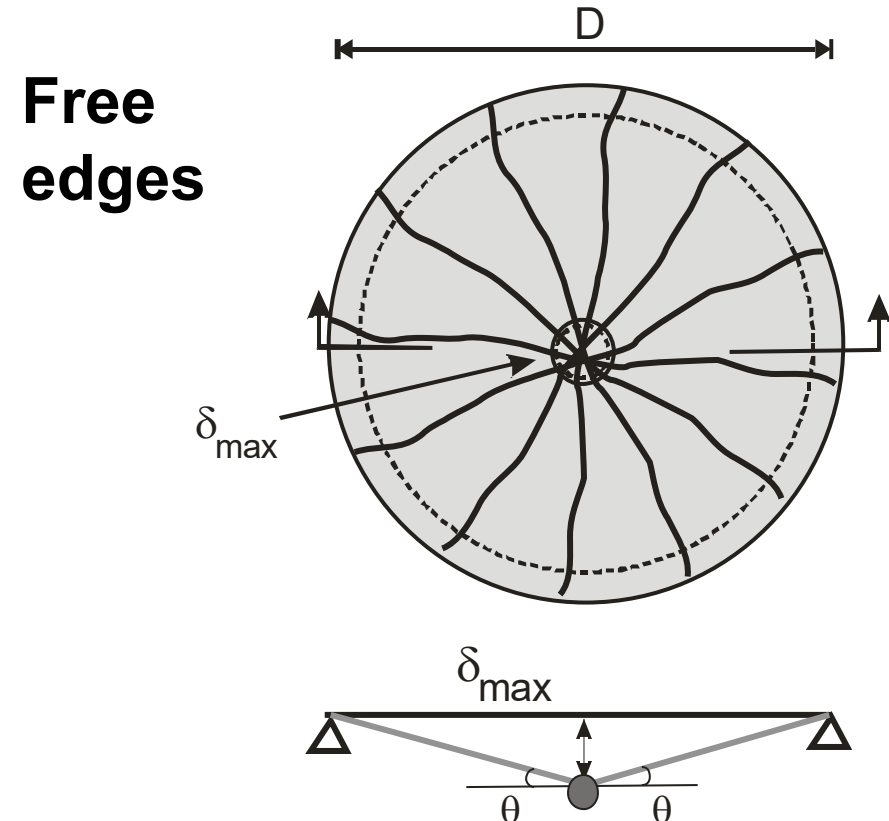


# Plastic analysis, Round Panel- Fixed or Free



$$W_{int} = W_{ext} \quad P_{ult} = 4\pi M_P$$

$$4\pi M_P = \frac{2\pi R}{2} \frac{q_{ult} R}{3}$$

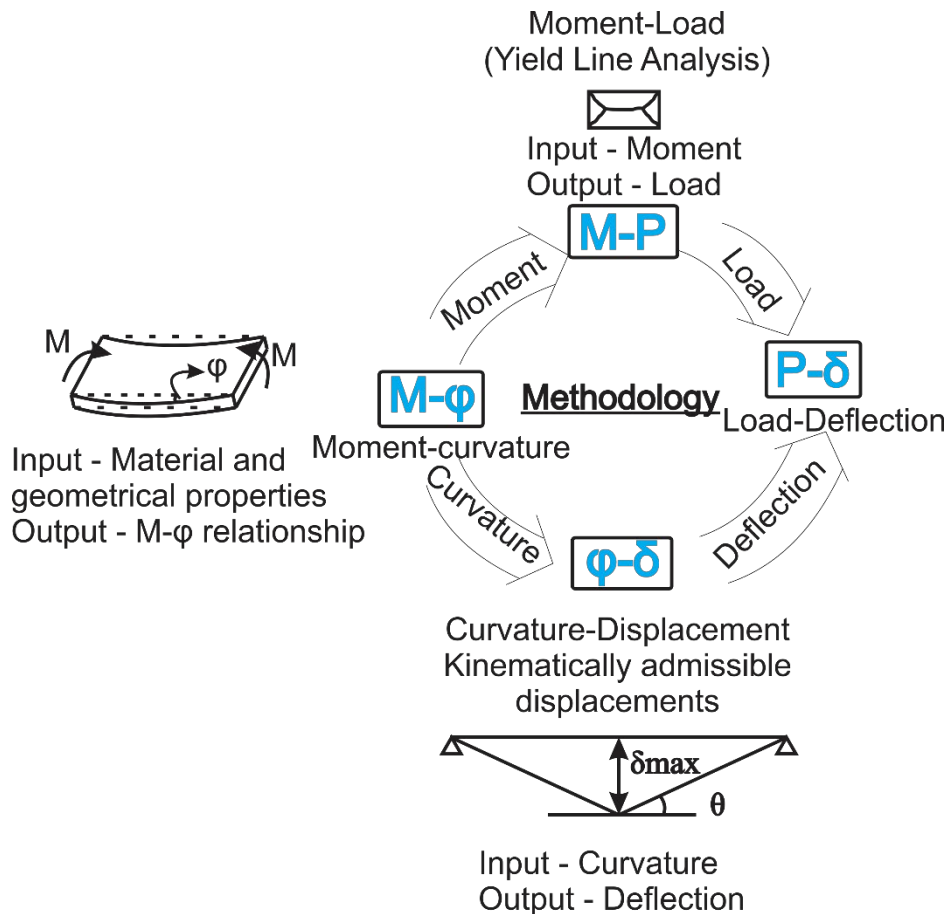


**Concentrated Load**  $P_{ult} = 2\pi M_P$

**Distributed Load**  $M_P = \frac{q_{ult} R^2}{12}$

# 2-D Member: Slab and Panel

## Model Approach



## Type of Members

Round Panel



Rectangle Panel



Square Panel



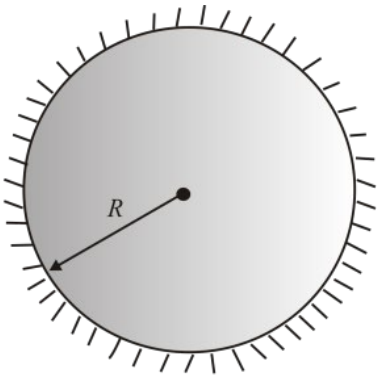
Boundary conditions { Simply supported  
Clamped

- Stage I: Elastic Solution
- Stage II : Yield Line Approach



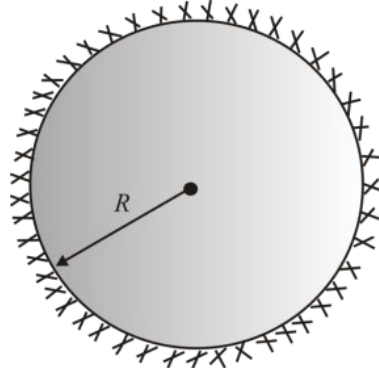
# Elastic Solution for Panels

Case 1.1



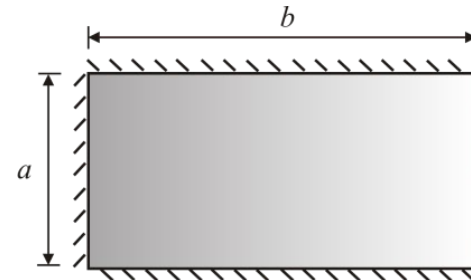
$$\frac{PR^2}{16\pi D} \left( \frac{3+\nu}{1+\nu} \right)$$

Case 1.2



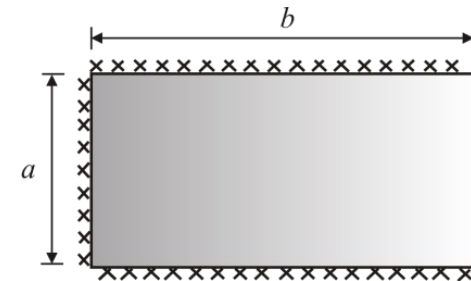
$$\frac{PR^2}{16\pi D}$$

Case 2.1



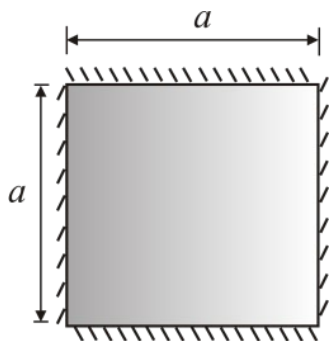
$$\propto \frac{qa^4}{D}$$

Case 2.2



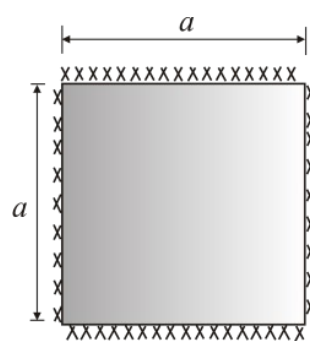
$$\frac{0.032(1-\nu^2)}{1 + \left(\frac{a}{b}\right)^4} \left( \frac{qa^4}{Et^3} \right)$$

Case 3.1



$$0.00406 \frac{qa^4}{D}$$

Case 3.2



$$\frac{0.032}{2} (1-\nu^2) \left( \frac{qa^4}{Et^3} \right)$$

Round Panel – Point load (P) at center  
Rectangular/Square panel— Uniform load (q)



Clamped support



Simply supported

Moment      Rotation



# Plastic Solution: Yield Line Moment– Applied Load

$$W_{\text{ext}} = W_{\text{int}}$$

$$\sum(N \delta) = \sum(m l \theta)$$

Work done in moving loads = Work done in rotating yield lines

Assumptions:

- Failure takes place according to the assumed pattern
- Yield lines are straight and end at slab boundary
- Yield lines at vertexes are at  $45^\circ$  to the edges in case of square and rectangular slabs
- Hogging moment about the yield lines and sagging moment about the supports are equal

$N$  – Load,  $\delta$  – Defl.,  $m$  – moment abt. YL,  $l$  – length of YL,  $\theta$  - Rotation

# Yield Line Moment – Applied Load Case study

Square panel with simply supported edges

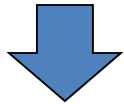
$$W_{\text{ext}} = W_{\text{int}}$$

$$\sum(N \delta) = \sum(m l \theta)$$

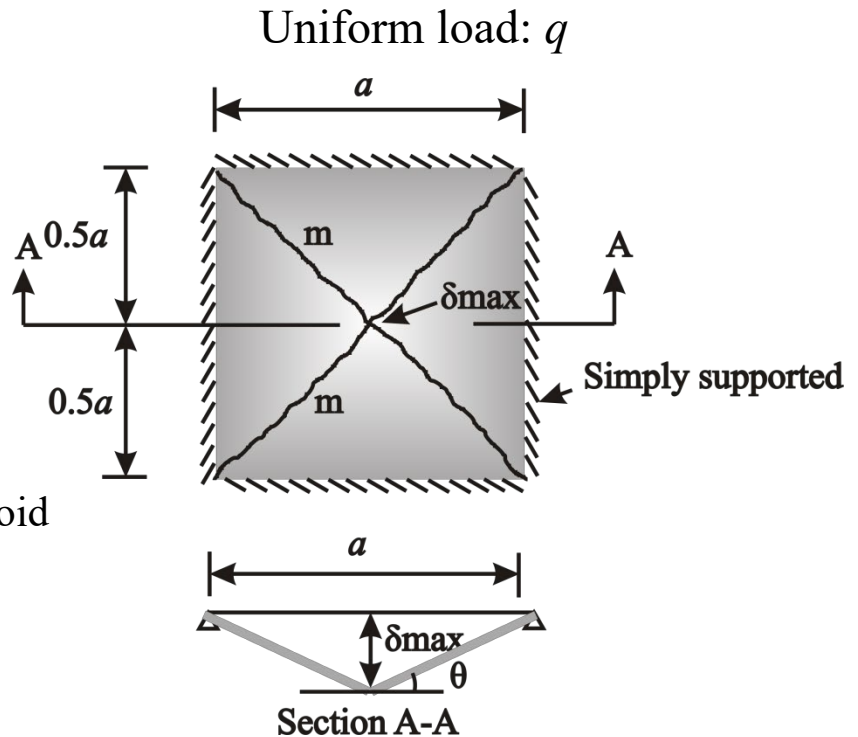
$$W_{\text{ext}} = \sum(N \delta) = 4 \times \left( q \frac{a^2}{4} \times \frac{1}{3} \right)$$

Equivalent point load      Deflection of the centroid

$$W_{\text{int}} = \sum(m l \theta) = 4 \times \left( m a \frac{1}{0.5a} \right)$$



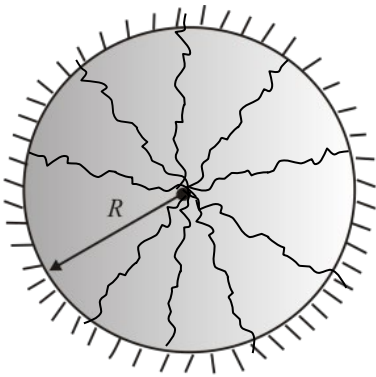
$$m = \frac{qa^2}{24}$$



$\delta_{\text{max}}$  is unit

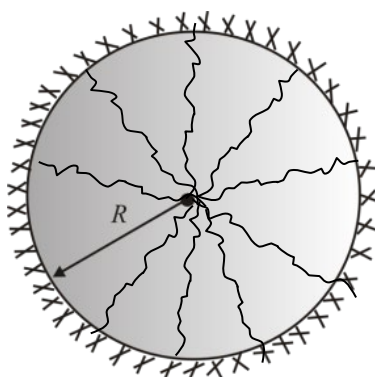
# Yield Line Moment – Applied Load

Case 1.1

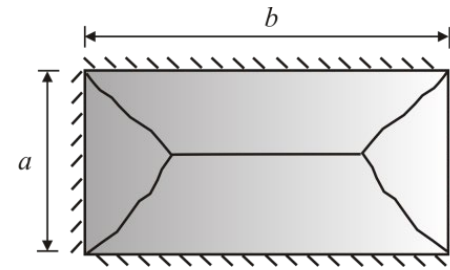


$$m = \frac{P}{2\pi}$$

Case 1.2

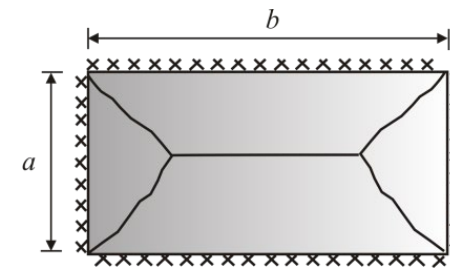


$$m = \frac{P}{4\pi}$$



Case 2.1

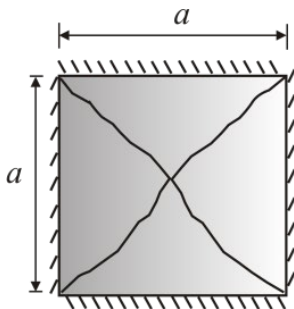
$$m = \frac{qb^2(3b-a)}{12(2b+2a)}$$



Case 2.2

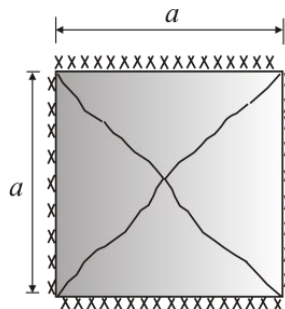
$$m = \frac{qb^2(3b-a)}{12(4b+4a)}$$

Case 3.1



$$m = \frac{qa^2}{24}$$

Case 3.2



$$m = \frac{qa^2}{48}$$

Round panel – Point load (P) at center  
Rectangular/Square panel— Uniform load(q)



Clamped support

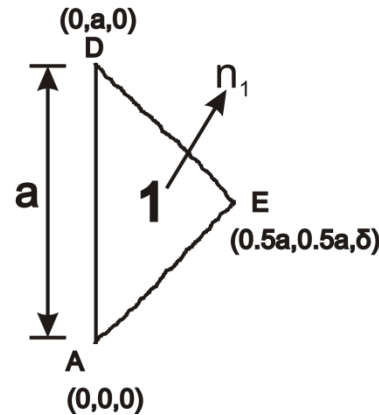
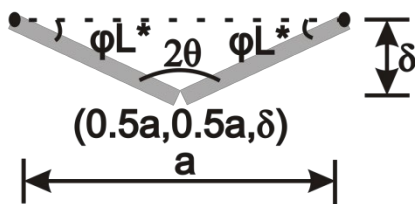
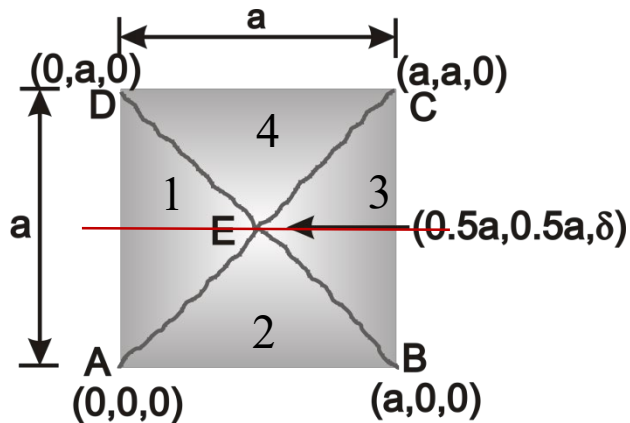
Simply supported

Moment      Rotation



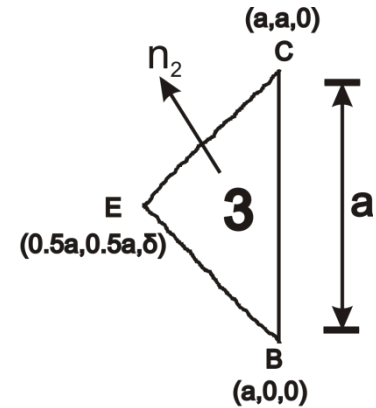
# Curvature-Deflection Relationship for Square panels

Square panel



Equation of plane # 1

$$(a\delta)x - 0.5a^2z = 0$$



Equation of plane # 3

$$(a\delta)x + 0.5a^2z - a^2\delta = 0$$

Angle between plane # 1 and #3

$$\cos 2\theta = \cos(\pi - 2\phi L^*) = \frac{a^2\delta^2 - 0.25a^4}{\sqrt{(a^2\delta^2 + 0.25a^4)}\sqrt{(a^2\delta^2 + 0.25a^4)}} = \frac{4\delta^2 - a^2}{4\delta^2 + a^2}$$

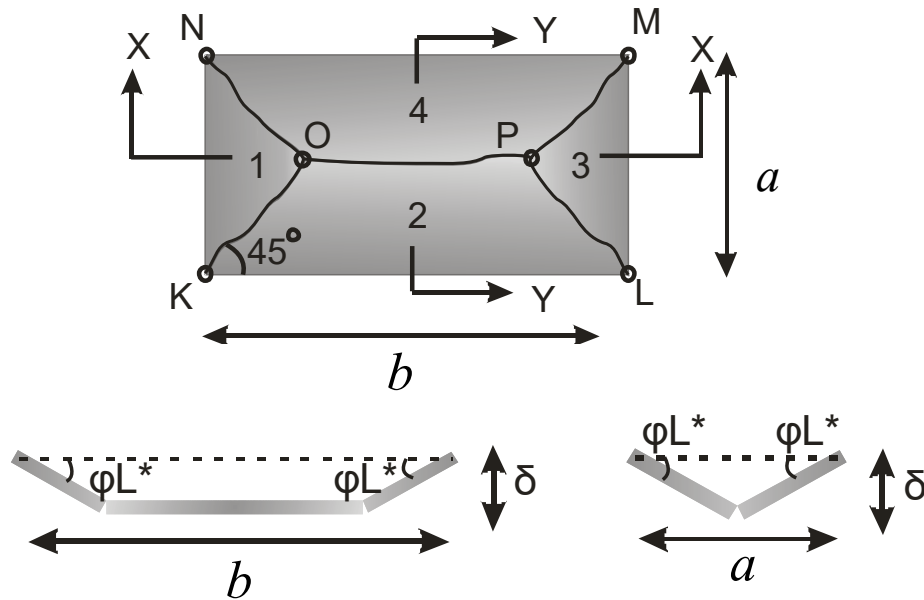
Simplifying

$$\delta = \frac{a}{2} \sqrt{\frac{1 - \cos 2\phi L^*}{1 + \cos 2\phi L^*}}$$

$L^*$  is the hinge length

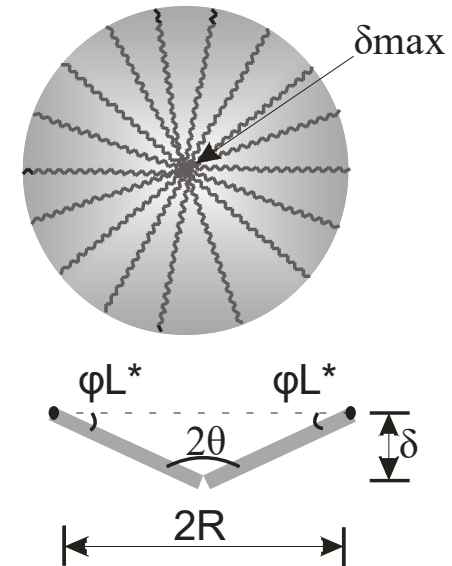
# Curvature-deflection relationship

Rectangular panel



$$\delta = \frac{a}{2} \sqrt{\frac{1 - \cos 2(\phi_{14} L^*)}{\cos 2(\phi_{14} L^*)}}$$

Round panel



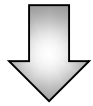
$$\delta = R \sqrt{\frac{2 - 2 \cos(\phi L^*)}{1 + 2 \cos(\phi L^*)}}$$

$\phi L^*$  - Rotation,  $L^*$  - Hinge length,  $L$  - Length,  $\phi$  - Curvature,  $\delta$  - Deflection

# Experimental Verification

Input

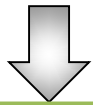
Dimension:  $b, d, L$   
Material properties:  $E, \epsilon_{cr}$



Calculate

$$M_{cr} = bd^2 E \epsilon_{cr} / 6$$

$$\phi_{cr} = 2\epsilon_{cr} / d$$



Select position  $x$

Assume maximum  $q'$  and  $\eta$   
Loop from  $q'=0$  to  $q'_{max}$



If  $q' > 1$



Stage II  
Calculate  
 $M(i), P(i), m', \xi(i)$



If Region ①  
 $0 < x \leq \xi(i)$   
 $\delta(i) = \delta_{21}$



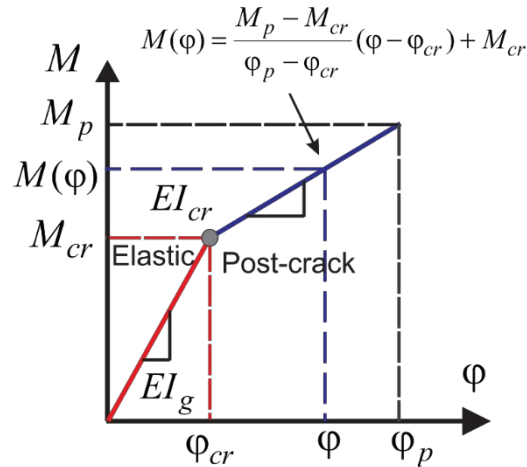
If Region ②  
 $\xi(i) < x < L/2$   
 $\delta(i) = \delta_{22}$

If  $q' \leq 1$

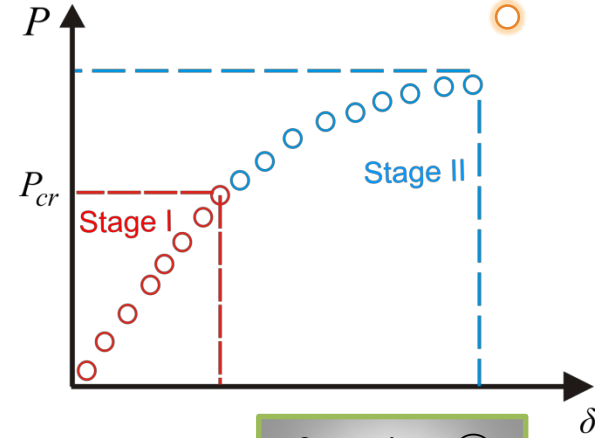


Stage I  
Calculate  
 $M(i), P(i), \delta(i) = \delta_1$

Moment-curvature



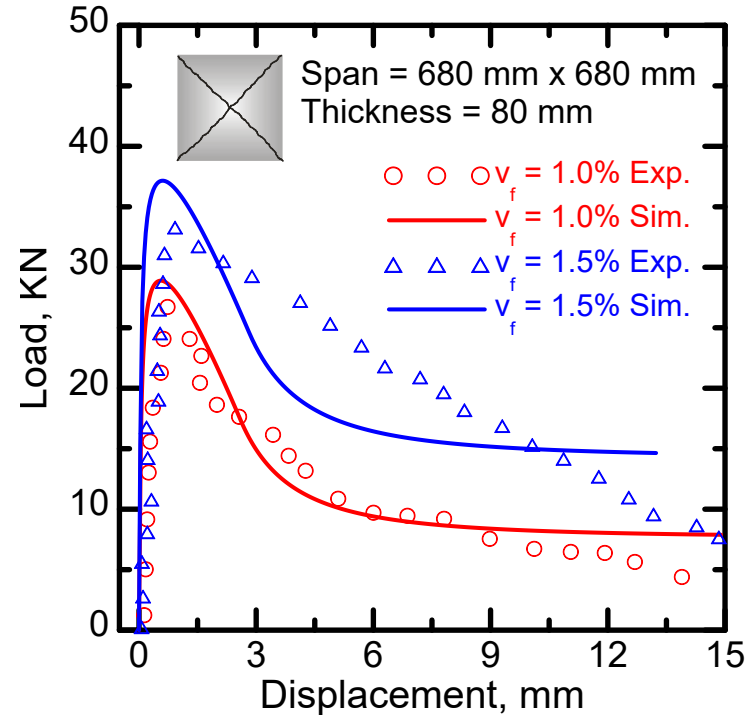
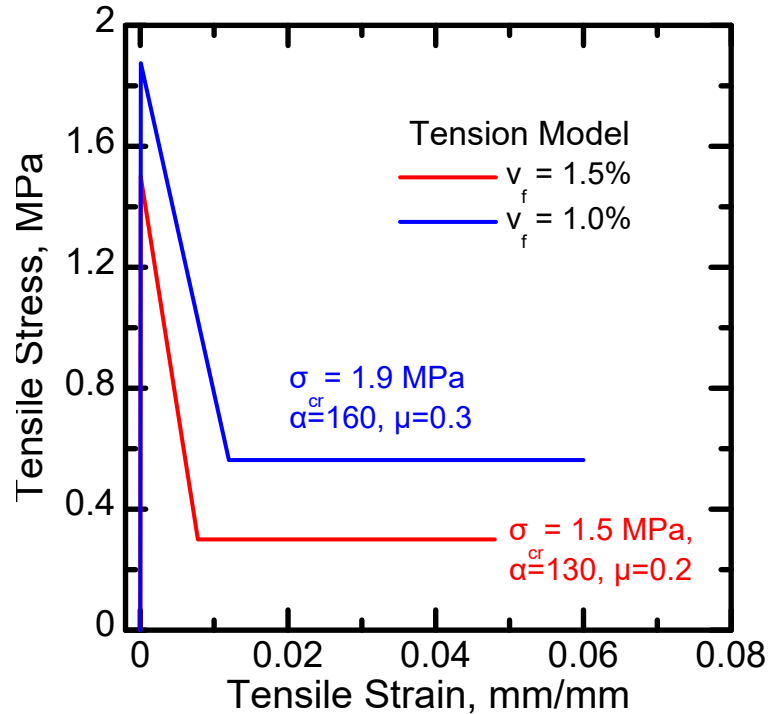
Load - deflection



# Experimental Verification

## Data Set 2 – Square Slab

Square slab with point load at center  
680 mm x 680 mm, thickness is 80 mm  
Steel fiber  $v_f = 1.0\%$  and  $1.5\%$





# Simplified Design Approach

- **Step 1:** Determine the full range moment-curvature relationship using the closed-form equations
- **Step 2:** Obtain simplified relations using polynomial curve fit

Stage 1:  $m = \kappa$

Stage 2.1:  $m = 2.805 - \frac{1.882}{\sqrt{\kappa}}$  (ascending)

$m = (10^{-4})\kappa^2 - 7.966(10^{-3})\kappa + 2.45$  (descending)

Stage 3.1:  $m = 0.592 + \frac{3949}{\kappa^2}$

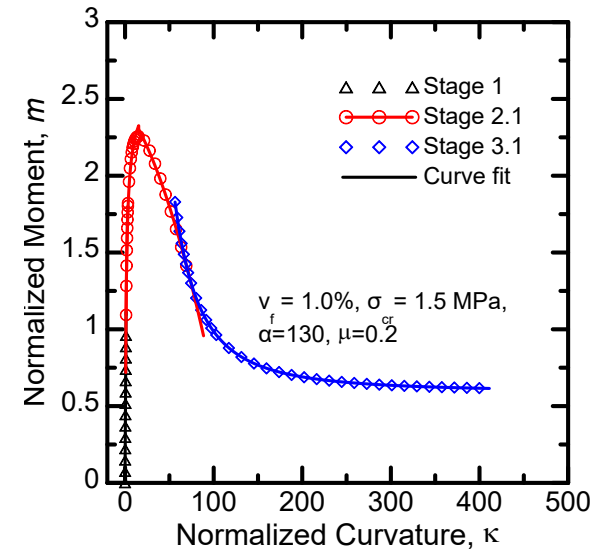
For clear span of 680 mm, thickness 80 mm, cracking stress as 1.5 MPa, moment–curvature relationship of Stage 2.1 can be shown as

$$M = \left( 2.806 - \frac{1.882}{\sqrt{\phi / (1.5 \cdot 10^{-6})}} \right) \frac{1.5 \cdot 1 \cdot 80^2}{6} = \left( 4489.6 - \frac{3.570}{\sqrt{\phi}} \right) N \cdot mm \quad (\text{ascending})$$

$$M = [-7.111(10^7)\phi^2 - 8497.1\phi + 3.920](10^3) N \cdot mm \quad (\text{descending})$$

Similar expression can also be calculated for Stage 3.1:

$$M = \left[ 946.72 + \frac{1.422(10^{-5})}{\phi^2} \right] N \cdot mm$$



Curve fit of the moment-curvature relationship

# Cont'd

- Step 3:** The boundary conditions are simply supported on all sides. The moment-load relationship for a point load for square slab is as follows:

$$P = 8m_0$$

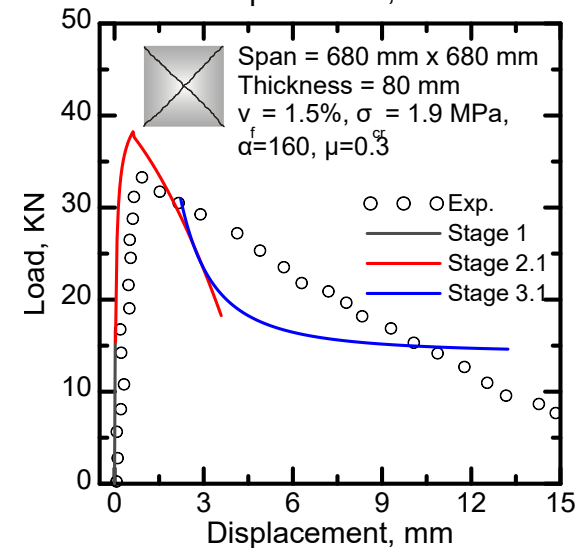
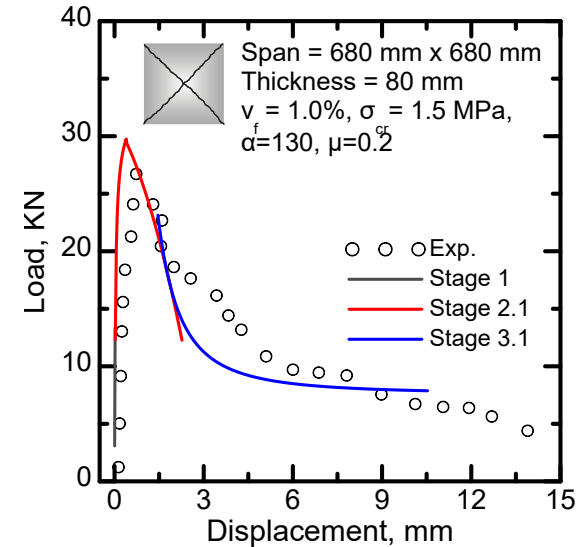
- Step 4:** The curvature-deflection relationship for a square slab in cracked stages (2.1 and 3.1) are derived as

follows:

$$\text{Elastic stage: } \delta = \frac{0.00406PL^2}{D}, \quad D = \frac{Et^3}{12(1-\nu^2)}$$

$$\text{Cracked stage: } \delta = \frac{L}{2} \frac{\sqrt{\cos \phi L^* (1 - \cos \phi L^*)}}{\cos \phi L^*}$$

- Step 5:** Calculate the load-deflection result using the results from Steps 3 and 4.



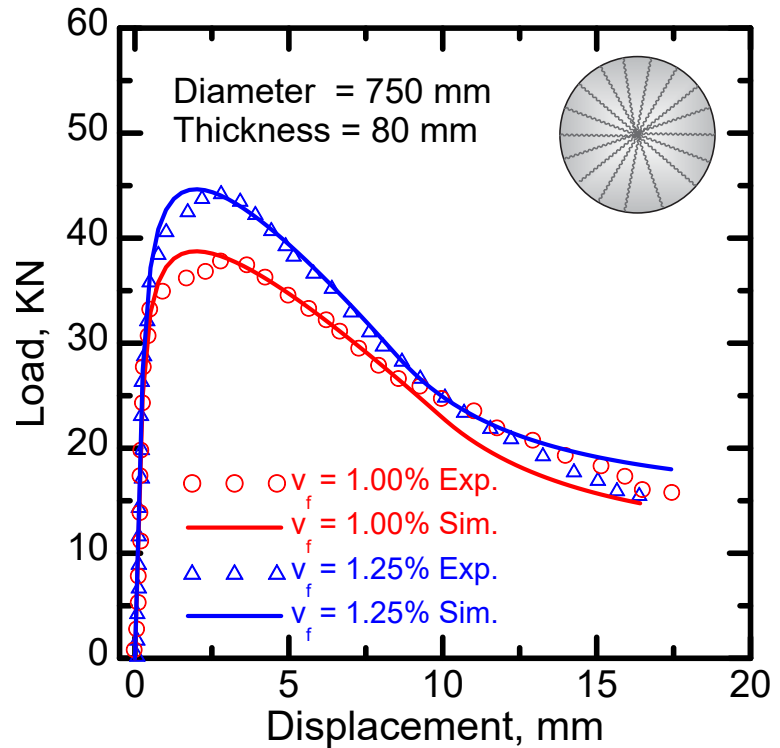
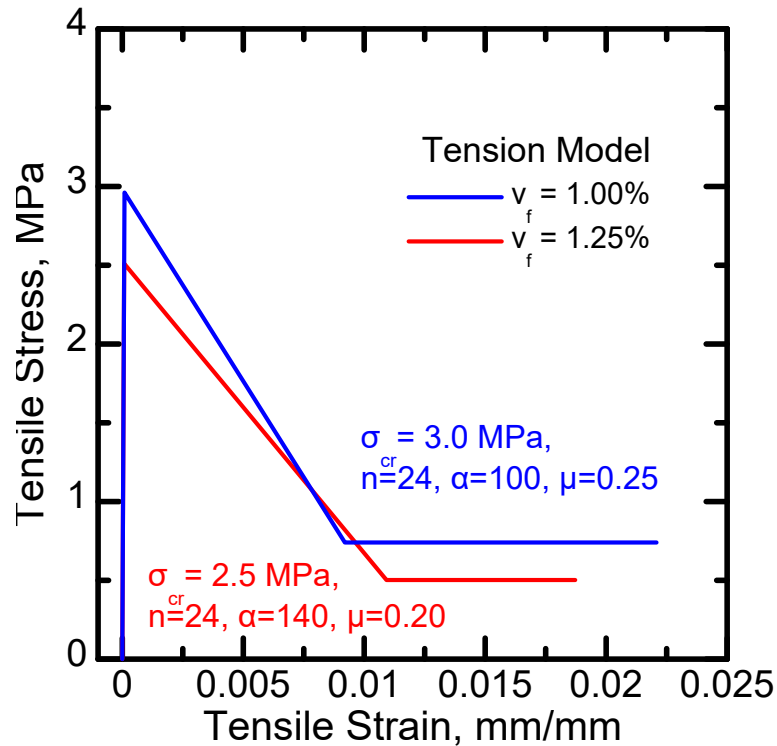
# Experimental Verification

## Data Set 3 – Round Panel

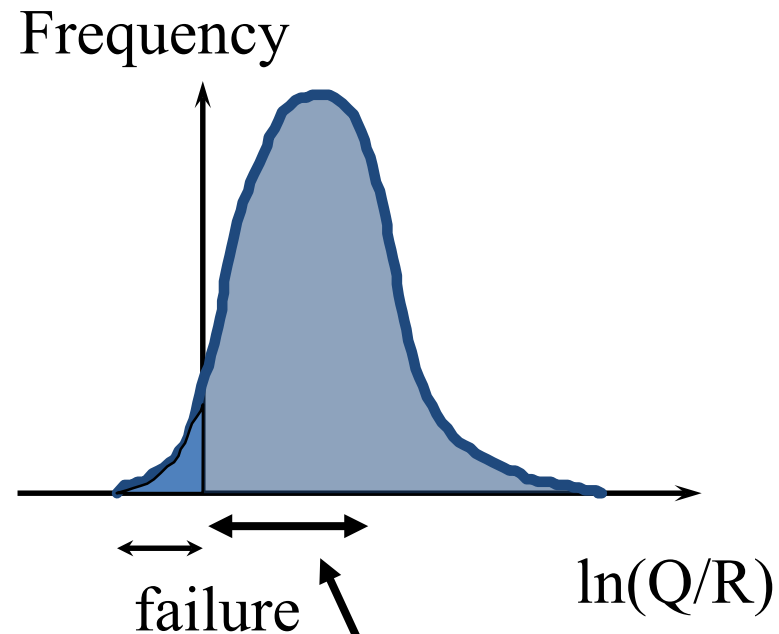
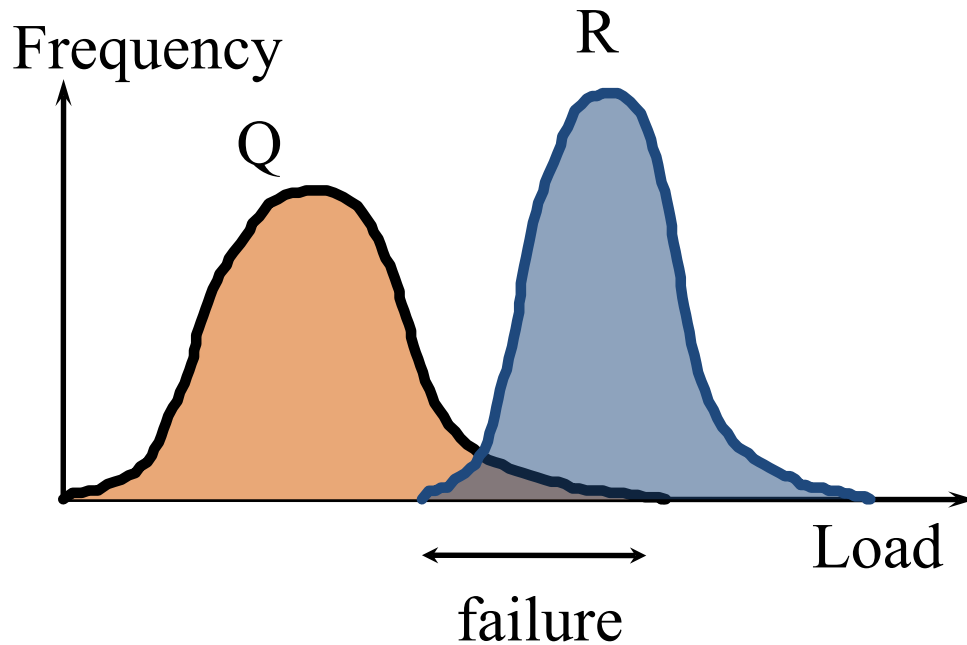
Round panel with point load at center

Diameter is 750 mm, thickness is 80 mm

Steel fiber  $v_f = 1.00\%$  and  $1.25\%$



# Probability based model



$V_R$  = Coefficient of variation of R  
 $V_Q$  = Coefficient of variation of Q

$$\beta = \frac{\ln(R/Q)}{\sqrt{V_R^2 + V_Q^2}}$$

$\beta$ , number of standard deviations

# Conclusions

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- Moment curvature response of fiber reinforced concrete composites can be derived explicitly
- The material model can be used to predict the flexural behavior of:
  - strain hardening materials
  - strain softening materials
    - deflection hardening
    - deflection softening
- Direct application of uniaxial tensile stress strain response under-predicts the flexural response for material that is brittle.
- Ratio of Flexural strength to tensile strength can be predicted using the present model.

# Conclusions

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- Closed form solutions of the moment curvature response provide a direct and universal method to design and analyze FRC materials and structures.
- The higher the brittleness of the material the higher differences are observed in correlating the tensile and flexural responses.
- When used in conjunction with the moment-area method and crack localization rules, moment- curvature responses can be used to predict load deflection response of various loading geometries.
- There are direct applications for design and analysis of FRC structures.

# References used

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- 1. Maalej, M. and Li, V.C. "Flexural/Tensile Strength Ratio in Engineering Cementitious Composites" *Journal of Materials in Civil Engineering*, 6(4), 1999 pp. 513-513
- 2. Marikunte, S., and Aldea, C., and Shah, S.P., "Durability of Glass Fiber Reinforced Cement Composites: Effect of Silica Fume and Metakaolin," *Adv Cement Based Materials*, V.5, No.3-4, Apr.-May. 1997, pp. 100-108.
- 3. Lim, T.Y., Paramasivam, P., and Lee, S.L. (1987). "Analytical Model for Tensile Behavior of Steel-Fiber Concrete." *ACI Material Journal*, 84(4), 286-551.
- 4. Lim, T.Y., Paramasivam, P., and Lee, S.L. (1987). "Bending Behavior of Steel-Fiber Concrete Beams." *ACI Structural Journal*, 84(6), 524-536.
- 5. Dupont, D., "Modeling and Experimental Validation of The Constitutive Law (s-e) and Cracking Behavior of Steel Fiber Reinforced Concrete", PhD Dissertation, Catholic University of Leuven, 2003
- 6. Soranakom, C., and Mobasher, B., "Closed Form Solutions for Flexural Response of Fiber Reinforced Concrete Beams," *Journal of Engineering Mechanics*, August, 2007.

|                                      | Stage I                | Stage II                                  | Stage III                                     |
|--------------------------------------|------------------------|-------------------------------------------|-----------------------------------------------|
|                                      | $(0 < \lambda \leq 1)$ | $(1 < \lambda \leq \omega)$               | $(\lambda > \omega)$                          |
| $\frac{F_{c2}}{bdE\varepsilon_{cr}}$ | -                      | -                                         | $-\frac{\omega k(\omega - \lambda)}{\lambda}$ |
| $\frac{F_{c1}}{bdE\varepsilon_{cr}}$ | $\frac{\lambda}{4}$    | $\frac{\lambda k}{2}$                     | $\frac{\omega^2 k}{2\lambda}$                 |
| $\frac{F_{t1}}{bdE\varepsilon_{cr}}$ | $\frac{\lambda}{4}$    | $\frac{k}{2\lambda}$                      | $\frac{k}{2\lambda}$                          |
| $\frac{F_{t2}}{bdE\varepsilon_{cr}}$ | -                      | $\mu - \frac{\mu(\lambda + 1)k}{\lambda}$ | $\mu - \frac{\mu(\lambda + 1)k}{\lambda}$     |



# Normalized height of comp. and tension zones for each stage of tensile strain at bottom fiber, $\beta$

| Normalized height  | Stage 1<br>( $0 < \beta \leq 1$ )<br>and<br>( $\lambda \leq \omega$ ) | Stage 2.1<br>( $1 < \beta \leq \alpha$ )<br>and<br>( $\lambda \leq \omega$ ) | Stage 2.2<br>( $1 < \beta \leq \alpha$ )<br>and<br>( $\omega < \lambda \leq \lambda_{cu}$ ) | Stage 3.1<br>( $\beta > \alpha$ )<br>and<br>( $\lambda \leq \omega$ ) | Stage 3.2<br>( $\beta > \alpha$ )<br>and<br>( $\omega < \lambda \leq \lambda_{cu}$ ) |
|--------------------|-----------------------------------------------------------------------|------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------|-----------------------------------------------------------------------|--------------------------------------------------------------------------------------|
| $\frac{h_{c2}}{d}$ | -                                                                     | -                                                                            | $\frac{k\beta - \omega(1-k)}{\beta}$                                                        | -                                                                     | $\frac{k\beta - \omega(1-k)}{\beta}$                                                 |
| $\frac{h_{c1}}{d}$ | $k$                                                                   | $k$                                                                          | $\frac{\omega(1-k)}{\beta}$                                                                 | $k$                                                                   | $\frac{\omega(1-k)}{\beta}$                                                          |
| <hr/>              |                                                                       |                                                                              |                                                                                             |                                                                       |                                                                                      |
| $\frac{h_{t1}}{d}$ | $1-k$                                                                 |                                                                              | $\frac{1-k}{\beta}$                                                                         |                                                                       | $\frac{1-k}{\beta}$                                                                  |
| $\frac{h_{t2}}{d}$ | -                                                                     |                                                                              | $\frac{(1-k)(\beta-1)}{\beta}$                                                              |                                                                       | $\frac{(1-k)(\alpha-1)}{\beta}$                                                      |
| $\frac{h_{t3}}{d}$ | -                                                                     | -                                                                            | -                                                                                           |                                                                       | $\frac{(1-k)(\beta-\alpha)}{\beta}$                                                  |

# Normalized stress at vertices for each stage of normalized tensile strain at bottom fiber ( $\beta$ )

| Normalized stress               | Stage 1<br>( $0 < \beta \leq 1$ )<br>and<br>( $\lambda \leq \omega$ ) | Stage 2.1<br>( $1 < \beta \leq \alpha$ )<br>and<br>( $\lambda \leq \omega$ ) | Stage 2.2<br>( $1 < \beta \leq \alpha$ )<br>and<br>( $\omega < \lambda \leq \lambda_{cu}$ ) | Stage 3.1<br>( $\beta > \alpha$ )<br>and<br>( $\lambda \leq \omega$ ) | Stage 3.2<br>( $\beta > \alpha$ )<br>and<br>( $\omega < \lambda \leq \lambda_{cu}$ ) |
|---------------------------------|-----------------------------------------------------------------------|------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------|-----------------------------------------------------------------------|--------------------------------------------------------------------------------------|
| $\frac{f_{c2}}{E\epsilon_{cr}}$ | -                                                                     | -                                                                            | $\omega\gamma$                                                                              | -                                                                     | $\omega\gamma$                                                                       |
| $\frac{f_{c1}}{E\epsilon_{cr}}$ | $\frac{\gamma\beta k}{1-k}$                                           | $\frac{\gamma\beta k}{1-k}$                                                  | $\omega\gamma$                                                                              | $\frac{\gamma\beta k}{1-k}$                                           | $\omega\gamma$                                                                       |
| <hr/>                           |                                                                       |                                                                              |                                                                                             |                                                                       |                                                                                      |
| $\frac{f_{t1}}{E\epsilon_{cr}}$ | $\beta$                                                               |                                                                              | 1                                                                                           |                                                                       | 1                                                                                    |
| $\frac{f_{t2}}{E\epsilon_{cr}}$ | -                                                                     | $1 + \eta(\beta - 1)$                                                        |                                                                                             | $1 + \eta(\beta - 1)$                                                 |                                                                                      |
| $\frac{f_{t3}}{E\epsilon_{cr}}$ | -                                                                     | -                                                                            | -                                                                                           |                                                                       | $\mu$                                                                                |

# Normalized force component for each stage of normalized tensile strain at bottom fiber ( $\beta$ )

| Normalized force component           | Stage 1<br>( $0 < \beta \leq 1$ )<br>and<br>( $\lambda \leq \omega$ ) | Stage 2.1<br>( $1 < \beta \leq \alpha$ )<br>and<br>( $\lambda \leq \omega$ ) | Stage 2.2<br>( $1 < \beta \leq \alpha$ )<br>and<br>( $\omega < \lambda \leq \lambda_{cu}$ ) | Stage 3.1<br>( $\beta > \alpha$ )<br>and<br>( $\lambda \leq \omega$ ) | Stage 3.2<br>( $\beta > \alpha$ )<br>and<br>( $\omega < \lambda \leq \lambda_{cu}$ ) |
|--------------------------------------|-----------------------------------------------------------------------|------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------|-----------------------------------------------------------------------|--------------------------------------------------------------------------------------|
| $\frac{F_{c2}}{bdE\varepsilon_{cr}}$ | -                                                                     | -                                                                            | $\frac{\omega\gamma}{\beta}(\beta k + \omega k - \omega)$                                   | -                                                                     | $\frac{\omega\gamma}{\beta}(\beta k + \omega k - \omega)$                            |
| $\frac{F_{c1}}{bdE\varepsilon_{cr}}$ | $\frac{\beta\gamma k^2}{2(1-k)}$                                      | $\frac{\beta\gamma k^2}{2(1-k)}$                                             | $\frac{\omega^2\gamma}{2\beta}(1-k)$                                                        | $\frac{\beta\gamma k^2}{2(1-k)}$                                      | $\frac{\omega^2\gamma}{2\beta}(1-k)$                                                 |
| $\frac{F_{t1}}{bdE\varepsilon_{cr}}$ | $\frac{\beta}{2}(1-k)$                                                |                                                                              | $\frac{(1-k)}{2\beta}$                                                                      |                                                                       | $\frac{(1-k)}{2\beta}$                                                               |
| $\frac{F_{t2}}{bdE\varepsilon_{cr}}$ | -                                                                     |                                                                              | $\frac{(1-k)(\beta-1)(\eta\beta-\eta+2)}{2\beta}$                                           |                                                                       | $\frac{(1-k)(\alpha-1)(\eta\alpha-\eta+2)}{2\beta}$                                  |
| $\frac{F_{t3}}{bdE\varepsilon_{cr}}$ | -                                                                     | -                                                                            | -                                                                                           |                                                                       | $\frac{(1-k)(\beta-\alpha)\mu}{\beta}$                                               |

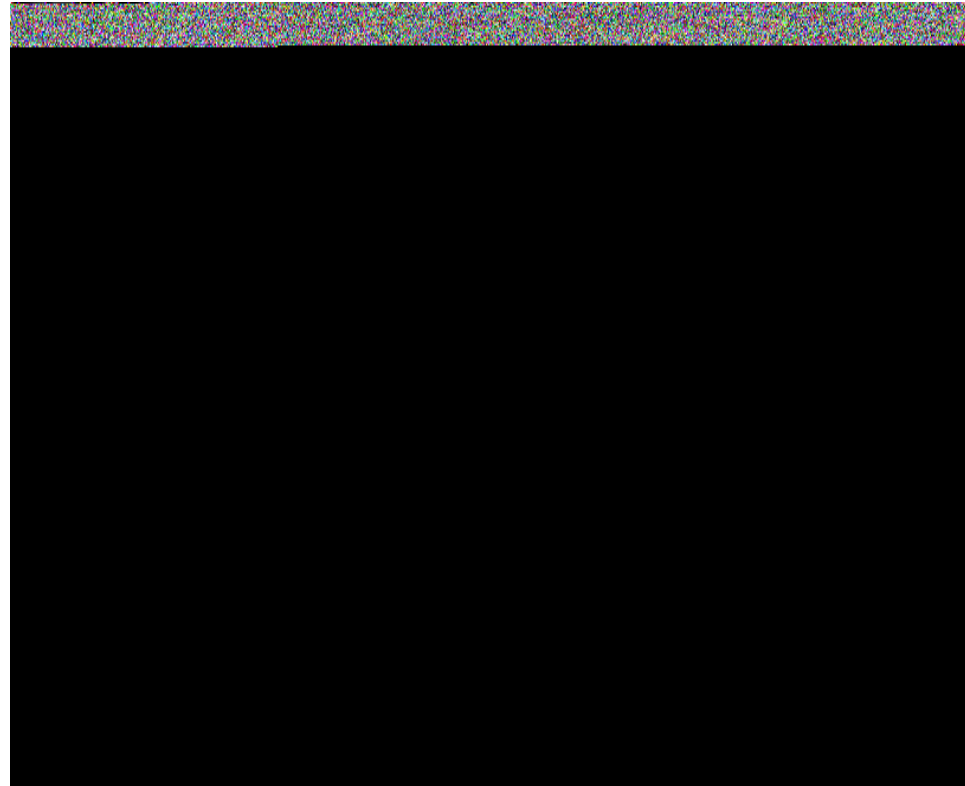
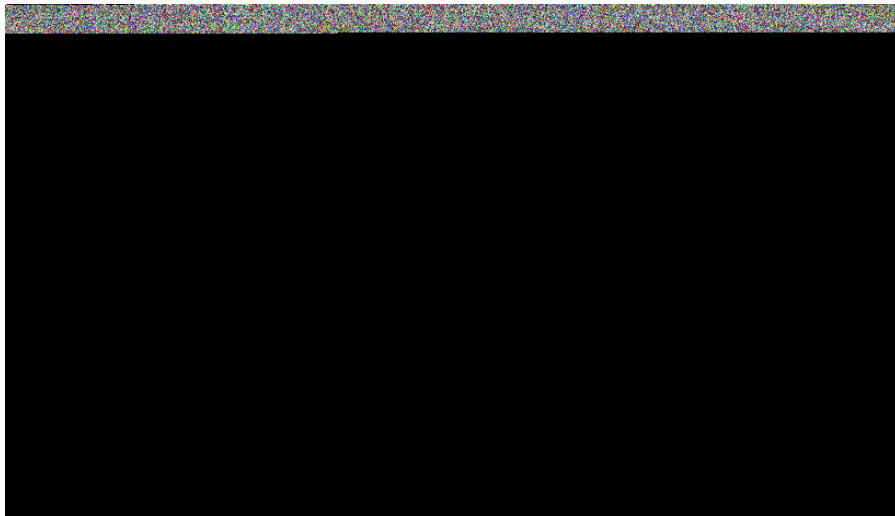
# Normalized moment arm for each stage of normalized tensile strain at bottom fiber ( $\beta$ )

| Normalized<br>moment<br>arm                 | Stage 1<br>( $0 < \beta \leq 1$ )<br>and<br>( $\lambda \leq \omega$ ) | Stage 2.1<br>( $1 < \beta \leq \alpha$ )<br>and<br>( $\lambda \leq \omega$ )             | Stage 2.2<br>( $1 < \beta \leq \alpha$ )<br>and<br>( $\omega < \lambda \leq \lambda_{cu}$ ) | Stage 3.1<br>( $\beta > \alpha$ )<br>and<br>( $\lambda \leq \omega$ )                        | Stage 3.2<br>( $\beta > \alpha$ )<br>and<br>( $\omega < \lambda \leq \lambda_{cu}$ ) |
|---------------------------------------------|-----------------------------------------------------------------------|------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|
| $\frac{y_{c2}}{d}$                          | -                                                                     | -                                                                                        | $\frac{2\omega(1-k)}{3\beta}$                                                               | -                                                                                            | $\frac{2\omega(1-k)}{3\beta}$                                                        |
| $\frac{y_{c1}}{d}$                          | $\frac{2}{3}k$                                                        | $\frac{2}{3}k$                                                                           | $\frac{\beta k + \omega(1-k)}{2\beta}$                                                      | $\frac{2}{3}k$                                                                               | $\frac{\beta k + \omega(1-k)}{2\beta}$                                               |
| <hr style="border-top: 1px dashed black;"/> |                                                                       |                                                                                          |                                                                                             |                                                                                              |                                                                                      |
| $\frac{y_{t1}}{d}$                          | $\frac{2}{3}(1-k)$                                                    |                                                                                          | $\frac{2}{3} \frac{(1-k)}{\beta}$                                                           |                                                                                              | $\frac{2}{3} \frac{(1-k)}{\beta}$                                                    |
| $\frac{y_{t2}}{d}$                          | -                                                                     | $\frac{2\eta\beta^2 - \eta\beta - \eta + 3\beta + 3}{3\beta(\eta\beta - \eta + 2)}(1-k)$ |                                                                                             | $\frac{2\eta\alpha^2 - \eta\alpha - \eta + 3\alpha + 3}{3\beta(\eta\alpha - \eta + 2)}(1-k)$ |                                                                                      |
| $\frac{y_{t3}}{d}$                          | -                                                                     | -                                                                                        | -                                                                                           | $\frac{(\alpha + \beta)}{2\beta}(1-k)$                                                       |                                                                                      |

# Drainage trough and cover

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- structural applications such as light weight drainage trough and covers



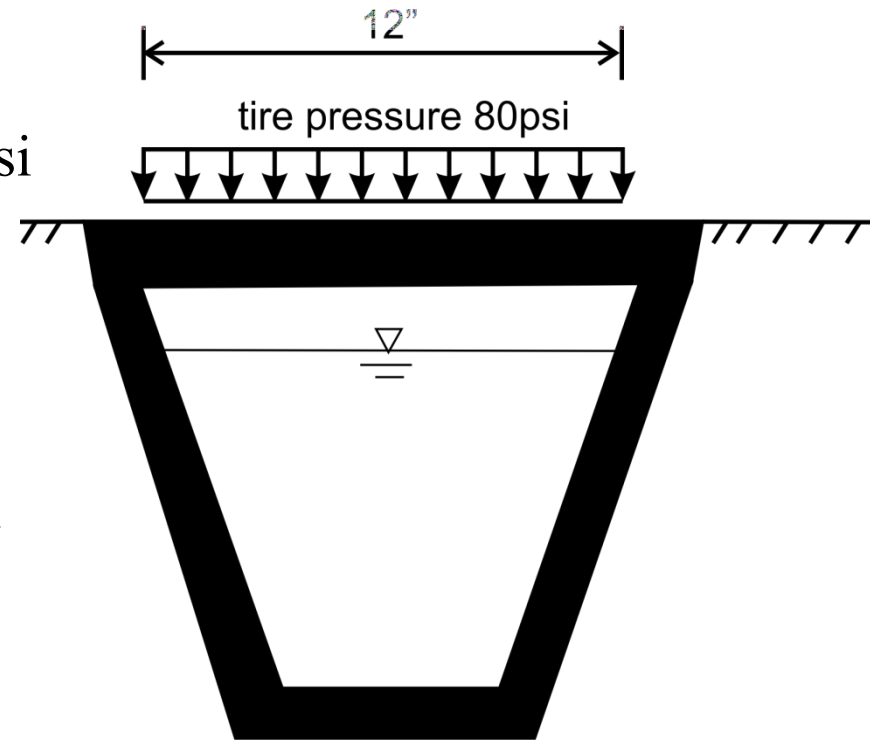
# Design example of strain hardening

Determine the moment capacity of the drainage trough cover made of nominal 3" thick. Can it withstand the truck tire pressure of 80 psi

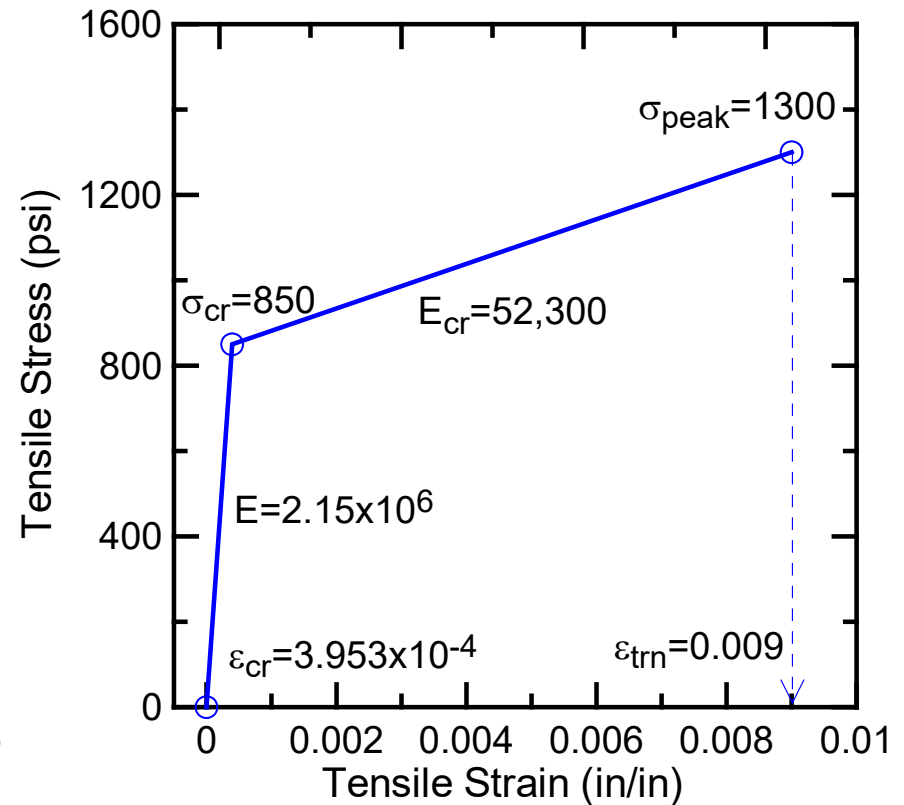
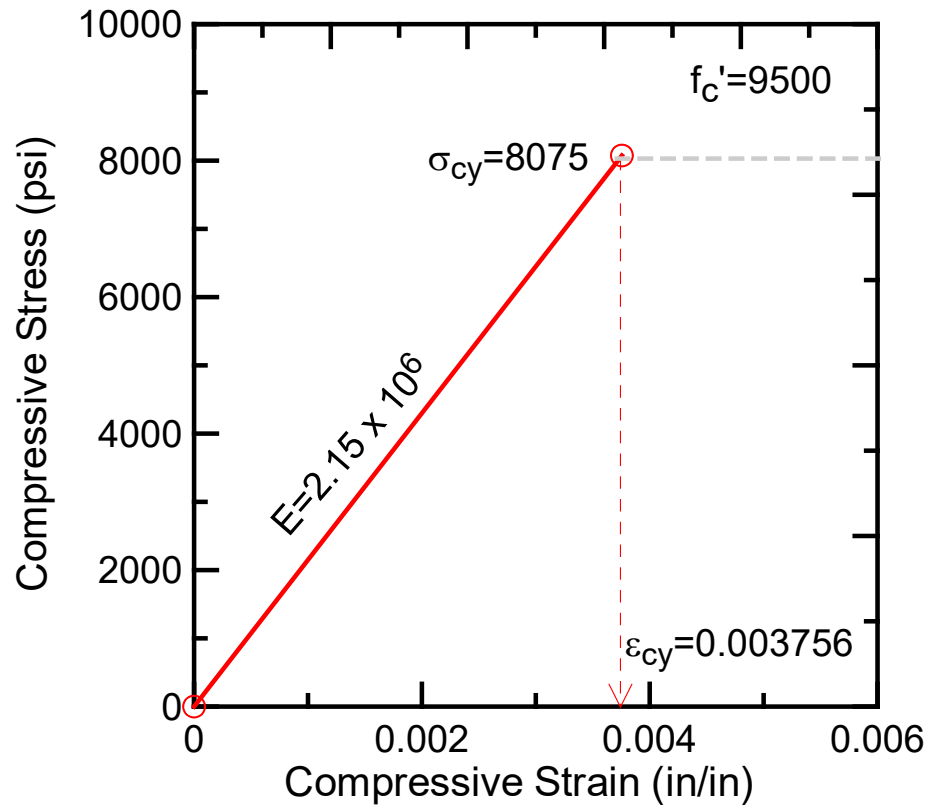
From manufactures, the GFRC used has

- Compressive strength,  $f_c' = 9,500$  psi
- Young modulus,  $E = 2.15 \times 10^6$  psi
- Cracking tensile strength,  $\sigma_{cr} = 850$  psi
- Peak tensile strength,  $\sigma_{peak} = 1,300$  psi
- Tensile strain at peak tensile strength,  
 $\epsilon_{trn} = 0.009$  in/in

The problem can be considered as a simple beam of one inch strip



# Material Properties used



# Design example of strain hardening

## 1. Calculate material parameters

- Cracking tensile strain, 
$$\varepsilon_{cr} = \frac{\sigma_{cr}}{E} = \frac{850}{2.15 \times 10^6} = 3.953 \times 10^{-4}$$

- Normalized tensile strain at peak tensile stress 
$$\alpha = \frac{\varepsilon_{trn}}{\varepsilon_{cr}} = \frac{0.009}{3.953 \times 10^{-4}} = 22.8$$

- Normalized post crack tensile modulus

$$\omega = \frac{0.85 \times f'_c}{\sigma_{cr}} = \frac{0.85 \times 9500}{850} = 9.5$$

$$\eta = \frac{E_{cr}}{E} = \frac{\sigma_{peak} - \sigma_{cr}}{(\varepsilon_{trn} - \varepsilon_{cr})E} = \frac{1300 - 850}{(0.009 - 0.0003953) \times 2.15 \times 10^6} = 0.0243$$

- Normalized compressive yield strain



# How to use design charts

- Use the maximum normalized tensile strain to get the normalized ultimate moment  $M'$
- First, use  $M'$ - $\beta$ & $\eta$  chart
  - draw a vertical from  $\beta=\alpha=22.8$  to the curve at  $\eta = 0.0243$
  - then, draw a horizontal line to the left, get  $M'=3.2$
- Check the normalized compressive stress by  $\lambda_{dv}$ - $\beta$ & $\eta$  chart
  - continue the vertical line to the curve at  $\eta = 0.0243$
  - draw a horizontal line to the right, get, get  $\lambda_{dv}=7.8$
  - $\lambda_{dv}=7.8 < \omega=9.5 \Rightarrow$  compressive strain is elastic  $\Rightarrow$  OK
- Thus,  $M'=3.2$  for this material used

# Design example of strain hardening

2. Calculate cracking moment  $M_{cr} = \frac{\sigma_{cr} b h^2}{6} = \frac{850 \times 1 \times 3^2}{6} = 1,275 \text{ lb-in/in}$

3. Calculate moment capacity  $\phi M_n = \phi M_{cr} M' = 0.65 \times 1275 \times 3.2 = 2,652 \text{ lb-in/in}$

4. Ignore self weight of the cover (negligible compared to the truck load)

$$w_u = 1.2(\approx 0) + 1.6(80) = 128 \text{ psi}$$

5. Calculate the ultimate moment at mid span

$$M_u = \frac{w_u l^2}{8} = \frac{128 \times 12^2}{8} = 2,304 \text{ lb-in/in} < \phi M_n 2,652 \text{ lb-in/in} \Rightarrow \text{OK}$$

Thus, the 3" thick is sufficient